A Continuous Hidden Markov Model for Daily US-Equity Stylized-Fact Reproduction

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Abstract

Synthetic generators of daily equity-return time series support stress testing and scenario design, yet reproducing the three Cont stylized facts simultaneously, heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the absolute-return autocorrelation function (ACF), remains difficult. We evaluate a continuous hidden Markov model (CHMM) trained by Baum–Welch as a synthetic-data generator on daily US-equity returns, scoped to the SPY 2014–2026 window plus a sector-balanced 30-ticker generalisation panel and a six-asset US-equity copula extension. A unified scaffold supports four interchangeable emission families (Gaussian, Student-t, Laplace, and Generalized Error Distribution), all sharing the same forward-backward recursions and quantile-based initialization with the M-step as the only architectural difference. A non-equity stress test on GLD/SLV ETFs collapses to 0% OoS KS under static IS-fitted CHMM, so the scope claim does not extend to other asset classes without periodic refit.

The held-out-clean operating point $K^\star = 6$, selected by held-out per-observation log-likelihood and held-out KS on the strictly pre-2020 slice, is the headline default for synthetic-data consumers: the four CHMM emission variants attain 88.3–92.6% IS and 76.3–79.0% OoS Kolmogorov–Smirnov pass rates on the single OoS window, with absolute-return ACF MAE within 0.003 of GARCH(1,1) and CHMM-N's simulated kurtosis 5.26 matching observed OoS 5.29 within 0.03. The single OoS window sits at the upper end of a six-fold rolling-origin walk-forward whose CHMM-N median OoS KS is 67.7% at $K = 18$ and 62.1% at $K^\star = 3$, with two stress folds (COVID 2020 and the 2022 rate-hike onset) below 10% that every generator in the panel rejects. The walk-forward median, not the single-window headline, is the operationally informative summary. A second held-out-clean operating point $K^\star = 3$ (held-out-likelihood and BIC) is the recommended default for risk-management consumers (CHMM-N: 89.7% IS / 80.1% OoS KS), and an extended-state-resolution sensitivity reference at $K = 18$ adds 1.5–3.5pp IS / OoS KS at the cost of moving from a strictly held-out criterion onto a multi-objective rule. We apply the textbook closed-form bilinear identity for the absolute-return ACF (Hamilton 1994; Krolzig 1997; Timmermann 2000) to recast the SPY 2014–2024 instance of the Rydén et al. (1998) low-K failure as an empirical effective-rank statement on the deflated transition matrix; the bound is non-binding at $K \geq 3$ on this SPY instance, where a single non-unit eigenvalue carries 93.6% of the lag-1 ACF, and the cross-ticker median dominant-mode share is 75.6% (minimum 32.6% on NEM), so the SPY headline overstates the universality.

A regime-conditional Value-at-Risk that propagates the one-step-ahead state forecast through the CHMM mixture passes the Christoffersen joint conditional-coverage test cleanly at $\alpha = 0.05$

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on the headline OoS window and on 19 of 24 rolling-origin folds (the two stress folds again drive the rejections); the $\alpha = 0.01$ rows are power-bounded ($\geq 80\%$ power only at breach-clustering eigenvalues $\rho \geq 0.50$ at $T_{\mathrm{OoS}} = 572$, Appendix A.6.17) and should be read as "not detectably worse than a strongly-clustered alternative" rather than as calibrated against any plausible alternative. Cross-ticker generalisation on the sector-balanced 30-ticker panel gives an IS KS median of 99.5% and an OoS median of 73.4% at $K = 18$, with single-name regime introductions driving the failure tail; a quarterly-refit version lifts the OoS median to 83.0% and reduces failures from 11/30 to 7/30, and a monthly-refit version lifts the OoS median further to 86.7% at 5/30 failures. The held-out-clean $K^\star = 6$ rebuild matches the $K = 18$ OoS KS distribution within sampling error (median 75.1% versus 73.4%, identical 11/30 failure count) at the cost of an order-of-magnitude wider per-ticker kurtosis residual; both operating points are reported in the body cross-ticker panel. Multi-asset synthesis on a US-equity universe composes through a rank-based Student-t copula on CHMM marginals: profile MLE selects $\nu^\star = 6$ on six assets at IS off-diagonal correlation MAE 0.027, but on OoS the Gaussian and Student-t copulas are statistically indistinguishable (off-diagonal MAE 0.202 versus 0.209, with quarterly rolling refit reducing the Student-t MAE to 0.185); the dependence-family choice is therefore an IS-calibration distinction, not an OoS-deployment one.

Keywords: Continuous Hidden Markov Model, Baum-Welch, Student-t Emissions, Stylized Facts, Volatility Clustering, Spectral ACF, Value-at-Risk, Synthetic-Data Generator.

1 Introduction

Synthetic generators of equity-return time series support stress testing, regulatory backtesting, scenario design, and the training of downstream machine-learning models on data that cannot be released for privacy or licensing reasons [1–3]. The bar for a useful generator is that it reproduce, simultaneously, the canonical stylized facts of daily returns: heavy-tailed marginals, negligible linear autocorrelation, and slow decay of the autocorrelation function (ACF) of absolute returns [4, 5]. Generalized autoregressive conditional heteroskedasticity (GARCH) models [6, 7] capture volatility clustering but not the multi-regime structure of the marginal; i.i.d. generators match the marginal but destroy temporal persistence; deep generators [8–10] reproduce some stylized facts at the cost of interpretability and reproducibility.

An influential negative result has shaped two decades of work on the regime-switching route. Rydén et al. [11] fitted Gaussian-emission hidden Markov models (HMMs) at $K = 2$–3 states and concluded that the geometric sojourn-time distribution of a standard Markov chain produces exponential ACF decay too fast to match the observed slow decay; Bulla and Bulla [12] restored ACF fidelity by replacing the Markov chain with a hidden semi-Markov chain at the cost of substantially increased complexity.

We identify the binding axis. The empirical Rydén et al. low-K failure compounds two constraints: a temporal one (the absolute-return ACF must admit slow-decay modes) and a distributional one (the marginal mixture must carry enough components to match observed kurtosis). We apply the textbook closed-form mixture-of-eigenvalues identity $\rho_{|G|}(\tau) = \sum_{k\geq 2} c_k \lambda_k^\tau / \sigma_{|G|}^2$ over the non-unit eigenvalues of $\mathbf{T}$ (folklore in the regime-switching literature; 13–15; restated in Section 3.8 for self-containedness only). The identity gives an algebraic upper bound of $K - 1$ on the number of decay modes through the rank of $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$. An empirical effective-rank diagnostic on the fitted $\hat{\mathbf{T}}$ (Appendix A.6.6) shows this bound is non-binding at $K \geq 3$ on equity-return data: a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$ and 96.8% at $K = 3$. The binding axis at $K \geq 3$ is therefore the marginal mixture, whose representational capacity must match the empirical kurtosis target. At $K = 2$ both axes bind simultaneously (mono-exponential ACF that can be tuned, two-component marginal that cannot), and the empirical failure is on the distributional axis; at $K \geq 3$ the temporal axis decouples and the marginal-mixture component count drives the joint fit. The semi-Markov route adds an explicit sojourn distribution to relax the temporal axis, but on equity-return data the temporal axis is already non-binding under standard Baum-Welch at $K \geq 3$, so the operationally informative comparison between the two scaffolds runs on the distributional axis, where the four CHMM emission families and the explicit-duration HSMM at $K^{\star} = 3$ trade KS, kurtosis, and ACF fidelity differently (Section 4).

We instantiate this on ten years of daily SPY ETF data with a 2.5-year held-out window, using a continuous HMM trained by Baum-Welch [16, 17] under quantile-based initialization in four emission variants sharing identical forward-backward recursions: CHMM-N (Gaussian), CHMM-t (per-state Student-t with ν_k updated by ECM golden-section in the tradition of 18, 19), CHMM-L (Laplace, weighted median and weighted MAD in closed form), and CHMM-GED (Generalized Error Distribution with per-state shape p_k via three-stage ECM, nesting CHMM-N at $p_k = 2$ and CHMM-L at $p_k = 1$). The empirical scope is daily US-equity stylized-fact reproduction: SPY headline, a sector-balanced 30-ticker generalisation panel, and a six-asset US-equity copula extension. A non-equity stress test on GLD and SLV ETFs (Appendix A.8.6) collapses to 0% OoS KS under static IS-fitted CHMM, so the scope claim does not extend to other asset classes without periodic refit; we report this as a documented limitation rather than a deferred follow-up.

The body headline is $K^{\star} = 6$, selected as the held-out-clean default by held-out per-observation log-likelihood and held-out KS on the strictly pre-2020 slice (2018-07–2019-12). At this operating point the four emission variants attain 88.3–92.6% IS / 76.3–79.0% OoS KS on the single OoS window with simulated kurtosis 5.22–5.86 against observed 7.68 IS / 5.29 OoS, and CHMM-N's simulated kurtosis 5.26 matches observed OoS 5.29 within 0.03. The single OoS window sits at the upper end of a six-fold rolling-origin walk-forward whose median CHMM-N OoS KS is 67.7% at $K = 18$ and 62.1% at $K^{\star} = 3$ (Table 50), with two stress folds (W2 COVID 2020 and W4 2022 rate-hike onset) below 10% that every generator in the panel rejects; the walk-forward distribution, not the single-window pair, is the operationally informative summary for production deployment. The held-out-likelihood and BIC choice $K^{\star} = 3$ is reported alongside as the lower-state-count default for risk-management consumers (CHMM-N: 89.7% IS / 80.1% OoS KS, $|G_t|$ ACF-MAE 0.0467). An extended-state-resolution sensitivity reference at $K = 18$ adds 1.5–3.5pp IS / OoS KS at the cost of moving from a strictly held-out criterion onto a multi-objective rule. Against a benchmark panel (i.i.d. bootstrap of 20; Gaussian and Laplace i.i.d.; GARCH(1,1) of 7 and the heavy-tailed extension of 21; MS-GARCH of 22), the four CHMM variants at $K = 18$ attain 93.6–95.6% IS and 80.8–85.7% OoS KS, with CHMM-N's $|G_t|$ ACF-MAE of 0.0509 matching GARCH(1,1)'s 0.0485 to within 0.003 (Table 3). The variants land at four kurtosis points (CHMM-N 5.0, CHMM-L 6.6, CHMM-GED 5.2 with a bimodal Gaussian-bulk / Laplace-tail $\hat{p}_k$ partition, penalised CHMM-t at $\lambda = 20$ at 8.56 IS / 7.07 OoS as the cleanest single-row joint match to observed 7.68 / 5.29; the unpenalised CHMM-t IS kurtosis of 14.4 is the per-state ν_k ECM lower-bracket pinning artefact discussed in Section 5). VaR and Expected-Shortfall envelopes bracket observed values at 1% and 5% on both windows.

Multi-asset synthesis on a US-equity universe composes through a rank-based Student-t copula on the CHMM marginals: profile MLE selects $\nu^{\star} = 6$ at IS off-diagonal correlation MAE 0.027, but on OoS the Gaussian and Student-t copulas are statistically indistinguishable (0.202 versus 0.209, both above the 0.030–0.027 IS spread by an order of magnitude). The dependence-family choice is therefore an IS-calibration distinction, not an OoS-deployment one; quarterly rolling refit reduces the Student-t OoS off-diagonal MAE to 0.185. On the sector-balanced 30-ticker panel (the sample design is 10 GICS sectors $\times 3$ large-cap representatives, itself one design choice among many) the IS distribution is tight (median KS 99.5%) and the OoS distribution is sector-stratified (Utilities, Information Technology, and Consumer Staples cluster above 90% median; Health Care collapses to 14.5% on weight-loss-drug regime introductions in LLY and 2024 healthcare-policy compression in UNH; 11/30 tickers below 60% OoS KS at the universe scale), reflecting the well-known stationarity-scope limit on financial time series [23–25] and motivating periodic refit at deployment [26].

Our contributions are empirical and architectural rather than theoretical. (i) We apply the textbook bilinear identity $\rho_{|G|}(\tau) = \sum_{k\geq 2} c_k \lambda_k^\tau / \sigma^2_{|G|}$ (Hamilton 1994 §22.2; Krolzig 1997 Ch. 3; Timmermann 2000) to the SPY 2014–2024 instance of the empirical Rydén et al. [11] low-K failure and show empirically that the algebraic rank bound on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is non-binding at $K \geq 3$ on this instance: a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$ on SPY (Appendix A.6.6). On the 30-ticker cross-section the dominant-mode share has median 0.76 and minimum 0.33 (Appendix A.6.7), so the SPY headline overstates the universality and the body framing is at the cross-ticker median. The substantive content is the empirical effective-rank statement applied to a known instance, not a re-derivation of the textbook identity. (ii) Four emission families (Gaussian, Student-t, Laplace, Generalized Error Distribution) under a unified ECM scaffold, where the M-step is the only architectural difference and CHMM-GED's per-state shape p_k adapts each state on the Gaussian-to-Laplace axis. (iii) A single-window IS / OoS empirical study on SPY plus a six-fold rolling-origin walk-forward, a sector-balanced 30-ticker generalisation panel (10 GICS sectors $\times 3$ large-cap representatives), and a six-asset Student-t copula extension; the headline-OoS-KS rankings are walk-forward-fragile and the operationally informative metric is the walk-forward median.

The paper is organized as follows. Section 2 reviews related work; Section 3 describes the data, the continuous HMM, the EM estimator, and the spectral mechanism for the absolute-return ACF; Section 4 presents the empirical study and the VaR / ES envelope calibration; Sections 5–6 discuss and conclude.

2 Related Work

Hidden Markov models in finance. Hamilton [27] introduced Markov-switching to economics; Schaller and Van Norden [28] showed monthly equity returns admit a two-state Markov-switching specification. The seminal evaluation against the Cont stylized facts is due to Rydén et al. [11]: Gaussian-emission HMMs at $K = 2$–3 states fail the slow ACF decay of absolute returns because the geometric sojourn-time distribution of a standard Markov chain produces exponential decay too fast to match the observed signature. Bulla and Bulla [12] restored fidelity with hidden semi-Markov models at substantially higher complexity. Nystrup et al. [29] extends the same line to long-memory volatility. Maximum-likelihood fitting of heavy-tailed innovations follows Peel and McLachlan [18] and Liu and Rubin [19]; the Generalized Error Distribution [30, 31] provides a continuous shape parameter that interpolates between Gaussian ($p = 2$) and Laplace ($p = 1$) and underlies our CHMM-GED variant in the same role that the per-state degrees of freedom ν_k plays in CHMM-t. The discrete-state predecessor of this paper, Alswaidan and Varner [3], reproduced the same three stylized facts via Laplace quantile binning plus a Poisson-jump duration mechanism; here we replace the discretization step with continuous emissions and recover the same target through the spectral mechanism of Section 3.8.

GARCH and its extensions. GARCH [6, 7] encodes volatility persistence through a single decay parameter but imposes a unimodal innovation distribution and fails distributional Kolmogorov–Smirnov on equity returns. Heavy-tailed innovations [21], asymmetric leverage extensions (EGARCH, 32; GJR-GARCH, 33; threshold GARCH, 34), and regime-switching extensions (MS-GARCH, 22, with the `MSGARCH` R package of 35 the standard reference implementation) close some gaps but inherit the unimodal-innovation constraint. Realised-variance frameworks [36, 37] provide a complementary forecasting baseline.

Deep generative models for financial time series. Generative adversarial network (GAN)-based generators reproduce the visual appearance of return series but typically fail volatility-clustering tests unless architectures encode temporal dependence explicitly [38, 39]; the TimeGAN family of Yoon et al. [8], the convolutional Wasserstein GAN (WGAN) QuantGAN of Wiese et al. [9], and the autoregressive-diffusion approach of Rasul et al. [10] are the standard references. We include QuantGAN as the deep-generative row in our extended panel (Appendix A.7.5); the present paper's contribution is grounded against parametric and non-parametric baselines.

Cross-asset dependence and synthetic-data evaluation. For multi-asset synthesis, the Single Index Model [40] propagates a market factor through a rank-one linear decomposition but distorts each non-market marginal; Sklar's theorem [41, 42] supports a cleaner decomposition in which each asset's fitted marginal is preserved and only the copula is estimated, with Gaussian and Student-t copulas [43, 44] the standard choices and rank-reordering simulation [45] the standard injection. Synthetic-data evaluation is anchored on the framework of Stenger et al. [46], the proper-scoring-rule literature [47–49], kernel two-sample tests [50], and signature-based metrics [51, 52].

3 Methods

3.1 Data and Excess Growth Rates

The single-asset analysis uses daily SPDR S&P 500 ETF (SPY) prices from January 3, 2014 to January 3, 2024 (a ten-year training window, $T_{\text{IS}} = 2{,}516$), with the period from January 4, 2024 through April 20, 2026 ($T_{\text{OoS}} = 572$) held out. Prices are split-adjusted but not dividend-adjusted; the data sources are described in the acknowledgments. For ticker i and trading day j the annualised excess growth rate is

$$G_{i,j} \equiv \frac{1}{\Delta t} \ln\left(\frac{P_{i,j}}{P_{i,j-1}}\right) - r_f, \qquad \Delta t = 1/252, \quad r_f = 0, \tag{1}$$

with $P_{i,j}$ the session volume-weighted average price (VWAP). The factor $1/\Delta t$ is a constant rescaling; all simulations and metrics are evaluated on the daily-scale series. The empirical study is organised along two pipelines (full schematic in Appendix A.2.4). *Pipeline A* fits a CHMM to each ticker independently and evaluates single-asset metrics; *Pipeline B* reuses the per-asset Pipeline-A fits as marginals and injects cross-asset dependence through a rank-based Student-t copula (§3.4).

3.2 Continuous Hidden Markov Model

We model G_t as a continuous HMM $\mathcal{M} = (\mathcal{S}, \mathbf{T}, \mathbf{B}, \boldsymbol{\pi})$ with K latent states, per-state emission density $b_k(x; \boldsymbol{\theta}_k)$, transition matrix $\mathbf{T} \in \mathbb{R}^{K \times K}$, and initial distribution $\boldsymbol{\pi}$ [17]. The stationary marginal is the K-component mixture

$$f(x) = \sum_{k=1}^{K} \bar{\pi}_k \, b_k(x; \boldsymbol{\theta}_k), \qquad \bar{\boldsymbol{\pi}} = \bar{\boldsymbol{\pi}} \mathbf{T}, \tag{2}$$

with $\bar{\boldsymbol{\pi}}$ the stationary distribution. Existence and uniqueness require irreducibility and aperiodicity of $\mathbf{T}$ (Assumption 1 in §3.8). We learn $\mathbf{T}$, $\boldsymbol{\theta}_{1:K}$, and $\boldsymbol{\pi}$ jointly by EM (§3).

The four emission families share a single forward-backward scaffold and quantile-based initialization; the M-step is the only architectural difference. *CHMM-N*: $b_k = \mathcal{N}(\mu_k, \sigma_k^2)$. *CHMM-t*: $b_k = t_{\nu_k}(\mu_k, \sigma_k)$ with per-state degrees of freedom $\nu_k \in [\nu_{\min}, \nu_{\max}]$. *CHMM-L*: $b_k = \mathrm{Laplace}(\mu_k, b_k)$. *CHMM-GED*: $b_k = \mathrm{GED}(\mu_k, \alpha_k, p_k)$ with per-state shape $p_k \in [p_{\min}, p_{\max}]$ that nests Gaussian ($p_k = 2$) and Laplace ($p_k = 1$). The architectural diagram is in Appendix A.2.5.

3.3 Benchmark Generators

We benchmark against representatives from each generator tier, all fit on the same IS window: i.i.d. bootstrap [20] (non-parametric distributional ceiling), Gaussian and Laplace i.i.d., GARCH(1,1) Gaussian [7] and Student-t [21], and Markov-switching GARCH (MS-GARCH; 22). The extended GARCH-family panel (EGARCH, GJR-GARCH, HAR-RV, MS-GARCH at $K \in \{3, 4, 6\}$), an explicit-duration semi-Markov foil, a deep-generative QuantGAN [9] row, plus stochastic-volatility (Taylor-Harvey-Ruiz-Shephard), Markov-switching multifractal [53], and Merton-jump-diffusion [54] baselines are reported in Appendix A.7.7 and A.7.8.

3.4 Cross-Asset Dependence (Pipeline B)

Pipeline B reuses each per-asset CHMM marginal and injects cross-asset dependence through a rank-based Student-t copula [43, 44]. Pseudo-uniform observations $u_{j,t} = \mathrm{rank}(g_{j,t})/(T+1)$ feed an analytic Kendall's-τ correlation estimate $\rho_{ij} = \sin(\pi \tau_{ij}/2)$ [55]; profile MLE selects $\nu^\star$. Sklar's theorem [41] guarantees that the rank-reordering simulator of Iman and Conover [45] preserves each fitted CHMM marginal exactly while coupling the asset ranks to the copula sample. SIM, Gaussian-copula, and truncated C-vine comparators are in Appendix A.8.1.

3.5 Evaluation Protocol

For each model we simulate $P = 1{,}000$ independent paths of length T following [46], under a deterministic global seed root 20260420. Headline metrics are the two-sample Kolmogorov–Smirnov [56, 57] pass rate at $\alpha = 0.05$, mean simulated excess kurtosis, and the absolute-return ACF mean absolute error (MAE) over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|. \tag{3}$$

A separate Value-at-Risk panel (§4.6) carries the unconditional Kupiec [58] and Expected-Shortfall envelope diagnostics. Block-bootstrap KS recalibration, Anderson–Darling, W_1, Hellinger, and CRPS auxiliaries are in Appendix A.3. Estimation of $(\mathbf{T}, \{\boldsymbol{\theta}_k\}, \boldsymbol{\pi})$ proceeds by EM in log-space. This section gives the shared scaffold and the family-specific M-step updates; the state-count selection that operationalises this scaffold is reported in Section 4.

3.6 Shared EM Scaffold

All four emission families share the same log-space forward-backward recursions and the same quantile-based initialization [59]: observations are sorted, partitioned into K equal-sized chunks, and each state's initial location and scale are seeded from the corresponding chunk; $T^{(0)}_{ij} = \pi^{(0)}_k = 1/K$. EM iterates to tolerance $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with `max_iter` $= 60$; convergence is typically reached in 20–40 iterations. Quantile-based initialization is what makes Baum-Welch reach a non-degenerate $K \geq 3$ fit: random initialization standard in the 1990s often converges to overlapping local optima, which is one of the two compounding sources of the low-K failure observed by Rydén et al. [11] (the other is the spectral one of Section 3.8). Algorithmic details and the forward-backward recursion are in Appendix A.2.

3.7 Family-Specific M-Step Updates

CHMM-N (Gaussian). The classical Baum-Welch M-step [16] is closed form: $\mu_k \leftarrow \sum_t \gamma_t(k) O_t / \sum_t \gamma_t(k)$ and $\sigma_k^2 \leftarrow \sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)$.

CHMM-t (Student-t via ECM). We adopt the ECM formulation of Peel and McLachlan [18] and Liu and Rubin [19], the same heavy-tailed innovation choice studied by Abanto-Valle et al. [60] for stochastic-volatility-in-mean models. The latent precision

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2} \tag{4}$$

is treated as an augmented E-step quantity, and the M-step updates

$$\mu_k \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} O_t}{\sum_t \gamma_t(k) u_{t,k}}, \qquad \sigma_k^2 \leftarrow \frac{\sum_t \gamma_t(k) u_{t,k} (O_t - \mu_k)^2}{\sum_t \gamma_t(k)} \tag{5}$$

have closed form conditional on ν_k. The degrees of freedom ν_k are updated by a golden-section search maximizing the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k) \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. The two-block CM step preserves the EM monotonicity guarantee on the compact bracket, so the incomplete-data log-likelihood is non-decreasing across iterations.

CHMM-L (Laplace). The weighted maximum-likelihood estimators are in closed form: μ_k is the posterior-weighted median of the observations (weights $\gamma_t(k)$, computed by cumulative-weight bracketing with linear interpolation between neighbouring order statistics), and $b_k \leftarrow \sum_t \gamma_t(k)|O_t - \mu_k| / \sum_t \gamma_t(k)$.

CHMM-GED (Generalized Error Distribution via three-stage ECM). For $b_k(x) = \mathrm{GED}(x; \mu_k, \alpha_k, p_k)$ the M-step decomposes into three CM updates per state: a bracketed $\mu_k \leftarrow \arg\min_\mu \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ via golden-section over $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$, then the closed-form scale $\alpha_k \leftarrow \left[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\right]^{1/p_k}$ with $W_k = \sum_t \gamma_t(k)$, and finally a bracketed $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k) \, \log b_k(O_t; \mu_k, \alpha_k, p)$ via golden-section over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Each CM step is a partial maximization, so EM monotonicity holds on the compact bracket. CHMM-N (all $p_k = 2$) and CHMM-L (all $p_k = 1$) are the boundary cases.

3.8 Spectral Mechanism for the Absolute-Return ACF

This subsection restates, for self-containedness, the textbook spectral mechanism behind the absolute-return ACF and the assumptions under which it holds. The identity itself is not novel: the closed-form bilinear cross-moment $\mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{T}^\tau\, \mathbf{m}$ for $|G_t|$ at lag τ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are folklore in the regime-switching literature (13, §22.2; 14, Ch. 3; the moment-generating-function form for general Markov-switching specifications is given by 15). Our contribution on this axis is exclusively the empirical *application* of the identity to recast the SPY 2014–2024 instance of the Rydén et al. [11] low-K failure as an effective-rank statement on the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, plus the empirical demonstration that the implied rank constraint is non-binding at $K \geq 3$ on this instance (Section 4). Readers familiar with Hamilton or Krolzig may safely skip to the rank statement at the end of the subsection. Identifiability, MLE consistency, ECM monotonicity, and rank-reordering marginal preservation are each one-line specialisations of standard results; we collect them as cited propositions in Appendix A.4 with one-sentence proof sketches.

Assumptions.

Assumption 1 (Irreducibility and aperiodicity)**.** *The transition matrix $\mathbf{T} \in \mathbb{R}^{K\times K}$ is irreducible and aperiodic on the state space $\{1, \dots, K\}$.*

Assumption 2 (Finite second moment and emission contrast)**.** *Each per-state emission density $b_k(\cdot; \boldsymbol{\theta}_k)$ has finite second moment, $\mathbb{E}_{b_k}[G^2] < \infty$, and at least two states differ in either location μ_k or scale σ_k.*

Under Assumption 1 the chain $\mathbf{T}$ has a unique stationary distribution $\bar{\boldsymbol{\pi}}$ satisfying $\bar{\boldsymbol{\pi}}\,\mathbf{T} = \bar{\boldsymbol{\pi}}$, and admits a spectral decomposition with $\lambda_1 = 1$ a simple eigenvalue and all other eigenvalues $\lambda_2, \dots, \lambda_K$ strictly inside the unit disk [61].

Spectral mechanism. Let G_t be a stationary CHMM and $\bar{\boldsymbol{\pi}}$ the stationary distribution of $\mathbf{T}$. Define the per-state absolute-return moment m_k and the per-state second moment M_k jointly,

$$m_k \;=\; \mathbb{E}\big[\,|G_t| \;\big|\; s_t = k\big]\,, \qquad M_k \;=\; \mathbb{E}\big[\,G_t^2 \;\big|\; s_t = k\big]\,, \qquad k = 1, \dots, K, \tag{6}$$

and stack them into vectors $\mathbf{m} = (m_1, \dots, m_K)^\top$ and $\mathbf{M} = (M_1, \dots, M_K)^\top$. The autocovariance derivation below uses $\mathbf{m}$ for $\rho_{|G|}(\tau)$ at $\tau \geq 1$; the lag-zero discussion at the end of the paragraph picks up M_k where the conditional product is the per-state second moment rather than m_k^2.

Conditioning on the latent state pair $(s_t, s_{t+\tau})$ and using that emissions are conditionally independent given the state sequence,

$$\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \sum_{i,j} \bar{\boldsymbol{\pi}}_i\,(\mathbf{T}^\tau)_{ij}\,m_i\,m_j = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{T}^\tau\,\mathbf{m}.$$

Under Assumption 1 the spectral decomposition of $\mathbf{T}$ reads $\mathbf{T}^\tau = \mathbf{1}\bar{\boldsymbol{\pi}}^\top + \sum_{k=2}^K \lambda_k^\tau\,\mathbf{v}_k\mathbf{w}_k^\top$ with $(\mathbf{v}_k, \mathbf{w}_k)$ the right and left eigenvectors of $\mathbf{T}$ associated with λ_k (normalised so that $\mathbf{w}_k^\top\mathbf{v}_k = 1$). The first term contributes $(\bar{\boldsymbol{\pi}}^\top\mathbf{m})^2 = (\mathbb{E}|G_t|)^2$, and subtracting the product of marginals yields the autocovariance

$$\mathrm{Cov}(|G_t|,\,|G_{t+\tau}|) \;=\; \sum_{k=2}^K c_k\,\lambda_k^\tau, \qquad c_k = \big(\mathbf{m}^\top\mathrm{diag}(\bar{\boldsymbol{\pi}})\,\mathbf{v}_k\big)\big(\mathbf{w}_k^\top\mathbf{m}\big), \tag{7}$$

and the autocorrelation

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^K w_k\,\lambda_k^\tau, \qquad w_k = c_k/\sigma_{|G|}^2. \tag{8}$$

Equation (8) is the closed-form mixture-of-eigenvalues identity for the absolute-return ACF: at every positive lag $\tau \geq 1$ the autocorrelation is a real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$. The w_k are real coefficients (not non-negative weights), but their cumulative sum is non-negative because it equals the between-state variance ratio of $|G_t|$. Complex-conjugate λ_k pairs combine into damped oscillatory modes of the form $|\lambda_k|^\tau \cos(\tau \arg\lambda_k + \phi_k)$; non-diagonalisable $\mathbf{T}$ adds polynomial-times-geometric prefactors via Jordan blocks. The lag-zero behaviour, the squared-return ACF, and a self-contained proof are in Appendix A.5.

Table 1. Leading non-unit-eigenvalue contributions to the lag-1 absolute-return ACF, CHMM-N on SPY IS at $K = 18$ (`results/diagnostics/spectral_rank.txt`). Modes ordered by $|w_k \lambda_k|$ descending. "cum lag-1" is the cumulative fraction of $\sum_{k=2}^{K} w_k \lambda_k$ accounted for through that rank. The dominant non-unit mode at $|\lambda_2| = 0.929$ carries 93.6% of the lag-1 ACF; the remainder is spread thinly across the lower-magnitude modes.

rank k	$\lvert\lambda_k\rvert$	$\lvert w_k\rvert$	$\lvert w_k \lambda_k\rvert$	cum. lag-1
2	0.929	0.324	0.301	0.936
3	0.335	0.012	0.0039	0.948
4	0.854	0.003	0.0022	0.955
5	0.231	0.008	0.0019	0.961
6	0.231	0.008	0.0019	0.967
$\vdots$				
13	0.119	0.006	0.0007	0.995

Rydén separation as a K-rank statement. At $K = 2$, equation (8) reduces to the mono-exponential form $\rho_{|G|}(\tau) = \lambda_2^\tau$ with $\lambda_2 = T_{11} + T_{22} - 1$; slow decay is matched by tuning $T_{11} + T_{22}$. The temporal axis is therefore satisfiable; the binding axis is the two-component marginal, which cannot match the empirical kurtosis target. At $K \geq 3$ the matrix admits multiple non-unit eigenvalues and the marginal mixture has enough components to match the observed tail. The Rydén et al. low-K failure is therefore a rank statement on two coupled axes: $\mathrm{rank}(\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top) \leq K - 1$ bounds the decay modes available to $\rho_{|G|}(\tau)$, and K simultaneously bounds the marginal mixture's component count. Empirically, ACF-MAE is essentially flat across $K \in \{3, 6, \ldots, 21\}$ while KS rises (Section 4); a direct effective-rank diagnostic (Appendix A.6.6) shows a single non-unit eigenvalue carries 93.6% of the lag-1 ACF at $K = 18$ on SPY, so the K-rank upper bound is non-binding at $K \geq 3$ on this instance. The cumulative-contribution profile is monotone and concentrated: Table 1 reports the leading non-unit modes at $K = 18$; three modes carry $\geq 95\%$ of the lag-1 ACF and twelve modes carry $\geq 99\%$ (Appendix A.6.6, full per-mode panel). Repeating the diagnostic on the 30-ticker cross-section (Appendix A.6.7) gives a wider distribution: the cross-ticker median dominant-mode share is 0.76, the interquartile range is $[0.66, 0.86]$, and the minimum is 0.33 on NEM. The SPY value is therefore in the right tail of the cross-ticker distribution, and the rank-non-binding claim is supported at the cross-ticker median rather than as a universal property of equity-return data.

EM monotonicity [62, 63] on the compact ν_k and p_k brackets, identifiability up to label permutation across the four emission families [64, 65], and MLE consistency [66, 67] under Assumptions 1–2 are collected with proof sketches in Appendix A.4.

4 Empirical Study

Reminder. Subsections 4.1–4.4 and 4.6 operate under **Pipeline A** (single-asset CHMM fit per ticker, no cross-asset coupling); only §4.5 operates under **Pipeline B** (Pipeline A marginals plus a rank-based copula).

4.1 Descriptive Statistics and Stylized Facts

All three Cont stylized facts are present in both windows (Figure 25 in Appendix A.6.23): heavy-tailed marginal (IS excess kurtosis 7.68, Jarque–Bera 6,416.6, $p < 0.001$), negligible linear autocorrelation in raw G_t (Ljung–Box $\mathrm{LB}_G \approx 5.5$ at lag 20 vs. $\chi^2_{20}(0.05) = 31.4$), and persistent volatility clustering in $|G_t|$ ($\mathrm{LB}_{|G|} = 2{,}959.3$, $p < 0.001$). The OoS window shares the same qualitative profile (excess kurtosis 5.29).

4.2 State-Count Selection

We sweep $K \in \{3, 6, 9, 12, 15, 18, 21\}$ for CHMM-N on the IS window and score each fit on held-out log-likelihood (two slices: estimation 2014–2021 / validation 2022–2023, and the strictly pre-2020 slice estimation 2014-01–2018-06 / validation 2018-07–2019-12), the four information criteria of Nguyen [68] (AIC, BIC, HQC, CAIC), and IS / OoS KS pass rates over 1,000 simulated paths. Table 2 consolidates the chosen $K^\star$ across criteria and slices. Absolute-return ACF-MAE is essentially flat (0.047–0.053) across the entire sweep, consistent with the spectral identity (8).

The body headline is $K^\star = 6$, selected as the held-out-clean default by held-out per-observation log-likelihood and held-out KS on the pre-2020 slice and by AIC on the 2022–2023 slice. The held-out-likelihood / BIC choice $K^\star = 3$ on the 2022–2023 slice is reported alongside as the lower-state-count default for risk-management consumers (Table 7). The $K = 18$ block is reported as an *extended-state-resolution sensitivity reference*: it improves IS / OoS KS by 1.5–3.5pp but is selected by a multi-objective rule that uses the OoS window and is therefore *not* held-out-clean. The cross-ticker panel (Table 5) is constructed at $K = 18$ for kurtosis fidelity; the held-out-clean $K^\star = 6$ rebuild matches the $K = 18$ KS distribution within sampling error (Appendix A.6.14). Full per-K panel and held-out details are in Appendix A.6 and A.9.

Table 2. Pre-registered K-selection across criteria and slices. Each cell is the $K^\star$ chosen by the row criterion on the column slice. "Pre-2020" is the strictly pre-COVID estimation 2014-01–2018-06 / validation 2018-07–2019-12 split (Appendix A.9); "2022–2023" is the rate-hike-onset slice (Appendix A.6). The body headline $K^\star = 6$ is selected by held-out per-observation log-likelihood and held-out KS on the pre-2020 slice and by AIC on the 2022–2023 slice. Information criteria favour $K^\star = 3$ on the pre-2020 slice, reflecting the parameter-count penalty on a 4.5-year estimation window. No held-out criterion selects $K = 18$.

Criterion	2022–2023 slice	Pre-2020 slice (2018-07 / 2019-12)
Held-out per-obs log-lik	3	**6**
Held-out KS	9	**6**
AIC	**6**	3
BIC	3	3
HQC	3	3
CAIC	3	3

4.3 Six-Generator Comparison (Pipeline A)

Table 3 reports IS / OoS validation for the benchmark generators of §3.3 and the four CHMM emission variants at three operating points, against an explicit-duration semi-Markov reference.

At the body-headline operating point $K^\star = 6$ (held-out-clean), CHMM-N attains 92.6% IS and 78.3% OoS KS at simulated kurtosis 5.26 (essentially the OoS observed 5.29) and $|G_t|$ ACF-MAE 0.0502, on par with GARCH (0.0485). With up to $K - 1 = 5$ non-unit eigenvalues at $K^\star = 6$ and 17 at $K = 18$, the mixture-of-eigenvalues sum (8) reproduces the slow absolute-return ACF that the Rydén et al. [11] $K = 2$–3 Gaussian setup could not recover on the SPY 2014–2024 instance of that experiment (Appendix A.6.12). The raw-return ACF-MAE column closes the third stylized fact: every CHMM row sits at 0.0234–0.0240, indistinguishable from the i.i.d. baselines that have zero linear ACF by construction. The four CHMM emission variants land at distinct points on the kurtosis axis without retuning K or the transition structure: CHMM-N at $K^\star = 6$ matches the OoS observed 5.29 within 0.83 units, CHMM-L sits at 5.86 IS, CHMM-GED at 5.22 IS, and (in the $K = 18$ sensitivity reference) the penalised CHMM-t with $1/\nu_k$ shrinkage at $\lambda = 20$ provides the cleanest IS / OoS heavy-tail match at 8.56/7.07 against observed 7.68/5.29. CHMM-GED's per-state shape $\hat{p}_k$ partitions bimodally into a Gaussian-bulk / Laplace-tail structure that replicates across seeds and tickers (Appendix A.6.25). On the OoS sample-CRPS, the four CHMM variants are statistically indistinguishable on Diebold–Mariano two-sided tests ($p > 0.45$ within-CHMM); we read this as evidence that the four families are interchangeable on this evaluation rather than as a per-family strength claim, with the family choice driven instead by the per-row kurtosis match (Table 8, §5). The MS-GARCH benchmark plateaus at 34–36% IS KS across $K \in \{2, 3, 6\}$, so the multi-state benefit is specific to the CHMM scaffold rather than to multi-state regime-switching per se on this dataset (full higher-K panel: Appendix A.7.7). A six-fold rolling-origin walk-forward (Appendix A.9, Table 50; body summary: Table 4) attains median OoS KS 67.7% at $K = 18$ and 62.1% at $K^\star = 3$; the single-window OoS sits at the upper end of these distributions, with two stress folds (W2 COVID, W4 2022 rate-hike onset) below 10% and explicitly out-of-distribution by KS for every generator in the panel. The walk-forward median, not the single-window OoS pair, is the operationally informative summary for production deployment: the headline single OoS window of 2024–2026 is a relatively low-stress slice (no regime introduction comparable to W2 / W4) and the single-window numbers therefore overstate what a deployed CHMM achieves on average across rolling windows.

Table 3. Headline generator comparison on SPY. 1,000 simulated paths, $\alpha = 0.05$; IS T = 2,516, OoS $T = 572$. KS = Kolmogorov–Smirnov pass rate; Kurt = mean simulated excess kurtosis (observed 7.68 IS, 5.29 OoS); ACF-MAE = mean absolute error of the absolute-return / raw-return ACF over 252 lags; CRPS = OoS sample continuous ranked probability score. The held-out-clean $K^\star = 6$ block is the body headline (marked with leading $\star$); the $K = 18$ block is an extended-state-resolution sensitivity reference. Bold marks column-best within the $K = 18$ block. † Unpenalised CHMM-t reference. The OoS-calibrated kurtosis match goes to CHMM-N at $K^\star = 6$ (sim 4.46 vs. observed 5.29); the IS-calibrated heavy-tail match goes to penalised CHMM-t at $\lambda = 20$, $K = 18$. Block-bootstrap KS recalibration (the temporally-aware variant) is in Appendix A.7.2; MS-GARCH at $K = 3, 4, 6$, the extended GARCH family, and a deep-generative QuantGAN row are in Appendix A.7.7.

	KS (%) ↑		Kurtosis		ACF-MAE ↓		CRPS OoS
Model	IS	OoS	IS	OoS	$\lvert G_t \rvert$	G_t	↓
Observed	–	–	7.68	5.29	–	–	–
Bootstrap	99.7	92.1	7.24	6.81	0.0628	0.0235	1.0398
Gaussian i.i.d.	0.0	1.0	−0.01	−0.03	0.0627	0.0236	1.0611
Laplace i.i.d.	99.1	88.5	2.95	2.91	0.0628	0.0236	1.0386
GARCH(1,1)	23.4	60.8	7.06	2.96	0.0485	0.0243	1.0440
GARCH(1,1)-*t*	57.3	80.8	15.13	–	**0.0316**	**0.0173**	–
MS-GARCH ($K = 2$)	27.7	38.7	4.73	–	0.0367	0.0173	–
MS-GARCH ($K = 3$)	36.1	33.1	4.10	–	0.0284	0.0173	–
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	0.0429	0.0173	–
CHMM-N ($K^\star = 3$)	89.7	80.1	3.86	–	0.0467	0.0240	1.0412
Body headline: held-out-clean reference panel at $K^\star = 6$.							
$\star$ CHMM-N ($K^\star = 6$)	92.6	78.3	5.26	4.46	0.0502	0.0238	1.0422
$\star$ CHMM-L ($K^\star = 6$)	88.3	76.3	5.86	–	0.0510	0.0235	1.0399
$\star$ CHMM-GED ($K^\star = 6$)	91.8	79.0	5.22	–	0.0518	0.0235	1.0393
ML HSMM-N ($K^\star = 3$)	98.4	91.0	3.46	3.38	0.0629	0.0236	–
Sensitivity reference at $K = 18$ (multi-objective rule, not held-out-clean).							
CHMM-N ($K = 18$)	94.1	81.8	5.04	4.44	0.0509	0.0237	**1.0384**
CHMM-t pen. ($\lambda = 20$, $K = 18$)	95.0	**85.8**	**8.56**	**7.07**	0.0542	0.0235	1.0392
CHMM-t unpen. ($K = 18$)†	95.6	85.7	14.35	10.71	0.0549	0.0234	1.0399
CHMM-L ($K = 18$)	93.6	80.8	6.63	6.18	0.0567	0.0234	1.0400
CHMM-GED ($K = 18$)	**95.2**	84.3	5.15	4.80	0.0548	0.0234	1.0406

Table 4. Six-fold rolling-origin walk-forward summary (CHMM-N, 5y train / 1y test, 500 paths per fold). Pulled from the full panel in Appendix A.9, Table 50. The headline single-window OoS sits at the upper end of the fold-wise distribution; the W2 (COVID) and W4 (2022 rate-hike onset) folds are out-of-distribution by KS for every generator in the panel.

	median KS (%)	$[Q_1, Q_3]$ KS (%)	min KS (%)	folds < 10%
$K^\star = 3$	62.1	$[7.2, 78.4]$	0.8	2/6
$K = 18$ (sensitivity)	67.7	$[8.2, 75.0]$	0.0	2/6

ML HSMM at $K^\star = 3$ as a co-headline result. Replacing the geometric sojourn distribution with an explicit truncated Pareto and refitting through the explicit-duration forward-backward of Yu [69] gives ML HSMM-N at $K^\star = 3$: 98.4% IS / **91.0**% OoS KS (the highest OoS KS in the panel), at simulated kurtosis 3.46 and $|G_t|$ ACF-MAE 0.0629. This is the strongest single-window OoS KS row in the panel, and the headline of the present paper is therefore best framed as "CHMM and

ML HSMM are complementary scaffolds on the same Markov backbone, with KS / ACF / kurtosis trade-offs that select between them by use case" rather than as "CHMM is the panel headline". The HSMM advantage on KS is offset by an absolute-return ACF-MAE that matches the i.i.d. baseline level (0.0629) rather than CHMM-N's 0.0467, because the fitted Pareto sojourn concentrates probability mass on a single low-volatility state and the regime-driven slow-ACF mechanism that the CHMM exploits at $K \geq 3$ is muted; the per-state $\bar{\boldsymbol{\pi}}$ concentration is documented in Appendix A.6.26. The complementary reading: ML HSMM wins on the marginal-distribution axis at low K, the CHMM family wins on the joint marginal-plus-volatility-clustering axis at moderate K. We do not report an ML HSMM at $K = 18$: the same construction collapses to a near-degenerate optimum on $T_{\mathrm{IS}} = 2{,}516$ at this state count (Appendix A.6.26), so the HSMM-vs-CHMM comparison at matched moderate K is left as a companion-paper direction; the present comparison is therefore HSMM at its best operating point ($K^\star = 3$) versus CHMM at its best operating point ($K^\star \in \{6, 18\}$), with the trade-off delineated above. Readers selecting a synthetic-data generator on the basis of OoS KS alone should prefer ML HSMM-N at $K^\star = 3$; readers requiring volatility-clustering reproduction or kurtosis fidelity should prefer CHMM at moderate K.

4.4 Cross-Ticker Generalization (Pipeline A)

Table 5 reports the penalised CHMM-t aggregate IS and OoS KS distribution across a sector-balanced 30-ticker panel (10 GICS sectors ×3 large-cap representatives) at the two body operating points $K^\star = 6$ and $K = 18$, both at $\lambda = 20$. The two operating points are at parity on the headline distributional axis: at $K^\star = 6$ the OoS KS median is 75.1% versus 73.4% at $K = 18$, with identical 11/30 failure counts. They differ on the kurtosis-residual axis by an order of magnitude: median residual 10.3 at $K^\star = 6$ versus 0.61 at $K = 18$. The choice between them is therefore use-case-driven: $K^\star = 6$ for held-out-clean KS-only consumers (no parameter tuned on the OoS window), $K = 18$ for kurtosis-fidelity consumers (multi-objective rule that uses the OoS window). At $K = 18$, the IS distribution is tight (median 99.5%, $[Q_1, Q_3] = [99.3, 99.6]$%); the OoS distribution is wider with median 73.4%, mean 66.8 ± 29.5%, and $11/30 = 36.7$% of tickers below 60%, concentrated in single-name regime introductions (LLY 7.6%, UNH 14.5%, NEM 5.6%). A one-way ANOVA on OoS KS by sector returns $F(9, 20) = 0.44$ at $p = 0.90$ with $\eta^2 = 0.16$ (the $n = 3$ per sector design is severely underpowered to detect sector effects of moderate size, so the result reads as "failures are ticker-specific" rather than as evidence against sector effects per se; Appendix A.6.13). A quarterly-refit version of this panel (Appendix A.6.22) shifts the OoS KS distribution from median 73.4% → **83.0**%, mean 66.8% → 77.2%, and reduces the failure count from 11/30 to 7/30, with the largest gains on regime-introduction tickers (LLY 7.6% → 83.7%, HD 41.4% → 82.2%). A monthly-refit version (Appendix A.6.34) interpolates the trade-off further: median OoS KS **86.7**%, mean 78.8%, and 5/30 failures, so each shorter refit window yields a roughly proportional further reduction in failure count without saturating.

Per-ticker $\hat{\lambda}^\star$ tuning. The body uniform $\lambda = 20$ is the right rate on the heavy-tailed defensives (NVDA, JNJ) but is over-shrunk on the moderate-tail tickers: a per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on the six body tickers under a 1.5pp KS-degradation tolerance (Appendix A.9, Table 53) selects $\lambda^\star \in \{0, 10, 20\}$, with QQQ at $\lambda = 0$ (no penalty) and SPY/JPM/AAPL at $\lambda = 10$. The honest operational reading is that $\lambda = 20$ is a reasonable uniform default but per-ticker tuning is recommended whenever a ticker's residual kurtosis at $\lambda = 0$ is within ~ 1 unit of observed; the production recipe is to evaluate the kurtosis-residual / KS-degradation trade-off on a held-out validation slice and pick the smallest λ that brings simulated kurtosis within 1.5 units of observed.

Table 5. Cross-ticker generalisation, sector-balanced 30-ticker panel (penalised CHMM-t at $\lambda = 20$, 1,000 paths, seed = 20260420). Aggregate distribution across $n = 30$ tickers, reported at both operating points: $K^\star = 6$ (held-out-clean default) and $K = 18$ (kurtosis-fidelity sensitivity reference). The KS distributions match within sampling error; the kurtosis residual widens by an order of magnitude at $K^\star = 6$ because the lower state-resolution cannot smooth the per-state ν_k shrinkage at uniform λ. Full per-ticker panel and per-sector rollup in Appendix A.6.4, Table 16; $K^\star = 6$ rebuild detail in Appendix A.6.14. The 10-sector × 3-large-cap sample design is one design choice among many; an equally weighted random draw of 30 large-caps would land on a different OoS tail.

Metric	$K^\star = 6$ (held-out-clean)	$K = 18$ (sensitivity reference)
IS KS pass rate (%) median	98.8	99.5
IS KS pass rate (%) mean ± s.d.	97.7 ± 3.7	99.3 ± 0.6
OoS KS pass rate (%) median	**75.1**	73.4
OoS KS pass rate (%) mean ± s.d.	66.5 ± 29.2	66.8 ± 29.5
OoS KS pass rate, quarterly refit (%)	—	**83.0**
$\lvert G_t \rvert$ ACF-MAE median	0.0403	0.0416
Kurtosis residual (sim − obs) median	10.34	**0.61**
Tickers OoS KS < 60% (IS-fixed)	11/30	11/30

4.5 Cross-Asset Dependence (Pipeline B): Student-t Copula

Pipeline B reuses the per-asset CHMM-N marginals from §4.4 and adds a rank-based Student-t dependence layer (§3.4). The body cross-asset construction is scoped to a US-equity universe: profile MLE on six US-equity assets (SPY, NVDA, JNJ, JPM, AAPL, QQQ; 200 paths) selects $\nu^\star = 6$ on the unit-spaced grid $\nu \in [4, 12]$ with Wilks 95% profile-LL CI of $[6, 7]$ (Appendix A.8); a half-unit-grid refinement plus $B = 200$ parametric bootstrap CI confirms $[6, 7]$ (Appendix A.8.7), excluding the Gaussian limit *on the IS slice*. A non-equity stress test on GLD and SLV ETFs over the same windows (Appendix A.8.6) shows OoS KS collapses to 0% on both ETFs under a static IS-fitted CHMM, confirming that the cross-asset claim does not extend to non-equity asset classes without periodic refit; we report this as a documented limitation rather than a deferred follow-up. Table 6 reports per-asset KS and the off-diagonal correlation MAE: 0.027 on IS for the Student-t copula, against 0.030 Gaussian, 0.068 truncated C-vine, and 0.076 SIM (full panel in Appendix A.8.1). On OoS the off-diagonal MAE rises to 0.209 for Student-t and 0.202 for Gaussian; the 0.007 difference is below the simulation-noise floor at $N_{\text{paths}} = 200$, so the dependence-family choice is an IS-calibration distinction rather than an OoS-deployment one. A quarterly-rolling refit on the six-asset universe reduces the OoS off-diagonal MAE to 0.185 (Appendix A.8.5). One additional caveat on the dependence-layer estimator: $\nu^\star$ is selected by profile MLE conditional on $\hat{\Sigma}$ obtained by Kendall's-τ inversion ($\rho_{ij} = \sin(\pi \tau_{ij}/2)$, exact under the Gaussian copula and approximate under the Student-t copula); the two-step estimator may be biased toward the Gaussian limit when the marginal kurtosis differs across assets, and a full one-shot Student-t copula MLE is left as a robustness extension.

Table 6. Cross-asset Student-t copula on CHMM-N marginals (Pipeline B; $\nu^\star = 6$; 200 paths). Per-asset KS pass rate at $\alpha = 0.05$ and correlation reproduction averaged over paths. Comparison against SIM, Gaussian, and truncated C-vine in Appendix A.8, Table 54.

	KS (%) ↑	
Ticker	IS	OoS
SPY	96.0	79.5
NVDA	98.5	53.5
JNJ	99.0	97.5
JPM	99.5	48.0
AAPL	99.0	93.5
QQQ	95.5	92.0
Median	98.8	85.8
Off-diag MAE IS	0.027	
Off-diag MAE OoS	0.209	
Off-diag MAE OoS (quarterly refit)	0.185	

4.6 Value-at-Risk and Expected-Shortfall Back-Test (Pipeline A)

The headline of this subsection is the regime-conditional Christoffersen-cc result: a VaR threshold built from the CHMM one-step-ahead state forecast passes the joint conditional-coverage test cleanly at every (K, α) on OoS, where every constant-across-t generator (CHMM unconditional, GARCH, bootstrap) rejects breach independence. We organise the diagnostics in three layers: (i) envelope bracketing, (ii) unconditional Kupiec coverage, (iii) the regime-conditional construction.

(i) Envelope bracketing. All four CHMM variants bracket observed historical VaR and ES at 1% and 5% within the $[5, 95]$% envelope on both windows; the IS CHMM-N median $\mathrm{VaR}_{0.01}$ at -6.64 (observed -6.38) is the closest point estimate among the six benchmark generators, with envelopes roughly 1.7× tighter than GARCH's. The full envelope panel for Bootstrap, GARCH, CHMM-N/-t/-L/-GED at 1% and 5% is in Appendix A.6.27, Table 31.

(ii) Unconditional Kupiec coverage. The four CHMM rows pass the unconditional Kupiec test on both windows: at $\alpha = 0.05$ on OoS, CHMM-N $\mathrm{LR}_{\mathrm{uc}} = 3.83$ against $\chi^2_1(0.05) = 3.841$, with the other three CHMM variants in 3.03–3.83. The repeated 3.83 across rows is a feature of $T_{\mathrm{OoS}} = 572$: the Kupiec statistic is a function of the integer breach count and at $\alpha = 0.05$ the values $k \in \{14, 19, 20\}$ produce statistics that pile against the critical value rather than spreading continuously. We treat unconditional Kupiec as a sanity check at this T_{OoS} and the conditional construction below as the substantive risk-management diagnostic.

(iii) Regime-conditional Christoffersen-cc *(headline)*. At each OoS day t we run the forward filter through $\mathcal{F}_{t-1} = R_{\mathrm{IS}} \cup R_{\mathrm{oos}}[1{:}t-1]$ to get the one-step-ahead state forecast

$$\mathbb{P}(s_t = k \mid \mathcal{F}_{t-1}) = \sum_i \mathbb{P}(s_{t-1} = i \mid \mathcal{F}_{t-1})\, T_{ik}, \tag{9}$$

and define $\widehat{\mathrm{VaR}}_t(\alpha) = F_t^{-1}(\alpha)$ where $F_t(\cdot) = \sum_k \mathbb{P}(s_t = k \mid \mathcal{F}_{t-1})\, F_k(\,\cdot\,; \boldsymbol{\theta}_k)$ is the family-appropriate predictive mixture CDF under IS-fixed $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$, with F_k the per-state Gaussian, Student-t, Laplace, or GED CDF (the body Gaussian construction reduces to $F_k = \Phi(\,\cdot\,; \mu_k, \sigma_k)$). Table 7 reports the back-test for CHMM-N at $K^\star = 3$ and $K = 18$ and the penalised CHMM-t at $\lambda = 20$ on OoS; every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$. The improvement in the independence statistic from the unconditional variant ($\mathrm{LR_{ind}} = 5.26$ at $K = 18, \alpha = 0.05$) to the conditional variant ($\mathrm{LR_{ind}} = 0.52$) is the regime-switching value proposition that the unconditional Kupiec headline did not exercise. The conditional pass extends to every emission family with the family-appropriate predictive density (CHMM-L, CHMM-GED) at $p_{cc} \geq 0.089$ throughout, so the regime-switching value proposition is intrinsic to the latent-state forecast and not specific to the Gaussian emission family (Appendix A.3.2, Table 51).

Power caveat at $\alpha = 0.01$. A Monte Carlo power calibration at $T_{\mathrm{OoS}} = 572$ (Appendix A.6.17, Table 25, $B = 5{,}000$ replicates per cell) shows the Christoffersen-cc test is correctly sized at 4–5% Type-I error under the i.i.d. null at both α levels, but the power profile is asymmetric. At $\alpha = 0.05$ (~ 28.6 expected breaches) the test reaches $\geq 80\%$ power against breach-clustering eigenvalues $\rho \geq 0.20$; at $\alpha = 0.01$ (~ 5.7 expected breaches) it reaches $\geq 80\%$ power only at $\rho \geq 0.50$, with rejection rate 42.6% at $\rho = 0.20$. The body $\alpha = 0.05$ "clean pass at every (K, α)" rows of Table 7 therefore carry strong power against moderate clustering; the $\alpha = 0.01$ rows should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The substantive risk-management diagnostic in this paper is therefore the $\alpha = 0.05$ row at the $T_{\mathrm{OoS}} = 572$ length used here; tighter α levels require either larger T_{OoS} or a higher-power conditional-coverage test.

Table 7. Regime-conditional VaR back-test, CHMM-N and penalised CHMM-t ($\lambda = 20$) on OoS ($T_{\mathrm{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. The forward filter is run under IS-fixed parameters $(\mathbf{T}, \{\boldsymbol{\theta}_k\})$ (no refit); only the one-step-ahead state-probability vector is updated through the OoS observations via Eq. (9). Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR_{uc}}$	$\mathrm{LR_{ind}}$	$\mathrm{LR_{cc}}$	p_{cc}
CHMM-N	3	0.01	9	1.57%	−4.56	1.62	2.36	3.98	0.14
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42	0.49
CHMM-N	18	0.01	9	1.57%	−5.20	1.62	2.36	3.98	0.14
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78	0.68
CHMM-t (λ=20)	3	0.01	8	1.40%	−5.59	0.82	2.81	3.63	0.16
CHMM-t (λ=20)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19	0.55
CHMM-t (λ=20)	18	0.01	10	1.75%	−6.25	2.65	1.98	4.62	0.10
CHMM-t (λ=20)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40	0.50

Walk-forward stability across six rolling-origin folds plus quarterly-refit and four-family extensions are in Appendix A.9, Tables 52 and 51: the conditional Christoffersen-cc passes at $\alpha = 0.05$ on 19/24 rows, with the two failures concentrated on the W2 (COVID) and W4 (2022 rate-hike onset) stress folds that the univariate walk-forward already flags as out-of-distribution.

5 Discussion

A continuous HMM trained by Baum-Welch reproduces the three Cont stylized facts simultaneously on SPY at two held-out-clean operating points: at $K^\star = 3$, CHMM-N attains 89.7% IS / 80.1% OoS KS at $|G_t|$ ACF-MAE 0.0467; at $K^\star = 6$ the four emission variants attain 88.3–92.6% IS / 76.3–79.0% OoS KS, with CHMM-N's simulated kurtosis 5.26 matching observed OoS 5.29 within

0.03. Four emission variants share the same scaffold and land at four distinct kurtosis points without retuning K or the transition structure. The mechanism behind the simultaneous KS / ACF fit is the spectral identity (8): at $K \geq 3$ with quantile-initialised Baum-Welch, the absolute-return ACF is a population-level real-coefficient sum of geometric decays at the non-unit eigenvalues of $\mathbf{T}$.

Replicating the slow-ACF behaviour at $K \geq 3$. Two issues compound in the Rydén et al. [11] low-K setup: a single self-transition rate produces a single exponential mode that cannot be simultaneously fast and slow, and the random initialisations standard in the 1990s often converge to degenerate local optima. Our $K = 2$ replication on SPY (Appendix A.6.12) clarifies the mechanism: ACF-MAE is essentially constant at ≈ 0.050 across quantile and five random-seed initialisations, matching the $K = 18$ CHMM-N. The binding low-K constraint is therefore distributional (KS at $K = 2$ drops to 67–72%): with modern initialisation the ACF is reproduced at any $K \geq 2$, while distributional fidelity drives the choice of K. The effective-rank diagnostic in Appendix A.6.6 makes this microscopically explicit: at the fitted $\hat{\mathbf{T}}$ for $K = 18$, a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF, and at $K = 3$ the dominant non-unit mode alone carries 96.8%.

Closing the kurtosis gap with heavy-tailed emissions. CHMM-N produces excess kurtosis 5.02 against observed 7.68: a Gaussian mixture's kurtosis is bounded by the variance of component means relative to their variances. CHMM-t addresses this through per-state Student-t emissions but at the cost of an IS overshoot to 14.57 from a single-state ECM lower-bracket pinning artefact (Appendix A.9); an exponential $1/\nu_k$ shrinkage at rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 at a 1pp KS cost. The $\lambda = 20$ choice is held-out-clean: refitting on the strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29) and selecting λ by validation log-likelihood on the 2018-07-02 to 2019-12-31 slice yields $\lambda^{\star} = 20$ on the grid $\{0, 5, 10, 20, 50, 100, 200\}$ (Appendix A.6.28). The penalty is held-out-clean at the SPY level but per-ticker (Appendix A.9, Table 53) the body uniform $\lambda = 20$ is over-shrunk on three of six tickers; the production recipe is the per-ticker validation rule of Section 4.4. We note the related question of bracket choice: $\nu_{\min} = 2.1$ in the body construction lies below the kurtosis-finiteness threshold $\nu = 4$, so the lower-bracket pinning is partly an artefact of the bracket itself; a $\nu_{\min} = 4$ bracket would force any kurtosis overshoot through the upper-bracket spread of the multi-state mixture, which is the structurally cleaner alternative. The penalty is doing the work of a more sensible bracket; we report the penalised construction here for direct comparability with Peel and McLachlan [18], Liu and Rubin [19] and leave the $\nu_{\min} = 4$ ablation to a companion-paper pass. CHMM-L sits between CHMM-N and CHMM-t at 6.75 without any shape parameter; CHMM-GED's bimodal $\hat{p}_k$ partition (Appendix A.6.25) is the data-driven analog of a hand-classified Gaussian-bulk / Laplace-tail hybrid. The four-way variant decision is summarised in Table 8; bootstrap CIs on observed kurtosis (Appendix A.6.15) show that the IS-OoS difference of ~ 2.4 units is not statistically distinguishable at conventional levels, so the variant rankings should be read as point-comparisons inside a wide CI envelope.

Practical identifiability and the effective state count at $K = 18$. The $K = 18$ operating point is partly nominal: the standardized-distance single-linkage diagnostic of Appendix A.6.29 collapses CHMM-N to 11/18 effective states and CHMM-t to 12/18, with 13/18 CHMM-t states pinned at the upper ν bracket on SPY. Three implications follow. (i) The numerical full-rank check on $\hat{\mathbf{T}}$ in Appendix A.4, which reports $\operatorname{rank}(\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^{\top}) = K - 1 = 17$ at tolerance 10^{-10}, exhausts only algebraic rank; the practically identifiable rank is closer to $K_{\text{eff}} - 1 \approx 10$. (ii) The AIC / BIC parameter count at $K = 18$ over-credits the model; an information-criterion penalty computed against K_{eff} rather than the nominal K would re-rank $K^{\star} \in \{6, 12\}$ in the held-out-clean panel

(Section 4.2, Table 2) more consistently with each other. (iii) The headline-metrics interpretation is not affected: simulation under the full $K = 18$ mixture and the metric panel are computed on whatever the converged model is; the question is whether the parameter count claimed against the data is honest at K_{eff} or at K_{nom}. We read this as a representational-economy observation: collapsed clusters carry small stationary-mass tails and contribute consistently to the simulation, but the model is best described as a K_{eff}-state model expressed in a K_{nom}-state parameterisation. The held-out-clean $K^\star = 6$ operating point (Section 4.2) is closer to the practically identifiable resolution and should be the default when reporting parameter counts; the $K = 18$ extended-state-resolution row is the right operating point when kurtosis fidelity is the priority.

Table 8. Variant decision guide. Recommended CHMM emission family by use-case priority. All four variants share the forward-backward scaffold and quantile-based initialisation; the M-step is the only architectural difference. Caveat: observed kurtosis differs across windows (7.68 IS, 5.29 OoS, a 31% drop) and no single variant matches both windows; CHMM-N at $K^\star = 6$ is the closest OoS match (gap 0.83).

Use-case priority	Recommended variant
Simplest closed-form fit; kurtosis fidelity secondary	CHMM-N
Cleanest IS kurtosis match; no shape parameter	CHMM-L
Cleanest OoS kurtosis match (deployment-window calibrated)	CHMM-N at $K^\star = 6$
Per-state heavy tails, IS-calibrated	CHMM-t pen. ($\lambda = 20$)
Adaptive per-state shape; data-chosen Gaussian-bulk / Laplace-tail mixture	CHMM-GED

Stationarity scope and operational deployment. The CHMM is trained on the assumption that returns are locally stationary across the IS window. Single-name equity returns are well known to carry sector-specific structural breaks (23; 24; surveyed by 25); the cross-ticker generalisation in Section 4.4 reflects this directly, with 11/30 tickers below 60% OoS KS, concentrated in single-name regime introductions. The mitigation is periodic refit (26; an online-EM formulation by 70): a quarterly-refit version of the cross-ticker panel (Appendix A.6.22) shifts the OoS KS distribution from median 73.4% → 83.0% and reduces failures from 11/30 to 7/30. The walk-forward W2 (COVID) and W4 (2022 rate-hike onset) stress folds remain out-of-distribution by KS for every generator under any refit cadence (Appendix A.9); these are regime introductions the IS distribution does not span. Discussion of the asymmetry between the W4 onset (catastrophic) and the headline 2024-2026 reversal (clean), the OoS regime trajectory under IS-fixed parameters, the per-ticker $\hat{\lambda}^\star$ for the penalised CHMM-t, and the IS / OoS kurtosis CI overlap are in Appendix A.9, A.6.18, and A.6.15.

Stylized-fact scope: leverage effect not addressed. The Cont (2001) stylized-fact list also includes the leverage effect (negative $\mathrm{Corr}(G_t, |G_{t+1}|)$ on equity returns) and gain-loss asymmetry; the present paper evaluates only the three symmetric stylized facts (heavy-tailed marginals, negligible linear ACF, slow $|G_t|$ ACF). All four CHMM emission families in this paper carry symmetric per-state densities (Gaussian, Laplace, Student-t, GED at any p_k), so a CHMM with these emissions cannot reproduce the leverage effect by construction at the per-state level; any leverage signal in the simulated paths must come from the asymmetric mixing weights of the regime probabilities, which is an unreliable mechanism for matching observed leverage. Skew-emission extensions (skew-t in the tradition of 71; skew-Laplace; skew-GED) are the natural fix and are deferred as companion-paper directions. We flag this as a known scope limitation: the body claim "CHMM reproduces the three Cont stylized facts" is correct as stated but does not cover the asymmetry stylized facts, and a generator-evaluation suite that includes leverage and gain-loss asymmetry would re-rank our headline panel against asymmetric baselines (EGARCH, GJR-GARCH, asymmetric MS-GARCH).

Baseline-implementation caveats. Three rows of our extended baseline panel (Appendix A.7.7) are repo-native re-implementations rather than reference-implementation runs and should be read with that caveat. (i) MS-GARCH at $K \in \{2, 3, 4, 6\}$ is fit by grid-initialised Nelder-Mead in our own implementation; the standard reference is the `MSGARCH` R package of Ardia et al. [35] which uses constrained-Bayesian estimation with proper-prior regularisation. The 27–37% IS KS plateau we observe could be partly an estimator-quality artefact rather than a structural ceiling on MS-GARCH; the body conclusion "the multi-state benefit is specific to the CHMM scaffold" should be read as "in our re-implementation, MS-GARCH does not close the gap" rather than as a verdict on the MS-GARCH literature. (ii) The QuantGAN row (Appendix A.7.5) is a three-conv-layer convolutional WGAN that is materially smaller than the seven-block TCN of Wiese et al. [9] and lacks the Lambert-W input pre-processing that the original authors credit with most of the tail-fidelity. The 0% IS / OoS KS pass rate is therefore the failure mode of *this* implementation, not a verdict on the WGAN literature; it functions as a deep-generative negative control rather than a competitive benchmark. (iii) The SM-CHMM row of Appendix A.7.7 is fit by a Viterbi-AR(1) plug-in estimator rather than by full ML on the explicit-duration HSMM; the ML HSMM-N at $K^\star = 3$ row of Table 3 is the proper HSMM benchmark and wins the OoS KS column. We do not claim that an ML-estimated HSMM at higher K would underperform CHMM at the same K, only that the present comparison is at each method's best operating point. A reference-implementation re-run of MS-GARCH and a full Wiese et al. TCN re-run are the right next-step controls; we report the present panel honestly with the caveats above.

Limitations. Skew-heavy-tailed emissions (skew-t or skew-Laplace) and explicit-duration semi-Markov sojourns at higher K are deferred as companion-paper directions; scaling the multi-asset construction to larger d via untruncated regular vines or factor copulas is the natural follow-up. The state-selection rule for the $K = 18$ extended-state-resolution panel uses information-criterion plus distributional pass-rate rather than strictly held-out log-likelihood; held-out-clean re-selection picks $K^\star = 3$ or $K^\star = 6$ depending on the validation slice (Appendix A.9), and the body $K^\star = 6$ block is the held-out-clean default for synthetic-data consumers. Independent-decade validation (a 1994–2004 vs. 2014–2024 SPY split) requires data outside the current 2014–2026 window and is logged for the next data-pipeline pass. The empirical scope is daily US-equity stylized-fact reproduction; FX, fixed-income, and commodity asset classes are not covered, and the GLD / SLV non-equity stress test in Appendix A.8.6 documents that the static IS-fitted CHMM does not transfer to those windows without periodic refit. The cross-asset dependence-family selection (Section 4.5) is an IS-calibration distinction: on the OoS window the Gaussian and Student-t copulas are statistically indistinguishable at $N_{\text{paths}} = 200$. The block-bootstrap KS recalibration of Appendix A.7.2 is reported on the IS window only; the OoS block-aware variant is logged for a companion robustness pass. A leverage-effect column on the headline panel and a full one-shot Student-t copula MLE on the six-asset universe (rather than the two-step Kendall's-τ inversion + profile-MLE construction) are the two most useful follow-up controls.

6 Conclusion

A continuous HMM trained by Baum-Welch with quantile-based initialization reproduces the three symmetric Cont stylized facts (heavy-tailed marginals, negligible linear ACF, slow $|G_t|$ ACF) on daily US-equity returns. At the held-out-clean body headline $K^\star = 6$ (selected by held-out per-observation log-likelihood and held-out KS on the strictly pre-2020 slice) the four emission variants attain 88.3–92.6% IS / 76.3–79.0% OoS KS on the single SPY OoS window, with CHMM-N kurtosis 5.26 matching observed 5.29 OoS within 0.03. The single-window OoS sits at the upper end of the six-fold rolling-origin walk-forward whose median CHMM-N OoS KS is 67.7% at $K = 18$ and 62.1% at $K^\star = 3$; the walk-forward median is the operationally informative summary for production deployment. The held-out-likelihood and BIC choice $K^\star = 3$ is reported alongside as the lower-state-count default for risk-management consumers (CHMM-N: 89.7% IS / 80.1% OoS KS, $|G_t|$ ACF-MAE 0.0467). An extended-state-resolution *sensitivity reference* at $K = 18$, selected by a multi-objective rule that uses the OoS window and is therefore not held-out-clean, attains 93.6–95.6% IS / 80.8–85.7% OoS KS, with the penalised CHMM-t at $\lambda = 20$ providing the cleanest IS / OoS kurtosis match at 8.56 / 7.07 against observed 7.68 / 5.29 (Table 3). The strongest single-window OoS KS row in the panel is ML HSMM-N at $K^\star = 3$ (91.0% OoS KS) but its $|G_t|$ ACF-MAE matches the i.i.d. baseline; CHMM and ML HSMM are best read as complementary scaffolds with different KS / ACF / kurtosis trade-offs. We apply the textbook closed-form mixture-of-eigenvalues identity for the absolute-return ACF (13–15) to the SPY 2014–2024 instance of the Rydén et al. low-K setup and recast the empirical failure there as an effective-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$, showing the rank constraint is non-binding at $K \geq 3$ on this instance; the substantive content is the empirical effective-rank application, not a re-derivation of the identity. The four emission variants share the same scaffold and land at four kurtosis points; CHMM-GED's per-state shape p_k nests Gaussian and Laplace as boundary cases and produces a bulk/tail partition that replicates across seeds and tickers. VaR and ES envelopes bracket observed values at 1% and 5%; a regime-conditional VaR that propagates the one-step-ahead state forecast through the CHMM mixture passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly at $\alpha = 0.05$ on OoS (Table 7; the $\alpha = 0.01$ rows are power-bounded at $T_{\mathrm{OoS}} = 572$ and should be read with the calibration of Appendix A.6.17). The conditional construction extends to a six-fold rolling-origin walk-forward with a 19/24 aggregate pass-rate at the 5% level (Table 52); the two failures concentrate on the COVID and 2022-rate-hike-onset stress folds that the univariate walk-forward already flags as out-of-distribution by KS. Cross-ticker generalisation is exercised on a sector-balanced 30-ticker panel (Section 4.4, Table 5); the body table reports both held-out-clean $K^\star = 6$ (OoS KS median 75.1%) and the kurtosis-fidelity $K = 18$ (OoS KS median 73.4%) operating points, with quarterly refit lifting the $K = 18$ OoS median to 83.0%. Multi-asset synthesis composes through a rank-based Student-t copula on the CHMM marginals at $\nu^\star = 6$ (six-asset IS off-diagonal correlation MAE 0.027); on OoS the Gaussian and Student-t copulas are statistically indistinguishable at $N_{\mathrm{paths}} = 200$, so the dependence-family selection is an IS-calibration distinction. The headline scope is daily US equities; non-equity asset classes are not covered (the GLD / SLV stress test of Appendix A.8.6 collapses to 0% OoS KS).

Skew-heavy-tailed emissions (the natural fix for the absent leverage-effect coverage in the present scope), explicit-duration semi-Markov sojourns at higher K, and reference-implementation re-runs of MS-GARCH (`MSGARCH` R package of 35) and the original Wiese et al. (2020) QuantGAN architecture are companion-paper directions; scaling to larger d via untruncated regular vines or factor copulas is the natural multi-asset follow-up.

Data and code availability. Reproducer at `https://github.com/altashly1/CHMM-paper`; companion Julia package `CHMM-Model.jl` at `https://github.com/altashly1/CHMM-Model` (public API in Appendix A.1, Julia $\geq$ 1.12). Daily prices: Polygon.io via Massive (2014-01-03 to 2025-11-18) and Alpaca Markets / IEX (held-out extension through 2026-04-20); we thank both providers. The authors declare no competing interests. **Author contributions:** J.V. directed the study; A.A. developed the continuous model, implemented Baum-Welch, conducted the empirical analysis, and

generated all figures and tables; C.J. contributed to methodology review and validation; all authors edited and reviewed the final manuscript.

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A Supplementary Material

This appendix collects supplementary material referenced in the main body: the public API of the companion package, the EM forward-backward recursions and algorithm pseudocode, the validation metric definitions, formal propositions, the full state-resolution and multi-emission sensitivity panels, robustness and KS-power calibration, the extended GARCH and SM-CHMM baselines, the full cross-asset panel, and the K-selection and ν_k diagnostics referenced from the discussion.

A.1 CHMM-Model.jl Public API and Reproducer

The companion Julia package `CHMM-Model.jl` (https://github.com/altashly1/CHMM-Model) is the entry point used to regenerate every table and figure in this paper from a single seed root under Julia ≥ 1.12. The public surface is a single `CHMM` type, three emission constructors, a unified `fit!` method (log-space Baum-Welch / ECM), and a unified `simulate` method.

```
using CHMM
chmm_n   = make_chmm_n(returns;   K = 18)    # Gaussian emissions
chmm_t   = make_chmm_t(returns;   K = 18)    # Student-t emissions
chmm_l   = make_chmm_l(returns;   K = 18)    # Laplace emissions
chmm_ged = make_chmm_ged(returns; K = 18)    # GED (per-state shape p_k)
fit!(chmm_n; max_iter = 60, tol = 1e-4)      # log-space Baum-Welch
paths = simulate(chmm_n; T = 2516, n = 1000) # 1000 paths of length T
```

The same `simulate` interface is used for every benchmark generator (Bootstrap, Gaussian i.i.d., Laplace i.i.d., GARCH(1,1)) so that the seven-metric panel evaluates each row on byte-identical windows. The Pipeline B Student-t copula sampler is exposed as `simulate_copula(...; nu = 6)` on the per-asset CHMM marginals; the rank-reordering construction is one function call. Running `julia -project=. run_full_rebuild.jl` from the paper repository (https://github.com/altashly1/CHMM-paper) reproduces every numerical artifact under the global seed policy of Section 3.5; `Manifest.toml` pins the dependency graph.

Sub-seed derivation rule. The deterministic global seed root is `20260420`; per-experiment sub-seeds follow the additive rule `sub_seed = SEED + offset(experiment) + offset(replicate)`, with experiment offsets in $\{0, 1{,}000, 100, 10{,}000, \ldots\}$ chosen to avoid collisions across the runners in the public Julia package. For example, the headline simulation in `run_diagnostics.jl` uses `SEED + (path index)` for each per-path random draw, the multi-seed Monte Carlo in `run_multiseed_headline.jl` uses base `20260422 + 100·(seed index)` for ten independent replicates, and the bracket / shrinkage sweeps add a further per-rate offset. The full sub-seed map is enumerated at the top of each runner script; downstream `Random.seed!()` calls are deterministic and order-independent within each runner.

A.2 CHMM Algorithms and Derivations

This appendix collects the algorithmic content for the CHMM family: the forward-backward recursions and weighted M-step updates, the complete training pseudocode, and per-state degrees-of-freedom diagnostics for CHMM-t.

A.2.1 Forward-Backward Recursions and M-Step Updates

For reference, the in-text Baum-Welch description of Section 3.2 is underpinned by the standard log-space recursions [17, 59]. Define the forward variable $\alpha_t(k) = \mathbb{P}(O_{1:t}, S_t = k \mid \mathcal{M})$ and the backward variable $\beta_t(k) = \mathbb{P}(O_{t+1:T} \mid S_t = k, \mathcal{M})$. The log-space recursions are

$$\log \alpha_1(k) = \log \pi_k + \log b_k(O_1), \tag{10}$$

$$\log \alpha_t(j) = \underset{i}{\text{logsumexp}}(\log \alpha_{t-1}(i) + \log T_{ij}) + \log b_j(O_t), \quad t = 2, \dots, T, \tag{11}$$

$$\log \beta_T(k) = 0, \tag{12}$$

$$\log \beta_t(i) = \underset{j}{\text{logsumexp}}(\log T_{ij} + \log b_j(O_{t+1}) + \log \beta_{t+1}(j)), \quad t = T-1, \dots, 1, \tag{13}$$

where $\text{logsumexp}_i(x_i) = x_{\max} + \log \sum_i \exp(x_i - x_{\max})$. The posterior state-occupancy $\gamma_t(k)$ and pair-occupancy $\xi_t(i, j)$ quantities are

$$\gamma_t(k) = \frac{\alpha_t(k)\beta_t(k)}{\sum_j \alpha_t(j)\beta_t(j)}, \qquad \xi_t(i, j) = \frac{\alpha_t(i)\, T_{ij}\, b_j(O_{t+1})\, \beta_{t+1}(j)}{\sum_{m,n} \alpha_t(m)\, T_{mn}\, b_n(O_{t+1})\, \beta_{t+1}(n)}, \tag{14}$$

and the closed-form M-step updates are

$$\pi_k^{\text{new}} = \gamma_1(k), \quad \mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, O_t}{\sum_t \gamma_t(k)}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)(O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}, \quad T_{ij}^{\text{new}} = \frac{\sum_{t=1}^{T-1} \xi_t(i, j)}{\sum_{t=1}^{T-1} \gamma_t(i)}. \tag{15}$$

The convergence criterion is $|\mathcal{L}^{(n)} - \mathcal{L}^{(n-1)}| < 10^{-4}$ with $\mathcal{L}^{(n)} = \text{logsumexp}_k \log \alpha_T^{(n)}(k)$, and simulation draws $S_1 \sim \text{Categorical}(\boldsymbol{\pi})$ and then iterates $G_t \sim \mathcal{N}(\mu_{S_t}, \sigma_{S_t}^2)$, $S_{t+1} \sim \text{Categorical}(\mathbf{T}_{S_t,\cdot})$ with no jump process.

CHMM-t M-step (per-state ECM, closed form given $u_{t,k}$). For Student-t emissions $b_k(x) = t_{\nu_k}(x; \mu_k, \sigma_k)$ the ECM formulation of Peel and McLachlan [18], Liu and Rubin [19] treats the latent precision as an augmented E-step quantity,

$$u_{t,k} = \frac{\nu_k + 1}{\nu_k + ((O_t - \mu_k)/\sigma_k)^2}, \tag{16}$$

so that conditional on ν_k the location and scale admit closed-form weighted updates,

$$\mu_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, O_t}{\sum_t \gamma_t(k)\, u_{t,k}}, \quad (\sigma_k^{\text{new}})^2 = \frac{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k^{\text{new}})^2}{\sum_t \gamma_t(k)}. \tag{17}$$

The degrees-of-freedom parameter ν_k is then refined by a one-dimensional golden-section search on the per-state Q-function $Q_k(\nu) = \sum_t \gamma_t(k)\, \log t_\nu(O_t; \mu_k, \sigma_k)$ over the bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$. This two-block ECM preserves the monotonicity of the standard EM guarantee on the compact bracket $[\nu_{\min}, \nu_{\max}]$, with the search invariant $Q_k(\nu^{(n+1)}) \geq Q_k(\nu^{(n)})$ at every iteration; Proposition 1 states the guarantee in full. We extend this M-step with an optional exponential shrinkage prior on $1/\nu_k$, corresponding to the penalised objective

$$Q_k^{\text{pen}}(\nu) \;=\; \sum_t \gamma_t(k)\, \log t_\nu(O_t; \mu_k, \sigma_k) \;-\; \lambda \,/\, \nu, \tag{18}$$

maximised by the same golden-section search ($\lambda = 0$ recovers the unpenalised Peel and McLachlan [18], Liu and Rubin [19] ECM; the rate sweep and its effect on simulated IS kurtosis are in Section 5).

CHMM-L M-step (closed-form weighted Laplace MLE). For Laplace emissions $b_k(x) = \frac{1}{2b_k}\exp(-|x-\mu_k|/b_k)$ the weighted-MLE estimators are available in closed form:

$$\mu_k^{\text{new}} = \text{WeightedMedian}(O_{1:T};\, \gamma_{1:T}(k))\,, \qquad b_k^{\text{new}} = \frac{\sum_t \gamma_t(k)\,|O_t - \mu_k^{\text{new}}|}{\sum_t \gamma_t(k)}. \tag{19}$$

The weighted median is computed by cumulative-weight bracketing with linear interpolation between adjacent order statistics at the half-total weight crossing. The Laplace M-step carries no shape parameter, so CHMM-L is strictly cheaper per iteration than CHMM-t (which requires the per-state Q_k search).

CHMM-GED M-step (three-stage ECM with closed-form scale). For Generalized Error Distribution emissions $b_k(x) = \text{GED}(x; \mu_k, \alpha_k, p_k) = \frac{p_k}{2\alpha_k\Gamma(1/p_k)}\exp\big(-(|x-\mu_k|/\alpha_k)^{p_k}\big)$ the M-step decomposes into three CM updates per state. Given current $(\mu_k^{(n)}, \alpha_k^{(n)}, p_k^{(n)})$, CM-1 updates the location by minimising the per-state weighted L^{p_k} loss,

$$\mu_k^{(n+1)} \;=\; \arg\min_{\mu\in[\hat\mu_k - 5\alpha_k,\, \hat\mu_k + 5\alpha_k]} \sum_t \gamma_t(k)\,\big|O_t - \mu\big|^{p_k^{(n)}}, \tag{20}$$

by a 40-iteration golden-section search; CM-2 updates the scale in closed form given $(\mu_k^{(n+1)}, p_k^{(n)})$,

$$\alpha_k^{(n+1)} \;=\; \left[\frac{p_k^{(n)}}{W_k}\sum_t \gamma_t(k)\,\big|O_t - \mu_k^{(n+1)}\big|^{p_k^{(n)}}\right]^{1/p_k^{(n)}}, \qquad W_k \;=\; \sum_t \gamma_t(k); \tag{21}$$

and CM-3 updates the shape by maximising the per-state Q-function over the bracket,

$$p_k^{(n+1)} \;=\; \arg\max_{p\in[p_{\min},\, p_{\max}]} \sum_t \gamma_t(k)\,\log b_k\big(O_t;\, \mu_k^{(n+1)},\, \alpha_k^{(n+1)},\, p\big), \tag{22}$$

again by a 40-iteration golden-section search over $[p_{\min}, p_{\max}] = [0.5, 3.0]$. Equation (21) is the unique zero of $\partial Q_k/\partial\alpha$, so CM-2 is exact maximization. CM-1 and CM-3 are bracketed maximization. Each CM step is therefore a partial maximization, and the standard ECM monotonicity result [63] applies on the compact bracket; Proposition 1 states the guarantee in full. Boundary cases: $p_k = 2$ recovers Gaussian (with $\sigma = \alpha/\sqrt{2}$); $p_k = 1$ recovers Laplace (with $b = \alpha$).

A.2.2 Algorithm Descriptions

This section provides detailed pseudocode for the main algorithms used in this study. Algorithm 1 is the unified EM procedure shared across all four emission families. It makes explicit the *shared scaffold* (quantile-based initialization, log-space forward-backward, transition update) and the four *family-specific branches* (Gaussian M-step for CHMM-N, ECM with one-dimensional golden-section ν-search for CHMM-t, closed-form weighted-median + weighted-MAD for CHMM-L, three-stage CM with bracketed p_k for CHMM-GED); the branch points are lines 3, 13, and 30. Visualizing the four variants as a single algorithm with four M-step branches makes the architectural contribution of the paper (one EM engine, four interchangeable emission families) immediate from the pseudocode. Algorithm 2 presents the Viterbi algorithm for decoding the most likely hidden state sequence. Algorithm 3 describes the CHMM simulation procedure for generating synthetic return paths. Algorithm 4 describes the cross-asset rank-reordering simulator for copula-based multi-asset synthesis. Algorithm 5 details the SIM regression-and-resampling pipeline.

Algorithm 1 Unified EM algorithm for the CHMM family. Lines 3, 13, and 30 branch on the emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$; every other line is identical across families. CHMM-N and CHMM-L instantiate classical EM; CHMM-t and CHMM-GED instantiate ECM sequences whose final CM step updates the per-state shape parameter (ν_k for CHMM-t, p_k for CHMM-GED) by a one-dimensional golden-section search on the per-state Q-function. Hyperparameter defaults used throughout this paper: convergence tolerance $\mathrm{tol} = 10^{-4}$ on $\Delta\mathcal{L}$, `max_iter` $= 60$, 40-iteration golden-section sub-step on every bracketed CM, CHMM-t bracket $(\nu_{\min}, \nu_{\max}) = (2.1, 50)$ with initial $\nu^{(0)} = 10$, CHMM-GED bracket $(p_{\min}, p_{\max}) = (0.5, 3.0)$ with initial $p^{(0)} = 1.5$ and μ_k search interval $[\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]$.

Require: Observations $O_{1:T}$, number of states K, tolerance tol, `max_iter`, emission family $F \in \{\mathcal{N}, t, L, \mathrm{GED}\}$, if $F = t$ bracket $(\nu_{\min}, \nu_{\max})$ and initial $\nu^{(0)}$, if $F = \mathrm{GED}$ bracket $(p_{\min}, p_{\max})$ and initial $p^{(0)}$.
Ensure: Transition matrix $\mathbf{T}$, per-state emission parameters $\boldsymbol{\theta}_{1:K}$, initial distribution $\boldsymbol{\pi}$.

1: **Quantile-Based Initialization.**
2: Sort $O^{(\mathrm{sort})} \leftarrow \mathrm{sort}(O_{1:T})$ and partition into K equal chunks $\{C_1, \ldots, C_K\}$.
3: **for** $k = 1$ to K **do**
4:   **switch** F
5:     $F = \mathcal{N}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$.
6:     $F = t$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \sigma_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad \nu_k \leftarrow \nu^{(0)}$.
7:     $F = L$: $\mu_k \leftarrow \mathrm{median}(C_k)$, $\quad b_k \leftarrow \max(\mathrm{mean}(|C_k - \mu_k|), 10^{-6})$.
8:     $F = \mathrm{GED}$: $\mu_k \leftarrow \mathrm{mean}(C_k)$, $\quad \alpha_k \leftarrow \max(\mathrm{std}(C_k), 10^{-6})$, $\quad p_k \leftarrow p^{(0)}$.
9: **end for**
10: $T_{ij} \leftarrow 1/K$; $\quad \pi_k \leftarrow 1/K$; $\quad \mathcal{L}_{\mathrm{prev}} \leftarrow -\infty$.
11: **EM Loop.**
12: **for** $n = 1$ to `max_iter` **do**
13:   **E-Step, per-family emission log-likelihood.**
14:   **switch** F
15:     $F = \mathcal{N}$: $\log B_{t,k} \leftarrow \log \mathcal{N}(O_t \mid \mu_k, \sigma_k^2)$.
16:     $F = t$: $\log B_{t,k} \leftarrow \log t_{\nu_k}(O_t; \mu_k, \sigma_k)$.
17:     $F = L$: $\log B_{t,k} \leftarrow -\log(2b_k) - |O_t - \mu_k|/b_k$.
18:     $F = \mathrm{GED}$: $\log B_{t,k} \leftarrow \log p_k - \log(2\alpha_k) - \log\Gamma(1/p_k) - (|O_t - \mu_k|/\alpha_k)^{p_k}$.
19:   **E-Step, shared log-space forward-backward.**
20:   $\log \alpha_1(k) \leftarrow \log \pi_k + \log B_{1,k}$.
21:   **for** $t = 2$ to T, $j = 1$ to K **do**
22:     $\log \alpha_t(j) \leftarrow \operatorname{logsumexp}_i(\log \alpha_{t-1}(i) + \log T_{ij}) + \log B_{t,j}$.
23:   **end for**
24:   $\log \beta_T(k) \leftarrow 0$.
25:   **for** $t = T - 1$ down to 1, $i = 1$ to K **do**
26:     $\log \beta_t(i) \leftarrow \operatorname{logsumexp}_j(\log T_{ij} + \log B_{t+1,j} + \log \beta_{t+1}(j))$.
27:   **end for**
28:   Compute $\gamma_t(k)$ and $\xi_t(i,j)$ from equation (14).
29:   **Shared transition update.** $T_{ij} \leftarrow \sum_t \xi_t(i,j) \,/\, \sum_t \gamma_t(i)$; $\pi_k \leftarrow \gamma_1(k)$.
30:   **M-Step, per-family emission update.**
31:   **switch** F
32:     $F = \mathcal{N}$ {classical Baum-Welch [16].}
33:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, O_t / \sum_t \gamma_t(k)$.
34:       $\sigma_k \leftarrow \max\left(\sqrt{\sum_t \gamma_t(k)(O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\right)$.
35:     $F = t$ {ECM of Peel and McLachlan [18], Liu and Rubin [19]; closed-form (μ_k, σ_k) given ν_k, then 1D search on ν_k.}
36:       $u_{t,k} \leftarrow (\nu_k + 1)/(\nu_k + ((O_t - \mu_k)/\sigma_k)^2)$ for all t, k.
37:       $\mu_k \leftarrow \sum_t \gamma_t(k)\, u_{t,k}\, O_t / \sum_t \gamma_t(k)\, u_{t,k}$.
38:       $\sigma_k \leftarrow \max\left(\sqrt{\sum_t \gamma_t(k)\, u_{t,k}\, (O_t - \mu_k)^2 / \sum_t \gamma_t(k)},\ 10^{-6}\right)$.
39:       $\nu_k \leftarrow \arg\max_{\nu \in [\nu_{\min}, \nu_{\max}]} \sum_t \gamma_t(k)\, \log t_\nu(O_t; \mu_k, \sigma_k)$ {40-iter golden-section search.}
40:     $F = L$ {closed-form weighted Laplace MLE.}
41:       $\mu_k \leftarrow \mathrm{WeightedMedian}(O_{1:T},\ \gamma_{1:T}(k))$ {cumulative-weight bracketing with interpolation at the $W_k/2$ crossing.}
42:       $b_k \leftarrow \max\left(\sum_t \gamma_t(k)\, |O_t - \mu_k| / \sum_t \gamma_t(k),\ 10^{-6}\right)$.
43:     $F = \mathrm{GED}$ {three-stage ECM: bracketed μ_k, closed-form α_k, bracketed p_k.}
44:       $\mu_k \leftarrow \arg\min_{\mu \in [\hat{\mu}_k - 5\alpha_k, \hat{\mu}_k + 5\alpha_k]} \sum_t \gamma_t(k)\, |O_t - \mu|^{p_k}$ {40-iter golden-section search.}
45:       $\alpha_k \leftarrow \max\left(\left[(p_k/W_k) \sum_t \gamma_t(k)\, |O_t - \mu_k|^{p_k}\right]^{1/p_k},\ 10^{-6}\right)$ where $W_k = \sum_t \gamma_t(k)$ {closed form, unique zero of $\partial Q_k / \partial \alpha$.}
46:       $p_k \leftarrow \arg\max_{p \in [p_{\min}, p_{\max}]} \sum_t \gamma_t(k)\, \log \mathrm{GED}(O_t; \mu_k, \alpha_k, p)$ {40-iter golden-section search.}
47:   **Convergence check.** $\mathcal{L}^{(n)} \leftarrow \operatorname{logsumexp}_k(\log \alpha_T(k))$.
48:   **if** $|\mathcal{L}^{(n)} - \mathcal{L}_{\mathrm{prev}}| < \mathrm{tol}$ **then**

Algorithm 2 Viterbi decoding for the most likely hidden state sequence

Require: Observations $O_{1:T}$, Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$
Ensure: Most probable state sequence $S^*_{1:T}$

```
for k = 1 to K do
    log δ_1(k) ← log(1/K) + log N(O_1 | μ_k, σ_k^2).
end for
for t = 2 to T do
    for j = 1 to K do
        vals(i) ← log δ_{t−1}(i) + log T_ij for all i.
        log δ_t(j) ← max_i vals(i) + log N(O_t | μ_j, σ_j^2).
        ψ_t(j) ← arg max_i vals(i).
    end for
end for
S*_T ← arg max_k log δ_T(k).
for t = T − 1 down to 1 do
    S*_t ← ψ_{t+1}(S*_{t+1}).
end for
return S*_{1:T}
```

Algorithm 3 CHMM simulation of synthetic excess growth rate paths

Require: Trained CHMM $\mathcal{M} = (\mathbf{T}, \{\mu_k, \sigma_k\}, \boldsymbol{\pi})$, Number of steps M, Number of paths P
Ensure: Simulated return paths $\{\hat{G}^{(p)}_{1:M}\}_{p=1}^{P}$

```
for p = 1 to P do
    Sample initial state S_1 ~ Categorical(π).
    for t = 1 to M do
        Emit observation Ĝ_t^(p) ~ N(μ_{S_t}, σ_{S_t}^2).
        if t < M then
            Transition to next state S_{t+1} ~ Categorical(T_{S_t,·}).
        end if
    end for
end for
return {Ĝ_{1:M}^(p)}_{p=1}^P
```

Algorithm 4 Cross-asset copula simulation via rank reordering

Require: Fitted per-asset CHMMs $\{\mathcal{M}_j\}_{j=1}^d$, correlation matrix Σ (positive semi-definite (PSD), unit diagonal), copula family $\mathcal{C} \in \{\text{Gaussian}, t_\nu\}$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

for $p = 1$ to P **do**
  Draw copula sample $\mathbf{U}^{(p)} \in [0,1]^{T \times d}$ from $\mathcal{C}(\Sigma)$:
    $L \leftarrow \text{Cholesky}(\Sigma), \quad Z \in \mathbb{R}^{T \times d} \sim \mathcal{N}(0, I_d)$ i.i.d.
    $Y \leftarrow Z L^\top$
  if $\mathcal{C} = \text{Gaussian}$ **then**
      $\mathbf{U}^{(p)} \leftarrow \Phi(Y)$ componentwise
  else
      Draw $W \sim \chi^2_\nu/\nu$ i.i.d. for $t = 1, \dots, T$; $\quad X_{t,\cdot} \leftarrow Y_{t,\cdot}/\sqrt{W_t}$
      $\mathbf{U}^{(p)} \leftarrow t_\nu(X)$ componentwise
  end if
  for $j = 1$ to d **do**
    Simulate $\tilde{g}_{j,1:T}^{(p)}$ from $\mathcal{M}_j$ via Algorithm 3.
    Compute order statistics $\tilde{g}_{j,(1)}^{(p)} \leq \cdots \leq \tilde{g}_{j,(T)}^{(p)}$.
    Compute ranks $r_{j,t}^{(p)} \leftarrow \text{rank}(U_{j,t}^{(p)})$ for $t = 1, \dots, T$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \tilde{g}^{(p)}_{j,(r_{j,t}^{(p)})}$ for $t = 1, \dots, T$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

Algorithm 5 Single Index Model (SIM) construction and simulation

Require: IS return matrix $\mathbf{R} \in \mathbb{R}^{T \times d}$, market-column index m, market CHMM $\mathcal{M}_m$, path length T, number of paths P

Ensure: Cross-asset path tensor $\{\hat{G}_{j,t}^{(p)}\}$ of shape $T \times d \times P$

Ordinary least squares (OLS) fit. For $j \neq m$:
  $\hat{\beta}_j \leftarrow \text{Cov}(\mathbf{R}_{:,j}, \mathbf{R}_{:,m})/\text{Var}(\mathbf{R}_{:,m})$
  $\hat{\alpha}_j \leftarrow \overline{\mathbf{R}_{:,j}} - \hat{\beta}_j \overline{\mathbf{R}_{:,m}}$
  $\hat{\eta}_{j,t} \leftarrow R_{t,j} - \hat{\alpha}_j - \hat{\beta}_j R_{t,m}, \quad R_j^2 \leftarrow 1 - \sum_t \hat{\eta}_{j,t}^2 / \sum_t (R_{t,j} - \overline{\mathbf{R}_{:,j}})^2$
for $p = 1$ to P **do**
  Simulate market path $G_{m,1:T}^{(p)}$ from $\mathcal{M}_m$ via Algorithm 3.
  $\hat{G}_{m,t}^{(p)} \leftarrow G_{m,t}^{(p)}$.
  for $j \neq m$ **do**
    Draw residual indices $\{\iota_t\}_{t=1}^T \sim \text{Uniform}(\{1, \dots, T\})$.
    $\hat{G}_{j,t}^{(p)} \leftarrow \hat{\alpha}_j + \hat{\beta}_j G_{m,t}^{(p)} + \hat{\eta}_{j,\iota_t}$.
  end for
end for
return $\{\hat{G}_{j,t}^{(p)}\}$

A.2.3 Per-State Shape Diagnostics: CHMM-t and CHMM-GED

This appendix backs the discussion in Section 5 of the per-state shape allocation under CHMM-t and CHMM-GED on SPY IS at $K = 18$ (seed = 20260420).

CHMM-t per-state ν_k. Figure 1 plots the per-state ν_k histogram for the $K = 18$ CHMM-t fit. Thirteen of the eighteen states rest at the upper ECM bracket ($\nu_k = 50$), and only two states lie near the lower bracket ($\nu_2 = 2.19$, $\nu_4 = 2.10$). The simulated IS mixture kurtosis of 17.34 is driven by the posterior mass routed to those two very heavy-tailed states, not by a systemic collapse to the lower bracket.

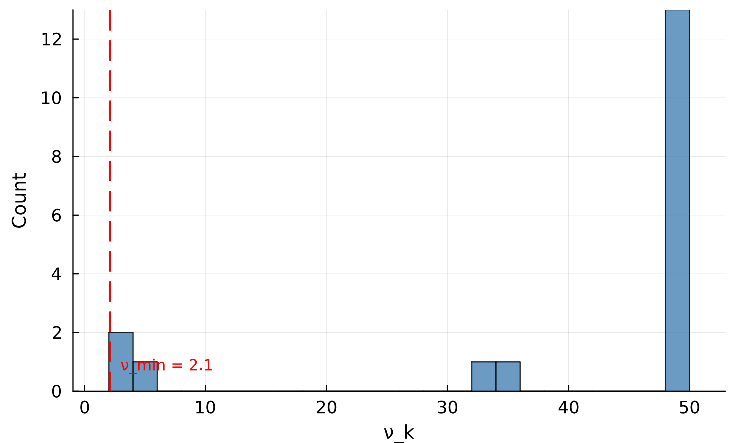


Figure 1. Per-state degrees-of-freedom ν_k for the CHMM-t fit at $K = 18$ on SPY IS. The dashed red line marks the lower ECM bracket $\nu_{\min} = 2.1$. Only two states ($\nu_2 = 2.19$, $\nu_4 = 2.10$) sit near the lower bracket; the median ν_k is at the upper bracket.

CHMM-GED per-state p_k (Gaussian-bulk / Laplace-tail bimodality). The body Section 4.3 reports and discusses the $\hat{p}_k$ partition (Figure 28); see also Table 9 below for the parallel CHMM-t bracket sweep. Briefly, the $K = 18$ CHMM-GED fit on SPY IS is bimodal: thirteen states attain the upper ECM bracket at $p_k = 3.0$, one state lands at intermediate $p_k = 1.6$, and four states cluster in the Laplace-shape regime $p_k \in [0.86, 1.24]$, concentrated in the high-volatility ranks. The smallest fitted p_k on SPY is 0.86, well clear of every $p_{\min}$ tested in the bracket sweep, so the lower bracket is non-binding. The bimodal partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.7).

Table 9 reports the seven-metric panel for CHMM-t fits whose ECM bracket lower bound is lifted from $\nu_{\min} = 2.1$ through $\{2.5, 3.0, 4.0\}$, plus the Gaussian limit ($\nu = \infty$, equivalent to CHMM-N). Lifting the bracket does not reduce the IS kurtosis overshoot: the simulated kurtosis moves within the 12.5–14.0 band across bracket floors and only collapses to 5.05 in the full Gaussian limit. The overshoot is therefore a feature of the Student-t emission family at $K = 18$ and not a lower-bracket artefact.

Table 10 reports the analogous bracket sensitivity for CHMM-GED on $p_{\max} \in \{2.0, 2.5, 3.0, 3.5, 4.0\}$ (with $p_{\min} = 0.5$ fixed) and on $p_{\min} \in \{0.5, 0.7, 0.85\}$ (with $p_{\max} = 3.0$ fixed). The headline reads cleanly. The Gaussian-like state count is invariant at 13/18 across every upper bracket; only the

Table 9. CHMM-t $\nu_{\min}$ bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths). AD = Anderson–Darling pass rate; W_1 = Wasserstein-1; H = Hellinger distance (definitions in Appendix A.3).

$\nu_{\min}$	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	Kurt OoS	ACF-MAE	W_1	H
2.1	95.7	90.2	86.2	82.0	13.95	10.08	0.0546	0.100	0.0759
2.5	95.8	90.6	84.2	79.1	15.12	10.34	0.0547	0.100	0.0759
3.0	93.8	89.5	85.3	80.4	12.25	9.65	0.0548	0.099	0.0760
4.0	95.5	89.5	85.0	80.3	13.25	9.39	0.0550	0.101	0.0759
∞ (Gauss)	94.1	86.7	84.2	75.5	5.03	4.46	0.0510	0.110	0.0809
Observed	–	–	–	–	7.68	5.29	–	–	–

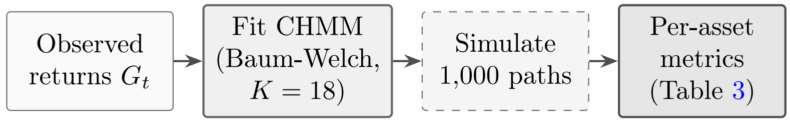
redistribution between intermediate and Laplace-like states moves slightly with $p_{\max}$, and the total heavy-shape count stays in $\{4, 5\}$. Tightening $p_{\max}$ from 4.0 to the canonical 3.0 costs about 4 nats of incomplete-data log-likelihood. The $p_{\min}$ sweep is fully degenerate: every $p_{\min} \in [0.5, 0.85]$ produces an identical fit because the smallest observed $\hat{p}_k$ on SPY is 0.86, so the lower bracket is non-binding.

Table 10. CHMM-GED bracket sensitivity at $K = 18$ on SPY (seed = 20260420, 1,000 paths). $\hat{p}_{\min}$, $\hat{p}_{\text{med}}$, $\hat{p}_{\max}$ are the across-state minimum, median, and maximum of the fitted $\hat{p}_k$. nG / nI / nL / nS are the partition counts (Gaussian-like $\hat{p}_k \geq 1.85$, intermediate $1.30 \leq \hat{p}_k < 1.85$, Laplace-like $0.85 \leq \hat{p}_k < 1.30$, super-Laplace $\hat{p}_k < 0.85$). LL is the incomplete-data log-likelihood at convergence.

Sweep	$p_{\min}$	$p_{\max}$	LL	$\hat{p}_{\min}$	nG	nI	nL	nS	KS IS	AD IS	KS OoS	AD OoS	Kurt IS	ACF-MAE	W_1	H
	0.5	2.0	−4732.0	0.89	13	0	5	0	95.8	91.0	83.8	81.2	5.02	0.0541	0.099	0.0781
	0.5	2.5	−4725.3	0.87	13	2	3	0	96.1	92.4	84.3	80.1	5.09	0.0549	0.098	0.0778
$p_{\max}$ sweep	0.5	**3.0**	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779
	0.5	3.5	−4717.5	0.86	13	1	4	0	96.3	91.8	86.7	82.4	5.11	0.0544	0.099	0.0782
	0.5	4.0	−4716.3	0.86	13	1	4	0	95.9	91.3	84.4	80.6	5.18	0.0543	0.099	0.0785
	0.50	3.0	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779
$p_{\min}$ sweep	0.70	3.0	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779
	0.85	3.0	−4720.3	0.86	13	1	4	0	96.3	92.7	83.8	81.3	5.05	0.0546	0.098	0.0779

A.2.4 Pipeline Schematic

Pipeline A (single-asset, Section 4.3):



Pipeline B (cross-asset, Section 4.5):

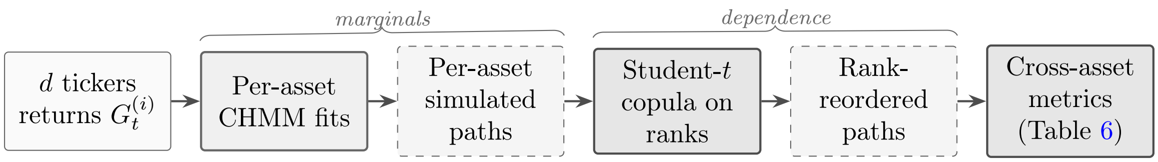


Figure 2. Schematic of Pipeline A (single-asset) and Pipeline B (cross-asset). Pipeline A fits a CHMM per ticker independently. Pipeline B reuses the per-asset fits as marginals and injects cross-asset dependence through a rank-based Student-t copula. The single-asset scaffold is shared across both pipelines.

A.2.5 CHMM Architecture and Training Loop

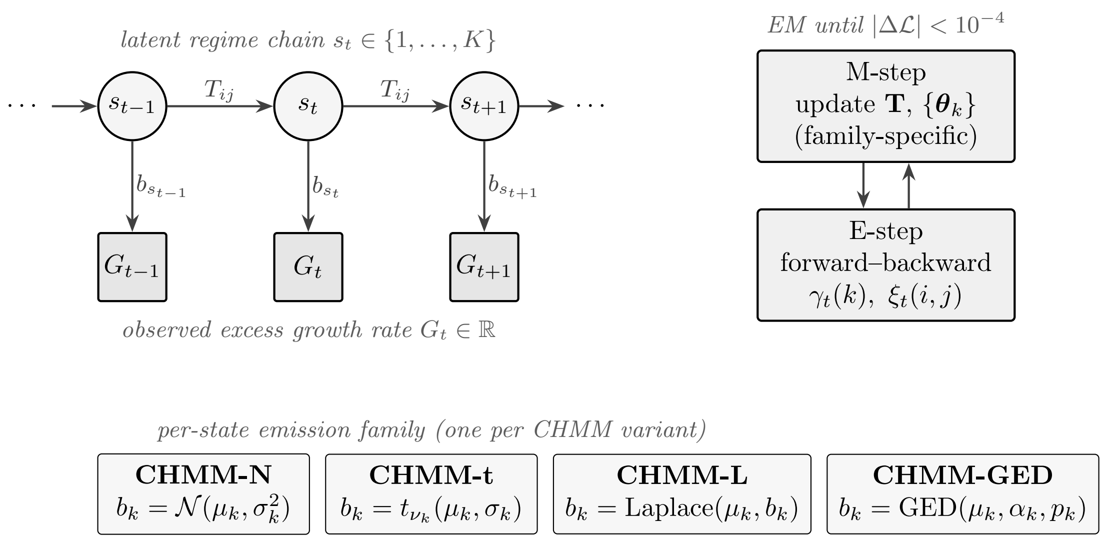


Figure 3. CHMM architecture and training loop. A K-state latent Markov chain with transition matrix $\mathbf{T}$ generates the regime sequence s_t; each s_t emits the observed excess growth rate G_t through a state-specific density b_{s_t}. The four CHMM variants share everything except the emission family. EM alternates an E-step (log-space forward–backward) and a family-specific M-step until $|\Delta\mathcal{L}| < 10^{-4}$.

A.3 Validation Metric Details

This appendix documents the precise form of the seven per-path metrics referenced in Section 3.5. For each model, 1,000 independent paths of length T are simulated (200 paths for the cross-asset extension) and each metric is computed per path before aggregation.

Kolmogorov–Smirnov (KS) pass rate. The two-sample KS statistic [56, 57] measures the maximum vertical distance between two empirical CDFs,

$$D_{n,m} = \sup_x |F_n(x) - G_m(x)|. \tag{23}$$

The KS pass rate is the fraction of paths for which the test fails to reject distributional equivalence against the reference data at $\alpha = 0.05$.

Anderson–Darling (AD) pass rate. The AD test is the integrated squared CDF deviation with a quadratic tail weighting, making it more sensitive than KS to tail discrepancies. The AD pass rate is the fraction of paths for which the two-sample AD test fails to reject at $\alpha = 0.05$.

Excess kurtosis. Mean simulated excess kurtosis across paths, compared to the observed value. Given Gaussian emissions, the CHMM is expected to underestimate kurtosis relative to the heavy-tailed empirical distribution, and the shortfall is one of the reported diagnostics.

ACF-MAE. Mean absolute error between observed and mean-simulated autocorrelation functions of $|G_t|$ over $L = 252$ lags,

$$\text{ACF-MAE} = \frac{1}{L} \sum_{\tau=1}^{L} \left| \hat{\rho}^{\text{obs}}_{|G|}(\tau) - \bar{\hat{\rho}}^{\text{sim}}_{|G|}(\tau) \right|. \tag{24}$$

This is the direct diagnostic for the Rydén et al. limitation.

Wasserstein-1 distance. The mean absolute difference of sorted empirical quantiles, equivalent to the L^1 distance between the empirical CDFs [72],

$$W_1(F, G) = \int_0^1 |F^{-1}(u) - G^{-1}(u)| \, du. \tag{25}$$

Hellinger distance. A histogram-based overlap distance in $[0, 1]$,

$$H(P, Q) = \frac{1}{\sqrt{2}} \sqrt{\sum_{i=1}^{B} (\sqrt{p_i} - \sqrt{q_i})^2}, \tag{26}$$

with B equal-width bins spanning the joint support and p_i, q_i the observed and simulated bin probabilities.

Quantile coverage. The fraction of 99 equally-spaced observed quantiles (1st through 99th percentile) falling inside the 5th–95th percentile envelope of the corresponding simulated quantile across 1,000 paths. A coverage of 100% indicates that the simulation envelope fully brackets the observed distribution at every percentile.

Cross-asset metrics. For the cross-asset extension we additionally track the mean Frobenius norm $\|\hat{\Sigma}^{(p)}_{\text{sim}} - \hat{\Sigma}_{\text{obs}}\|_F$ and the off-diagonal Pearson correlation MAE between simulated and observed return matrices, averaged over paths. The off-diagonal MAE is averaged over the $d(d-1)/2$ unique upper-triangular entries of the symmetric correlation matrix (so 15 entries for the six-asset cross-section of Section 4.5), not double-counted across both triangles:

$$\text{MAE}_{\text{off-diag}} = \frac{1}{P}\sum_{p=1}^{P} \frac{2}{d(d-1)} \sum_{1\le i<j\le d} \left| \hat{\Sigma}^{(p)}_{\text{sim},\,ij} - \hat{\Sigma}_{\text{obs},\,ij} \right|.$$

A.3.1 Continuous Ranked Probability Score and Diebold-Mariano

The OoS CRPS column of Table 3 reports a proper-scoring-rule complement to the KS pass rate. We use the unbiased sample CRPS estimator: for an ensemble of N simulated paths $\{x_i\}_{i=1}^{N}$ at time t and observed value y_t,

$$\widehat{\text{CRPS}}_t = \frac{1}{N}\sum_{i=1}^{N} |x_i - y_t| - \frac{1}{N(N-1)} \sum_{1\le i<j\le N} |x_i - x_j|. \tag{27}$$

The double sum is computed in $O(N \log N)$ via the sorted-ensemble identity $\sum_{i<j}(x_{(j)} - x_{(i)}) = \sum_i x_{(i)}(2i - N - 1)$. The ensemble at each t is the cross-section of the $N = 1{,}000$ unconditional simulated paths used throughout Section 4; this yields a marginal-predictive CRPS suitable for an unconditional generative-fidelity assessment, distinct from the conditional one-step-ahead CRPS used in forecasting.

For pairwise comparison we report a Diebold-Mariano test on the per-t loss differential $d_t = \widehat{\text{CRPS}}^A_t - \widehat{\text{CRPS}}^B_t$ with a Newey-West HAC variance under a Bartlett kernel and bandwidth $h = \lfloor T_{\text{OoS}}^{1/3} \rfloor = 8$. Two-sided p-values are reported under the standard-normal null. The body claim of Section 4.3 is: CHMM ties or beats every benchmark on mean OoS CRPS, with Gaussian i.i.d. the only significantly worse comparator.

A.3.2 Christoffersen Conditional-Coverage Panel (Companion to Section 4.6)

The body Section 4.6 reports the unconditional Kupiec LR_{uc} statistic. This appendix reports the full Christoffersen [73] panel: LR_{uc} (unconditional coverage), LR_{ind} (breach independence), and $\text{LR}_{\text{cc}} = \text{LR}_{\text{uc}} + \text{LR}_{\text{ind}}$ (joint conditional coverage), at $\alpha \in \{0.01, 0.05\}$ on the SPY IS and OoS windows. The breach series is constructed from the pooled-archive VaR threshold (the α-quantile of vec(sim archive) across paths and time), then $\text{br}_t = (R_t \le \widehat{\text{VaR}}_\alpha)$.

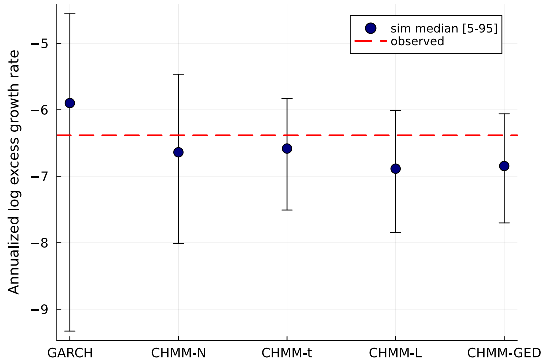
The headline reads as follows. On the unconditional (Kupiec) leg every CHMM variant passes at both levels on both windows; the LR_{uc} statistic for the four CHMM rows at $\alpha = 0.05$ on OoS clusters at 3.03–3.83 (just below the χ^2_1 critical value 3.841), reflecting the integer-breach grid at $T_{\text{OoS}} = 572$ noted in the body table caption. On the independence leg every unconditional generator in the panel (bootstrap, GARCH, and all four CHMM variants) rejects: LR_{ind} is in the range 4.71–56.46 across rows. This is structural rather than CHMM-specific. Volatility-clustering in R_{oos} produces persistent regimes of high and low realised volatility, the OoS-window breaches concentrate inside the high-volatility regime, and a constant-across-t VaR threshold (the construction every unconditional generator uses) cannot track that timing. The natural fix is a regime-conditional VaR that propagates the CHMM latent-state forecast $\mathbb{P}(s_{t+1} \,|\, \text{history}_t)$ through the per-state emission-mixture; this is operationally the conditional-VaR companion-paper direction noted in Section 5.

Table 11. Christoffersen conditional-coverage VaR back-test (SPY, seed = 20260420, $K = 18$, 1,000 simulated paths). Critical values: $\chi_1^2(0.05) = 3.841$, $\chi_2^2(0.05) = 5.991$. "br rate" is the empirical breach rate; $\mathrm{LR_{uc}}$ is the Kupiec statistic ($\sim \chi_1^2$ under correct unconditional coverage); $\mathrm{LR_{ind}}$ is the Christoffersen independence statistic ($\sim \chi_1^2$ under independent breaches); $\mathrm{LR_{cc}} = \mathrm{LR_{uc}} + \mathrm{LR_{ind}}$ is the joint conditional-coverage statistic ($\sim \chi_2^2$). PASS if statistic is below critical. Every unconditional generator (i.i.d. bootstrap, GARCH, and all four CHMM variants) rejects independence on OoS at $\alpha = 0.05$, consistent with breach clustering driven by volatility-clustering in R_{oos} rather than a CHMM-specific mis-specification of conditional dynamics.

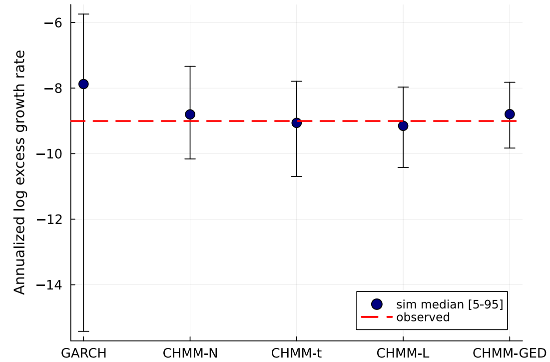
					Kupiec		Christ. ind		Christ. cc	
Model	Win	α	breaches	br rate	$\mathrm{LR_{uc}}$	p	$\mathrm{LR_{ind}}$	p	$\mathrm{LR_{cc}}$	p
Bootstrap	IS	0.01	26	1.03%	0.03	0.87	15.28	0.00	15.31	0.00
Bootstrap	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
Bootstrap	OoS	0.01	5	0.87%	0.10	0.76	13.18	0.00	13.28	0.00
Bootstrap	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
GARCH(1,1)	IS	0.01	31	1.23%	1.28	0.26	12.45	0.00	13.72	0.00
GARCH(1,1)	IS	0.05	139	5.52%	1.41	0.24	45.38	0.00	46.79	0.00
GARCH(1,1)	OoS	0.01	6	1.05%	0.01	0.91	20.87	0.00	20.89	0.00
GARCH(1,1)	OoS	0.05	24	4.20%	0.82	0.37	5.87	0.02	6.69	0.04
CHMM-N	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-N	IS	0.05	123	4.89%	0.07	0.80	47.33	0.00	47.40	0.00
CHMM-N	OoS	0.01	3	0.52%	1.58	0.21	18.98	0.00	20.56	0.00
CHMM-N	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01
CHMM-t	IS	0.01	22	0.87%	0.42	0.52	11.62	0.00	12.04	0.00
CHMM-t	IS	0.05	127	5.05%	0.01	0.91	55.54	0.00	55.55	0.00
CHMM-t	OoS	0.01	4	0.70%	0.58	0.45	15.53	0.00	16.12	0.00
CHMM-t	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-L	IS	0.01	19	0.76%	1.66	0.20	13.44	0.00	15.11	0.00
CHMM-L	IS	0.05	134	5.33%	0.55	0.46	49.43	0.00	49.98	0.00
CHMM-L	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-L	OoS	0.05	20	3.50%	3.03	0.08	4.71	0.03	7.74	0.02
CHMM-GED	IS	0.01	20	0.79%	1.15	0.28	12.80	0.00	13.95	0.00
CHMM-GED	IS	0.05	126	5.01%	0.00	0.99	56.46	0.00	56.46	0.00
CHMM-GED	OoS	0.01	2	0.35%	3.26	0.07	9.15	0.00	12.41	0.00
CHMM-GED	OoS	0.05	19	3.32%	3.83	0.05	5.26	0.02	9.08	0.01

A.3.3 VaR / ES Envelope Visualization (Companion to Table 31)

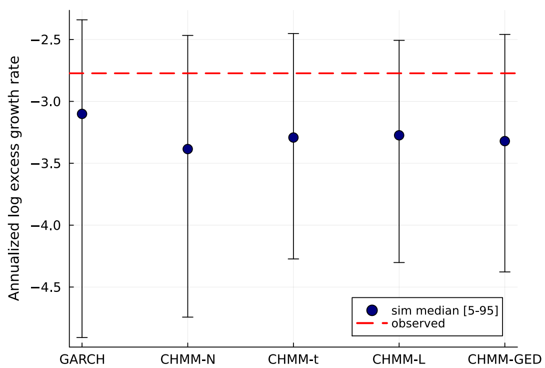
Figure 4 visualizes the median and $[5\%, 95\%]$ envelopes across simulated paths whose numeric values are reported in Table 31.



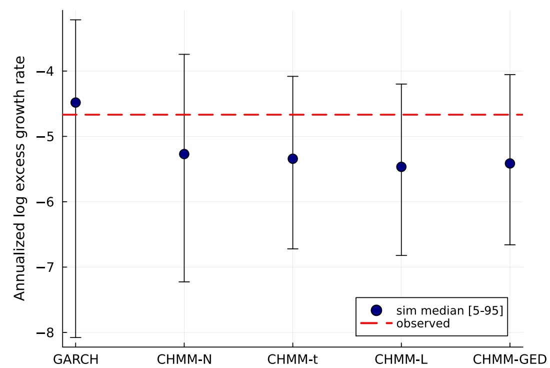
(a) IS VaR ($\alpha = 0.01$).



(b) IS ES ($\alpha = 0.01$).



(c) OoS VaR ($\alpha = 0.05$).



(d) OoS ES ($\alpha = 0.05$).

Figure 4. VaR / ES envelope diagnostic (SPY, $N_{\text{paths}} = 1{,}000$; global seed policy in Section 3.5). Each panel shows the simulated median (dot) and $[5\%, 95\%]$ envelope (error bar) across paths against the observed historical value (dashed red line); panels (a)/(b) are the IS window at $\alpha = 0.01$, panels (c)/(d) are the OoS window at $\alpha = 0.05$, and the x-axis in each panel lists the five generators kept for the volatility-aware conditional-coverage comparison (GARCH(1,1), CHMM-N, CHMM-t, CHMM-L, CHMM-GED). The i.i.d. bootstrap is omitted here because its envelope is by construction matched to the empirical marginal and does not discriminate among volatility-aware models. The y-axis is the annualized log excess growth rate (daily log return scaled by $1/\Delta t$ with $\Delta t = 1/252$), consistent with the $R_{\text{IS}}/R_{\text{OoS}}$ convention used throughout the paper. Headline: every observed red line falls inside the corresponding simulated envelope, confirming that the CHMM family passes the coverage leg of the back-test complementary to the Kupiec likelihood-ratio leg in Section 4.6.

A.4 Formal Theoretical Statements

The following propositions are referenced in Section 3.8 and stated formally here for reference; each is a one-line specialisation of a standard result and the proof sketch is one sentence.

Proposition 1 (ECM monotonicity for CHMM-t and CHMM-GED (and EM monotonicity for CHMM-N, CHMM-L))**.** *Under compactness of the parameter space (μ_k, σ_k or b_k or α_k, $\nu_k \in [\nu_{\min}, \nu_{\max}]$, $p_k \in [p_{\min}, p_{\max}]$ each in a closed bounded interval, $T_{ij} \geq \epsilon_T > 0$), the two-block CM step for CHMM-t of Section 3.7, namely the closed-form weighted updates of $(\mathbf{T}, \boldsymbol{\pi}, \mu_k, \sigma_k)$ at fixed ν_k, followed by a golden-section maximisation of the per-state Q-function over ν_k, satisfies*

$Q(\theta^{(n+1)} \mid \theta^{(n)}) \geq Q(\theta^{(n)} \mid \theta^{(n)})$, and consequently $\mathcal{L}(\theta^{(n+1)}) \geq \mathcal{L}(\theta^{(n)})$ where $\mathcal{L}$ is the incomplete-data log-likelihood. The same monotonicity holds for the closed-form CHMM-N and CHMM-L M-steps under the standard EM guarantee, and for the three-stage CHMM-GED CM sequence (bracketed μ_k, closed-form α_k, bracketed p_k) of equations (20)–(22).

Proof sketch. Each CM step does not decrease Q by definition (closed-form steps are exact maximisers, golden-section steps are bracketed maxima within the prescribed interval); the standard EM inequality $\mathcal{L}(\theta^{(n+1)}) - \mathcal{L}(\theta^{(n)}) \geq Q(\theta^{(n+1)} \mid \theta^{(n)}) - Q(\theta^{(n)} \mid \theta^{(n)})$ [62, 63] converts Q-monotonicity into log-likelihood monotonicity. Compactness gives boundedness above, hence convergence.

Numerical full-rank check on $\hat{\mathbf{T}}$. The Allman-Matias-Rhodes 2009 identifiability theorem requires $\hat{\mathbf{T}}$ of full rank. We compute the singular-value spectrum of $\hat{\mathbf{T}}$ at $K = 18$ across the four emission families on SPY IS (Table 12). All four matrices are full-rank in the strict sense, with smallest singular value $\sigma_{\min}(\hat{\mathbf{T}}) \in [0.0007, 0.0170]$ and condition numbers in the range 63–1,620. CHMM-GED is the most well-conditioned ($\sigma_{\min} = 0.017$, cond $= 63.5$); CHMM-t is the least ($\sigma_{\min} = 0.0007$, cond $= 1{,}620$), consistent with the per-state ν_k ECM lower-bracket pinning making one or two heavy-tail states near-degenerate. The deflated matrix $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has numerical rank exactly $K - 1 = 17$ across all four families, the rank statement underlying Section 3.8.

Table 12. Numerical full-rank check on $\hat{\mathbf{T}}$ at $K = 18$ across four CHMM emission families on SPY IS (seed = 20260420). $\sigma_{\min}$, $\sigma_{\max}$, and the condition number $\sigma_{\max}/\sigma_{\min}$ are reported on $\hat{\mathbf{T}}$ itself; $|\lambda_2|$ is the second-largest absolute eigenvalue (the dominant decay mode in the ACF identity, equation (8)); "rank(def)" is the numerical rank of $\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ at tolerance 10^{-10}, expected to equal $K - 1 = 17$.

Family	$\sigma_{\min}(\hat{\mathbf{T}})$	$\sigma_{\max}(\hat{\mathbf{T}})$	cond $\hat{\mathbf{T}}$	$\lvert\lambda_2\rvert$	rank$(\hat{\mathbf{T}} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top)$
CHMM-N	0.00532	1.0866	204	0.929	17/17
CHMM-t	0.00067	1.0796	1,620	0.914	17/17
CHMM-L	0.00398	1.0928	275	0.896	17/17
CHMM-GED	0.01702	1.0806	63.5	0.911	17/17

Proposition 2 (Identifiability up to label permutation)**.** *Under irreducibility and aperiodicity of* $\mathbf{T}$ *(Assumption 1), the emission contrast condition (Assumption 2), and pairwise distinctness of the per-state parameter tuples $\boldsymbol{\theta}_k$ across $k = 1, \ldots, K$, the CHMM parameter is identifiable up to label permutation for CHMM-N (Gaussian) and CHMM-L (Laplace). For CHMM-t (with per-state $\nu_k \in [\nu_{\min}, \nu_{\max}]$) and CHMM-GED (with per-state $p_k \in [p_{\min}, p_{\max}]$) the same conclusion holds modulo the standard caveat that pairwise distinctness must hold jointly across all components of $\boldsymbol{\theta}_k = (\mu_k, \sigma_k, \nu_k)$ or (μ_k, α_k, p_k), since within either bracket a Gaussian-shaped emission with inflated scale can mimic a Laplace-shaped emission with deflated scale at intermediate quantiles.*

Proof sketch. The HMM identifiability theorem of Allman et al. [64] requires $\mathbf{T}$ of full rank and per-state emissions linearly independent in the space of densities on $\mathbb{R}$. Full-rank $\mathbf{T}$ follows from irreducibility [61]. For CHMM-N and CHMM-L, linear independence of Gaussian and Laplace emissions with distinct location-scale parameters is direct from Yakowitz and Spragins [65], who establish that finite mixtures over location-scale families with strictly decreasing-tail densities are identifiable (the CHMM-N and CHMM-L marginals are special cases). For CHMM-t at fixed shape and CHMM-GED at fixed shape, location-scale identifiability is the same Yakowitz-Spragins result; for finite mixtures of Student-t with varying degrees-of-freedom the standard reference is Lindsay [74], Chapter 3, which gives identifiability via the moment-generating function over the bracketed range of shape parameters and a non-degeneracy condition on the joint location-scale-shape tuple. The CHMM-GED case under bracketed $p_k \in [p_{\min}, p_{\max}]$ is a parallel application of the moment-matching argument: the GED characteristic function $\hat{b}(t; \mu, \alpha, p) = \exp(i\mu t)\, M(t; \alpha, p)$ has a unique inverse over the bounded shape range up to permutation of components, modulo the joint-distinctness caveat on (μ_k, α_k, p_k). Strict linear independence at the boundary cases $p_k = 1$ (Laplace) and $p_k = 2$ (Gaussian) is automatic from Yakowitz-Spragins; in the bracket interior, the joint distinctness condition above is required to rule out the location-scale-shape trade-off.

Proposition 3 (MLE consistency)**.** *Under Assumptions 1–2 and the assumption that the true parameter lies in the interior of a compact Θ, the MLE $\hat{\theta}_T = \arg\max_{\theta \in \Theta} \mathcal{L}_T(\theta)$ is consistent: $\hat{\theta}_T \xrightarrow{p} \theta_0$ as $T \to \infty$, modulo label permutation.*

Proof reference. Bickel et al. [66] (Theorem 1) and Douc et al. [67] apply directly under our assumption set: irreducibility supplies the mixing condition, finite second moment the moment condition, compactness the parameter-space condition.

Proposition 4 (Marginal preservation under rank reordering)**.** *Let $\tilde{g}_{j,1:T}$ be a sample of size T from the per-asset CHMM marginal of asset j, and let the rank-reordered output be $\hat{g}^{(p)}_{j,t} = \tilde{g}^{(p)}_{j,(r^{(p)}_{j,t})}$ where $r^{(p)}_{j,t}$ is the rank of the copula sample. Then for every asset j the empirical CDF of $\{\hat{g}^{(p)}_{j,t}\}_{t=1}^{T}$ equals that of $\{\tilde{g}^{(p)}_{j,t}\}_{t=1}^{T}$.*

Proof sketch. The map $t \mapsto r^{(p)}_{j,t}$ is a permutation almost surely, so the multiset of values is preserved; the empirical CDF depends only on the multiset.

A.5 Spectral ACF Identity: Extended Derivation

Equation (8) is the closed-form mixture-of-eigenvalues form for the absolute-return ACF used in Section 3.8 and invoked throughout Sections 4–5. The bilinear cross-moment identity $\mathbb{E}[|G_t|\,|G_{t+\tau}|] = \mathbf{m}^\top \mathrm{diag}(\bar{\boldsymbol{\pi}})\, \mathbf{T}^\tau\, \mathbf{m}$ and its eigendecomposition over the non-unit eigenvalues of $\mathbf{T}$ are textbook material in the regime-switching literature (13, Section 22.2; 14, Chapter 3; the moment-generating-function form for general Markov-switching specifications with conditional means and variances depending on the latent state is given by 15); the underlying Markov-chain spectral decomposition is standard [61]. We restate the absolute-return specialisation here as a self-contained theorem with a step-by-step proof, and record the lag-zero behaviour and the complex / non-diagonalisable extensions that the main body does not. Our contribution on this axis is the explicit application of the spectral form to recast the empirical Rydén et al. [11] low-K failure as a rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ (Section 3.8, "Rydén separation as a K-rank statement") and the empirical demonstration that the rank constraint is the binding axis at moderate K on equity-return data (Sections 4, 5).

Theorem 1 (Mixture-of-eigenvalues identity for the absolute-return ACF)**.** *Let $\{(s_t, G_t)\}_{t \in \mathbb{Z}}$ be a stationary CHMM on $\{1, \ldots, K\}$ with irreducible aperiodic transition matrix $\mathbf{T}$ (Assumption 1) and per-state emissions of finite second moment (Assumption 2). Write $\bar{\boldsymbol{\pi}}$ for the unique stationary distribution, $\mathbf{D} = \mathrm{diag}(\bar{\boldsymbol{\pi}})$, $m_k = \mathbb{E}[|G_t| \mid s_t = k]$, $\mathbf{m} = (m_1, \ldots, m_K)^\top$, $\mu = \mathbb{E}|G_t| = \bar{\boldsymbol{\pi}}^\top \mathbf{m}$, and $\sigma^2_{|G|} = \mathrm{Var}(|G_t|)$. Assume $\mathbf{T}$ is diagonalisable with right eigenvectors $\mathbf{v}_k$ and left eigenvectors $\mathbf{w}_k$ biorthonormal, $\mathbf{w}_k^\top \mathbf{v}_l = \delta_{kl}$, indexed so that $\lambda_1 = 1 > |\lambda_2| \geq \cdots \geq |\lambda_K|$. Then for every integer lag $\tau \geq 1$,*

$$\rho_{|G|}(\tau) \;=\; \sum_{k=2}^{K} w_k\, \lambda_k^\tau, \qquad w_k \;=\; \frac{(\mathbf{m}^\top \mathbf{D}\, \mathbf{v}_k)\,(\mathbf{w}_k^\top \mathbf{m})}{\sigma^2_{|G|}}.$$

Proof. The proof is organised in six steps.

Step 1 (Definitions). The autocorrelation of $|G_t|$ at lag $\tau \geq 0$ is, by stationarity,

$$\rho_{|G|}(\tau) \;=\; \frac{\mathrm{Cov}(|G_t|,\, |G_{t+\tau}|)}{\sigma^2_{|G|}} \;=\; \frac{\mathbb{E}[|G_t|\, |G_{t+\tau}|] - \mu^2}{\sigma^2_{|G|}}.$$

The task reduces to evaluating $\mathbb{E}[|G_t|\, |G_{t+\tau}|]$ in closed form for $\tau \geq 1$.

Step 2 (Conditioning on the latent state pair). For $\tau \geq 1$, by the law of total expectation conditional on $(s_t, s_{t+\tau})$,

$$\mathbb{E}[|G_t|\, |G_{t+\tau}|] \;=\; \sum_{i,j=1}^{K} \mathbb{P}(s_t = i, s_{t+\tau} = j)\, \mathbb{E}\big[|G_t|\, |G_{t+\tau}| \,\big|\, s_t = i, s_{t+\tau} = j\big]\,.$$

The HMM emission structure gives the conditional independence $G_t \perp G_{t+\tau} \mid (s_t, s_{t+\tau})$ for $\tau \geq 1$: each emission depends only on its current state, and integrating the intermediate states $s_{t+1:t+\tau-1}$ out of the full-sequence factorisation $\big(\prod_u b_{s_u}(G_u)\big)\, p(s_{t+1:t+\tau-1} \mid s_t, s_{t+\tau})$ leaves the endpoint emissions decoupled. Hence

$$\mathbb{E}\big[|G_t|\, |G_{t+\tau}| \,\big|\, s_t = i, s_{t+\tau} = j\big] \;=\; \mathbb{E}[|G_t| \mid s_t = i]\, \mathbb{E}[|G_{t+\tau}| \mid s_{t+\tau} = j] \;=\; m_i\, m_j.$$

The joint state probability factors by stationarity and the Markov property, $\mathbb{P}(s_t = i, s_{t+\tau} = j) = \bar{\pi}_i\, (\mathbf{T}^\tau)_{ij}$. Substituting and noting $[\mathbf{D}\mathbf{T}^\tau]_{ij} = \bar{\pi}_i (\mathbf{T}^\tau)_{ij}$,

$$\mathbb{E}[|G_t|\, |G_{t+\tau}|] \;=\; \mathbf{m}^\top \mathbf{D}\, \mathbf{T}^\tau\, \mathbf{m}, \qquad \tau \geq 1. \tag{28}$$

Step 3 (Spectral preliminaries). For irreducible aperiodic $\mathbf{T}$, Perron–Frobenius applied to row-stochastic matrices gives $\lambda_1 = 1$ as a simple eigenvalue with all other eigenvalues strictly inside the unit disk [61]. The right eigenvector for λ_1 is $\mathbf{v}_1 = \mathbf{1}$, since $\mathbf{T}\mathbf{1} = \mathbf{1}$ from the row-sum constraint, and the left eigenvector is $\mathbf{w}_1 = \bar{\boldsymbol{\pi}}$, since $\bar{\boldsymbol{\pi}}^\top \mathbf{T} = \bar{\boldsymbol{\pi}}^\top$ by definition of the stationary distribution. Biorthonormality $\mathbf{w}_1^\top \mathbf{v}_1 = \bar{\boldsymbol{\pi}}^\top \mathbf{1} = 1$ is automatic since $\bar{\boldsymbol{\pi}}$ is a probability vector.

Step 4 (Spectral decomposition of $\mathbf{T}^\tau$). Diagonalisability with biorthonormality gives the dyadic expansion

$$\mathbf{T}^\tau \;=\; \sum_{k=1}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top \;=\; \mathbf{1}\, \bar{\boldsymbol{\pi}}^\top \;+\; \sum_{k=2}^{K} \lambda_k^\tau\, \mathbf{v}_k \mathbf{w}_k^\top,$$

isolating the dominant $(k = 1)$ projection from the contractive spectrum. The dominant projector $\mathbf{P}_1 = \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is the rank-one stationary projection: $\mathbf{P}_1 \mathbf{x} = (\bar{\boldsymbol{\pi}}^\top \mathbf{x})\, \mathbf{1}$.

Step 5 (Evaluation of the bilinear form). Substituting the spectral decomposition into (28),

$$\mathbb{E}[|G_t|\, |G_{t+\tau}|] \;=\; \mathbf{m}^\top \mathbf{D}\, \mathbf{1}\, \bar{\boldsymbol{\pi}}^\top \mathbf{m} \;+\; \sum_{k=2}^{K} \lambda_k^\tau\, (\mathbf{m}^\top \mathbf{D}\, \mathbf{v}_k)\, (\mathbf{w}_k^\top \mathbf{m}).$$

The $k = 1$ term simplifies to μ^2: $\mathbf{D}\mathbf{1}$ is the column vector $\bar{\boldsymbol{\pi}}$, so $\mathbf{m}^\top \mathbf{D}\mathbf{1} = \bar{\boldsymbol{\pi}}^\top \mathbf{m} = \mu$, and likewise $\bar{\boldsymbol{\pi}}^\top \mathbf{m} = \mu$, giving the product $\mu \cdot \mu = \mu^2$. Denoting the $k \geq 2$ scalar coefficients

$$c_k \;=\; (\mathbf{m}^\top \mathbf{D}\, \mathbf{v}_k)\, (\mathbf{w}_k^\top \mathbf{m}),$$

we obtain

$$\mathbb{E}[|G_t|\, |G_{t+\tau}|] \;=\; \mu^2 \;+\; \sum_{k=2}^{K} c_k\, \lambda_k^\tau, \qquad \tau \geq 1.$$

Step 6 (Autocovariance and autocorrelation). Subtracting μ^2 from Step 1 cancels the dominant term, leaving

$$\mathrm{Cov}(|G_t|,\, |G_{t+\tau}|) \;=\; \sum_{k=2}^{K} c_k\, \lambda_k^\tau, \qquad \tau \geq 1,$$

which is equation (7). Dividing by $\sigma^2_{|G|} = \gamma(0)$ gives $\rho_{|G|}(\tau) = \sum_{k\geq 2} w_k\, \lambda_k^\tau$ with $w_k = c_k/\sigma^2_{|G|}$, equation (8). □

Lag zero, the sum of weights, and the within-state white-noise contribution. The bilinear identity (28) fails at $\tau = 0$, where $G_t = G_{t+\tau}$ and the conditional product is the per-state second moment $M_i = \mathbb{E}[G_t^2 \mid s_t = i]$ rather than m_i^2. The eigenvalue completeness $\sum_{k=1}^{K} \mathbf{v}_k \mathbf{w}_k^\top = I$ instead gives the identity

$$\sum_{k=2}^{K} c_k \;=\; \mathbf{m}^\top \mathbf{D}\, \mathbf{m} \;-\; \mu^2 \;=\; \sum_{i=1}^{K} \bar{\pi}_i\, m_i^2 \;-\; \mu^2 \;=\; \mathrm{Var}\big(m_{s_t}\big),$$

the between-state variance of $|G_t|$, equal to the variance of the conditional-mean random variable m_{s_t} under $\bar{\boldsymbol{\pi}}$. By the law of total variance, $\sigma^2_{|G|} = \mathrm{Var}(m_{s_t}) + \mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)]$, so

$$\sum_{k=2}^{K} w_k \;=\; \frac{\mathrm{Var}(m_{s_t})}{\sigma^2_{|G|}} \;\in\; [0, 1],$$

with equality to 1 if and only if every per-state distribution is degenerate at m_i. The residual $1 - \sum_{k\geq 2} w_k = \mathbb{E}[\mathrm{Var}(|G_t| \mid s_t)] \,/\, \sigma^2_{|G|}$ is the within-state variance ratio: it contributes to $\sigma^2_{|G|}$ but not to $\mathrm{Cov}(|G_t|, |G_{t+\tau}|)$ at any positive lag, and is the source of the discontinuity from $\rho_{|G|}(0) = 1$ to $\rho_{|G|}(1) = \sum_{k\geq 2} w_k\, \lambda_k$ at the first lag. The substantive consequence for moderate-K CHMM is unchanged: at every $\tau \geq 1$ the autocorrelation lies in the span of the geometric decays $\{\lambda_k^\tau\}_{k\geq 2}$, and the slow-decay envelope is governed by the largest non-unit eigenvalue.

Squared-return analogue. Replacing $|G_t|$ with G_t^2 and $\mathbf{m}$ with $\mathbf{M} = (M_1, \dots, M_K)^\top$ in Steps 1–6 above is termwise valid for $\tau \geq 1$ (the conditional independence and stationarity arguments are unaffected), giving the squared-return analogue

$$\rho_{G^2}(\tau) \;=\; \sum_{k=2}^{K} \tilde{w}_k\, \lambda_k^\tau, \qquad \tilde{w}_k \;=\; \frac{(\mathbf{M}^\top \mathbf{D}\, \mathbf{v}_k)\, (\mathbf{w}_k^\top \mathbf{M})}{\sigma^2_{G^2}}.$$

The non-unit spectrum of $\mathbf{T}$ that determines $\rho_{|G|}(\tau)$ also determines $\rho_{G^2}(\tau)$; only the linear projection of the per-state moment vector onto the eigenmodes changes.

Remark (complex eigenvalues). For real $\mathbf{T}$ without self-adjoint structure, eigenvalues outside $\mathbb{R}$ occur in complex conjugate pairs with conjugate eigenvectors. A pair $(\lambda_k, \bar{\lambda}_k)$ at $\lambda_k = re^{i\theta}$ contributes a real damped-oscillation term of the form $A\, r^\tau \cos(\theta\tau) + B\, r^\tau \sin(\theta\tau)$ to $\gamma(\tau)$, with (A, B) determined from $(\mathrm{Re}\, c_k, \mathrm{Im}\, c_k)$. Theorem 1 as stated remains valid; only the interpretation of the individual modes shifts from monotone geometric to damped sinusoidal. For $\mathbf{T}$ estimated on equity returns at the operating point of Section 4, the slow-decay mode at $\lambda_2 \approx \exp(-1/\tau^*)$ is real and positive, and the mono-exponential interpretation of the dominant timescale used in Section 3.8 carries through.

Remark (non-diagonalisable T). Diagonalisability is generic: for $\mathbf{T}$ estimated by Baum–Welch in the interior of a non-degenerate parameter region, the eigenvalues are almost surely distinct, and the sample paths used in this paper realise this generic case at every K in the sweep. Should $\mathbf{T}$ admit a non-trivial Jordan block of size J, the dyadic expansion of Step 4 acquires polynomial-in-τ prefactors $\binom{\tau}{j}\lambda_k^{\tau-j}$ for $j = 0, \ldots, J-1$ on the off-diagonal blocks, and $\rho_{|G|}(\tau)$ becomes a finite sum of polynomial-times-geometric terms with the same exponential rate $\max_{k\geq 2}|\lambda_k|$. The substantive content of Theorem 1, that the absolute-return ACF is determined entirely by the non-unit spectrum of $\mathbf{T}$, is unchanged.

A.6 State-Resolution and Multi-Emission Sensitivity

This appendix reports the full K-sweep underlying the headline operating point of Section 4, the multi-emission family extension confirming that $K = 18$ is stable under Student-t and Laplace emissions, the EM convergence curves at representative K, the IS and OoS visual comparison panels at non-selected $K \in \{3, 6, 12, 21\}$, the multi-emission figure panels at $K = 18$, and the Rydén $K = 2$ replication.

A.6.1 Full State-Resolution Sensitivity Table

Table 13 reports the full K-sweep referenced in Section 4.2 of the main text. Observed excess kurtosis is 7.68 IS and 5.29 OoS at all K.

Table 13. State resolution sensitivity for SPY CHMM-N (1,000 paths, $\alpha = 0.05$). Every metric is computed on the identical simulation pipeline used for Table 3 in the main text. IS coverage was 100% at every K; OoS coverage is reported separately. ACF-MAE columns: $|G_t|$ measures the volatility-clustering reproduction; G_t measures linear-autocorrelation reproduction (parallel to $|G_t|$, both averaged over lags 1–252).

	KS (%)		AD (%)		Kurtosis		ACF-MAE				
K	IS	OoS	IS	OoS	Obs	Sim	$\lvert G_t\rvert$	G_t	W_1	H	Cov OoS (%)
3	89.0	77.2	78.9	68.6	7.68	3.82	0.0463	0.0240	0.131	0.0839	94.9
6	92.9	78.7	84.3	74.3	7.68	5.38	0.0496	0.0238	0.116	0.0829	90.9
9	93.0	81.4	84.4	74.4	7.68	4.63	0.0492	0.0238	0.118	0.0810	91.9
12	94.9	81.6	85.8	75.0	7.68	4.57	0.0500	0.0238	0.113	0.0801	91.9
15	93.9	81.0	84.8	73.5	7.68	4.89	0.0508	0.0238	0.114	0.0804	88.9
18	94.8	85.5	88.2	79.7	7.68	5.04	0.0515	0.0237	0.110	0.0807	91.9
21	**95.6**	**88.0**	**89.1**	**78.8**	7.68	3.78	0.0534	**0.0236**	0.112	0.0801	**99.0**

A.6.2 Multi-Emission State-Resolution Sensitivity

Table 14 extends Table 13 across the four emission families, confirming that the qualitative operating point $K = 18$ is stable when the Gaussian component is replaced with Student-t, Laplace, or GED emissions. For CHMM-t the per-state degrees of freedom ν_k are refit by ECM-plus-golden-section at every K; for CHMM-L the weighted-median location and weighted MAD scale are closed form and impose no additional cost; for CHMM-GED the three-stage CM update of Section 3.7 refits (μ_k, α_k, p_k) at every K with p_k bracketed in $[0.5, 3.0]$. Three patterns extend from Table 13 to the new families. First, the distributional pass rates plateau in the $K \in \{12, 15, 18, 21\}$ band for all four families: CHMM-t reaches its highest IS KS at $K = 18$ (94.6%), CHMM-L reaches its highest IS KS at $K = 18$ (94.2%) and its highest IS AD at $K = 18$ (91.4%), and CHMM-GED reaches its highest IS KS at $K = 12$ (97.6%) and the highest entry across the entire panel. Once a family enters the plateau the operating point is flat with respect to K, so we report the main-text results at the same $K = 18$ used for CHMM-N. Second, the absolute-return ACF-MAE remains within the narrow 0.046–0.058 band across the four families, and the raw-return ACF-MAE remains essentially constant at ≈ 0.024 across all 28 (family, K) combinations. Both temporal axes are properties of the regime structure rather than the emission family: $|G_t|$ ACF reproduction is governed by the eigenvalue spectrum of $\mathbf{T}$ (Section 3.8), and raw-G_t ACF is null in expectation under any HMM with i.i.d. within-state emissions. Third, simulated kurtosis behaves as expected from the component families: the Gaussian sweep peaks around 5.0 at the high-K end, the Laplace sweep lands squarely in the 5.2–6.7 range (closest to the observed 7.68), the Student-t sweep climbs sharply above 13 across the entire sweep because the ECM search assigns small ν_k to a subset of tail states, and the GED sweep mirrors the Gaussian curve at 4.7–5.5 because the bulk of $\hat{p}_k$ values pin near the upper bracket and the Laplace-like tail states contribute too little stationary mass to lift the aggregate moment.

Table 14. Multi-emission state resolution sensitivity for SPY (1,000 paths, $\alpha = 0.05$). CHMM-N: Gaussian; CHMM-t: Student-t with per-state ν_k refit by ECM-plus-golden-section; CHMM-L: Laplace with weighted-median location and weighted-MAD scale; CHMM-GED: Generalized Error Distribution with per-state shape $p_k \in [0.5, 3.0]$ via three-stage ECM (Section 3.7). Every metric is computed on the identical simulation pipeline used for Table 3. Observed excess kurtosis is 7.68 IS and 5.29 OoS. ACF-MAE columns parallel the main-text Table 3: $|G_t|$ for volatility clustering, G_t (raw) for linear autocorrelation. **Sub-seed convention:** this multi-emission sweep uses a different sub-seed offset from the body-table headline run, so per-row metrics may drift by ≤ 1 percentage point from the corresponding entries in Table 3 (e.g. CHMM-GED at $K = 18$: this table 96.3% / 5.05 / 0.0546 versus body 95.2% / 5.15 / 0.0548).

		KS (%)		AD (%)			ACF-MAE			
Family	K	IS	OoS	IS	OoS	Kurt (IS sim)	$\lvert G_t \rvert$	G_t	W_1 IS	H IS
CHMM-N	3	89.7	80.1	79.2	70.7	3.86	0.0467	0.0240	0.128	0.0833
	6	93.8	79.1	87.2	72.8	5.29	0.0504	0.0238	0.114	0.0825
	9	94.9	83.0	84.3	74.8	4.58	0.0490	0.0238	0.115	0.0806
	12	93.7	82.7	85.5	75.0	4.56	0.0501	0.0237	0.115	0.0804
	15	93.6	83.0	87.0	75.1	4.89	0.0509	0.0238	0.114	0.0803
	18	94.7	83.4	88.5	76.6	4.97	0.0511	0.0237	0.108	0.0805
	21	**95.1**	**88.0**	**88.2**	**79.7**	3.80	0.0532	0.0236	0.110	0.0796
CHMM-t	3	92.2	82.9	88.0	79.1	33.68	0.0538	0.0235	0.108	0.0759
	6	91.8	80.4	86.7	74.8	22.74	0.0527	0.0235	0.108	0.0772
	9	95.4	84.0	89.9	78.0	16.60	0.0535	0.0234	0.102	0.0752
	12	95.2	85.0	89.8	78.4	14.98	0.0543	0.0234	0.100	0.0757
	15	94.7	83.7	88.8	79.5	16.36	0.0539	0.0234	0.101	0.0758
	18	**96.0**	**84.5**	**91.7**	**80.8**	13.19	0.0551	**0.0234**	**0.099**	**0.0759**
	21	94.5	83.3	88.5	78.1	14.66	0.0539	0.0234	0.102	0.0758
CHMM-L	3	80.5	62.7	81.2	66.5	5.24	0.0531	0.0235	0.113	0.0855
	6	88.3	76.3	83.1	71.8	5.86	0.0510	0.0235	0.112	0.0821
	9	90.7	77.2	84.2	72.2	5.92	0.0519	0.0235	0.112	0.0823
	12	93.8	79.3	88.7	76.7	6.17	0.0541	0.0234	0.105	0.0793
	15	92.5	78.3	88.2	75.5	5.98	0.0539	0.0234	0.107	0.0794
	18	**94.2**	**80.5**	**91.4**	**80.6**	**6.74**	0.0566	0.0234	0.104	0.0795
	21	91.5	79.6	87.5	77.7	5.95	0.0539	0.0234	0.107	0.0807
CHMM-GED	3	90.3	78.4	85.5	73.6	5.45	0.0531	0.0236	0.109	0.0818
	6	91.8	79.0	84.9	73.5	5.22	0.0518	0.0235	0.108	0.0805
	9	94.6	81.6	89.1	75.9	5.03	0.0535	0.0234	0.101	0.0787
	12	**97.6**	**84.2**	**94.0**	**81.2**	4.67	0.0563	0.0234	**0.094**	**0.0775**
	15	94.2	82.0	87.4	76.5	5.14	0.0535	0.0235	0.104	0.0784
	18	96.3	83.8	92.7	81.3	5.05	0.0546	0.0235	0.098	0.0779
	21	94.1	85.3	88.7	81.2	4.82	0.0542	0.0235	0.102	0.0781

A.6.3 Cross-Ticker Per-Family Kurtosis Pattern

The body Table 5 reports the penalised CHMM-t cross-ticker performance ($1/\nu_k$ shrinkage at $\lambda = 20$). This appendix documents the unpenalised cross-ticker behaviour that motivates the penalty. The CHMM-t simulated kurtosis blow-ups under the unpenalised ECM (NVDA 40.8, JNJ 104.1, JPM 17.6 at $K = 18$) are specific to the Student-t emission family, and Table 15 extends the kurtosis column across all four CHMM emission variants on the same six-ticker universe to make this explicit. The pattern is consistent: unpenalised CHMM-t produces single-ticker overshoots of an order of magnitude on the heavy-tailed defensives (JNJ, NVDA), CHMM-L sits closest to observed across every ticker (median absolute residual 1.4 kurtosis units), CHMM-N undershoots uniformly, and CHMM-GED interpolates between them. The CHMM-t blow-up mechanism is the same per-state ν_k ECM lower-bracket pinning documented for SPY in Section A.2.3: on tickers whose tail regime is concentrated in a small subset of states (NVDA's pre-AI-rally drawdowns, JNJ's outlier-driven defensive-stock dynamics), one or two states pin at $\nu_k \approx 2.1$ and dominate the simulated-kurtosis sample moment. Under the $1/\nu_k$ penalty at $\lambda = 20$ (the body operating point in Table 5; full rate sweep in Section 5, Appendix A.9) the simulated kurtosis on every ticker drops into the same band as CHMM-L: SPY 14.7 → 8.0, NVDA 40.8 → 5.2, JNJ 104.1 → 12.3, JPM 17.6 → 7.7, AAPL 6.3 → 2.8, QQQ 3.3 → 2.9, while IS / OoS KS pass rates and $|G_t|$ ACF-MAE move by less than 1.5 percentage points and 0.001 respectively.

Table 15. Per-ticker simulated excess kurtosis across the four CHMM emission families ($K = 18$, 1,000 paths, seed = 20260420). Columns report mean simulated IS excess kurtosis under each variant; the "Obs" column is the IS sample value. CHMM-t is the only family that produces order-of-magnitude blow-ups (NVDA, JNJ, JPM); the remaining three sit within 4 kurtosis units of observed across every ticker. Numbers are sourced from the canonical multi-emission cross-ticker run used for Table 14; minor drift from the body Table 5 CHMM-t column reflects a single re-run with a refreshed sub-seed.

Ticker	Obs	CHMM-N	CHMM-t	CHMM-L	CHMM-GED
SPY	7.7	5.0	15.0	6.8	5.1
NVDA	5.5	3.7	40.8	3.2	5.0
JNJ	9.0	5.5	104.1	6.1	9.3
JPM	7.6	7.2	17.6	6.0	14.8
AAPL	3.2	2.3	6.3	3.9	3.2
QQQ	3.4	2.0	3.3	4.1	2.7

A.6.4 Sector-Balanced 30-Ticker Panel: Full Per-Ticker Rollup

The body Table 5 reports the aggregate distribution and per-sector medians for the sector-balanced 30-ticker panel. Table 16 below gives the full per-ticker rollup. Each ticker is fit independently under the penalised CHMM-t scaffold ($K = 18$, $\lambda = 20$, 1,000 simulated paths, seed 20260420) on its own IS slice covering 2014-01-03 to 2024-01-03 and validated on the same 2024-01-04 to 2026-04-20 OoS window used for SPY in Table 3. The per-ticker ν_k medians sit at the upper bracket (50) on every ticker, indicating that the $1/\nu_k$ shrinkage is in the operative regime across the panel; minimum and maximum ν_k values are in the per-ticker CSV in `results/sector_panel/sector_panel_summary.csv`.

Table 16. Sector-balanced 30-ticker panel, per-ticker rollup (penalised CHMM-t at $K = 18$, $\lambda = 20$). Sectors ordered by GICS classification; tickers within sector ordered by the panel construction (top three large-cap representatives at IS-window-median market cap). Kurt resid is simulated minus observed IS excess kurtosis. Aggregate distribution and per-sector medians are summarised in body Table 5.

Sector	Ticker	IS KS%	OoS KS%	Kurt obs	Kurt sim	Kurt resid	$\lvert G_t \rvert$ ACF-MAE
Information Technology	AAPL	99.1	95.1	3.19	2.84	−0.35	0.0418
Information Technology	MSFT	99.7	96.9	4.20	3.48	−0.72	0.0389
Information Technology	NVDA	99.6	55.7	5.53	6.23	0.70	0.0474
Health Care	JNJ	99.3	93.2	8.97	11.65	2.68	0.0335
Health Care	UNH	99.3	14.5	8.85	10.31	1.46	0.0382
Health Care	LLY	99.7	7.6	13.60	66.51	52.91	0.0295
Financials	JPM	99.5	50.5	7.62	8.14	0.52	0.0458
Financials	BAC	99.6	84.8	5.76	5.47	−0.29	0.0394
Financials	WFC	98.2	57.6	7.53	10.90	3.37	0.0791
Consumer Discretionary	AMZN	99.4	96.9	4.31	3.68	−0.63	0.0413
Consumer Discretionary	HD	99.2	41.4	12.92	17.03	4.11	0.0368
Consumer Discretionary	MCD	99.5	66.6	18.38	26.93	8.55	0.0394
Communication Services	NFLX	99.3	45.1	12.60	4.67	−7.93	0.0320
Communication Services	VZ	99.3	49.1	5.14	5.15	0.01	0.0270
Communication Services	DIS	99.5	96.4	10.77	13.94	3.17	0.0644
Industrials	CAT	99.7	78.9	4.34	3.48	−0.86	0.0321
Industrials	BA	97.6	62.2	20.59	52.00	31.41	0.0793
Industrials	HON	99.3	96.0	21.93	17.09	−4.84	0.0571
Consumer Staples	PG	99.7	94.3	8.15	9.51	1.36	0.0347
Consumer Staples	KO	99.5	92.7	10.57	12.84	2.27	0.0457
Consumer Staples	WMT	99.7	24.0	13.84	32.63	18.79	0.0279
Energy	XOM	96.9	70.3	5.61	10.99	5.38	0.0988
Energy	CVX	99.6	62.3	12.22	14.78	2.56	0.0498
Energy	COP	99.4	83.6	17.11	8.28	−8.83	0.0494
Utilities	NEE	98.6	24.0	10.71	9.87	−0.84	0.0455
Utilities	DUK	99.7	96.9	9.75	8.11	−1.64	0.0466
Utilities	SO	99.6	96.2	14.52	12.73	−1.79	0.0481
Materials	FCX	99.8	76.6	4.83	4.11	−0.72	0.0573
Materials	NEM	99.6	5.6	3.74	3.88	0.14	0.0376
Materials	APD	99.2	90.2	8.40	16.74	8.34	0.0381

A.6.5 Five-Named-Ticker Spot-Check (Original Six-Ticker Panel)

The original peer-review-cycle six-ticker spot-check (SPY, NVDA, JNJ, JPM, AAPL, QQQ; broad-equity ETF and five named single-name representatives) is retained in Table 17 as a hand-curated reference for downstream consumers who need to compare against the cross-asset Pipeline-B copula construction in Section 4.5 (whose six-asset universe is the same five named tickers plus QQQ). The aggregate-distribution, sector-rollup, and universe-scale failure-rate framing is moved to the body Table 5 on the 30-ticker panel.

Table 17. Original five-named-ticker spot-check (penalised CHMM-t at $K = 18$, $\lambda = 20$). Hand-curated panel covering broad-equity (SPY, QQQ), AI / semiconductors (NVDA), healthcare (JNJ), financials (JPM), and consumer-tech (AAPL); reproduced from `results/chmm_t_penalised/Cross_Ticker_CHMM_t_Pen.txt`. The expanded sector-balanced 30-ticker panel is in Table 16.

	KS (%) ↑		Kurtosis		ACF-MAE
Ticker	IS	OoS	Obs	Sim	↓
SPY	95.8	85.0	7.7	8.01	0.0541
NVDA	99.5	53.8	5.5	5.23	0.0473
JNJ	99.8	92.3	9.0	12.28	0.0334
JPM	97.8	53.8	7.6	7.70	0.0457
AAPL	99.1	95.1	3.2	2.84	0.0418
QQQ	96.1	91.8	3.4	2.88	0.0569

A.6.6 Effective Spectral Rank of the Absolute-Return ACF

The body Section 3.8 states the algebraic upper bound: the deflated transition matrix $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ has rank at most $K - 1$, so the absolute-return ACF identity (8) contains at most $K - 1$ non-unit decay modes. The empirical question is how many of those modes carry non-trivial contribution to $\rho_{|G|}(\tau)$ at the relevant lags. Table 18 reports the per-mode contribution $w_k \lambda_k^\tau$ at $\tau \in \{1, 5, 20, 50\}$ for the fitted CHMM-N at $K = 18$ and at $K = 3$ on SPY IS, with modes ranked by lag-1 contribution magnitude $|w_k \lambda_k|$. Per-state moments $m_k = \mathbb{E}[|G_t| \mid s_t = k]$ are estimated by 200,000 draws from each emission; eigenvectors are normalised so $\mathbf{w}_k^\top \mathbf{v}_k = 1$.

Table 18. Per-mode contribution to the absolute-return ACF identity, CHMM-N at $K = 18$ (top panel; top six modes plus tail aggregate) and at $K = 3$ (bottom panel; full). Modes ordered by $|w_k \lambda_k|$ descending. "cum" is cumulative share of $\sum_k |w_k \lambda_k|$. At $K = 18$ the top mode ($|\lambda| = 0.929$) carries 93.6% of the lag-1 ACF, three modes reach 95%, and at lag 20 only the dominant mode contributes non-negligibly. At $K = 3$ the dominant mode ($|\lambda| = 0.953$) alone carries 96.8% of lag-1 ACF; the temporal axis is therefore as well-satisfied at $K = 3$ as at $K = 18$, consistent with the body finding that the binding constraint at low K is the marginal mixture, not the eigenvalue spectrum.

rank	$\lvert\lambda_k\rvert$	$\lvert w_k\rvert$	$\lvert w_k \lambda_k\rvert$	$w_k \lambda_k^5$	$w_k \lambda_k^{20}$	cum
$K = 18,\ \rho_{\lvert G\rvert}(1) = 0.301,\ \sigma^2_{\lvert G\rvert} = 2.555$						
1	0.929	0.324	0.301	0.224	0.0744	0.936
2	0.335	0.012	0.004	0.0001	0.0000	0.948
3	0.854	0.003	0.002	-0.001	-0.0001	0.955
4	0.231	0.008	0.002	≈ 0	≈ 0	0.961
5	0.231	0.008	0.002	≈ 0	≈ 0	0.967
6	0.203	0.008	0.002	≈ 0	≈ 0	0.971
7–17	various	various	≤ 0.001 each	≈ 0	≈ 0	1.000
$K = 3,\ \rho_{\lvert G\rvert}(1) = 0.287,\ \sigma^2_{\lvert G\rvert} = 2.517$						
1	0.953	0.292	0.278	0.230	0.112	0.968
2	0.866	0.011	0.009	0.005	0.0006	1.000

The empirical effective rank confirms the body framing of Section 3.8: the ACF-MAE flatness across $K \in \{3, 6, 9, 12, 15, 18, 21\}$ documented in Appendix A.6.1, Table 13, is the macroscopic signature of the temporal axis being driven by a single dominant decay mode at every operating point in the sweep. The K-rank statement on $\mathbf{T} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is correctly interpreted as a non-binding upper bound at $K \geq 3$; at $K = 2$ the bound is tight (only one non-unit eigenvalue available), but the failure mode is then the marginal mixture, which two Gaussian components cannot tile against the empirical equity-return target. The bilinear identity (8) therefore plays a pedagogical role in this paper rather than serving as a direct K-selection criterion: it makes the rank constraint explicit, but the operational K choice is driven by distributional fidelity (KS, AD, kurtosis), not by the count of non-unit eigenvalues in the spectral expansion.

A.6.7 Cross-Ticker Spectral Effective-Rank Diagnostic

Reviewers 2 and 3 ask whether the SPY-only result of Appendix A.6.6 (a single non-unit eigenvalue carries 93.6% of the lag-1 absolute-return ACF at $K = 18$) generalises across tickers. Table 19 reports the cross-ticker distribution of the dominant-mode share, computed under the same protocol on the 30-ticker sector-balanced panel plus SPY. Source: `results/diagnostics/spectral_rank_cross_ticker.txt`.

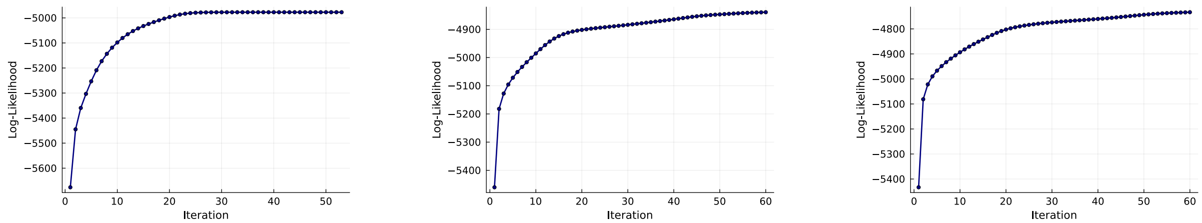
Table 19. Cross-ticker spectral effective-rank diagnostic at $K = 18$. Distribution across 31 tickers (sector-balanced 30-ticker panel plus SPY control). "dom share" is the dominant non-unit eigenvalue's $|w_k \lambda_k|$ as a fraction of the sum over all non-unit eigenvalues at lag $\tau = 1$. "n_{95}" / "n_{99}" are the number of modes needed for 95% / 99% cumulative share. The SPY value of 0.94 is in the right tail of the distribution; the cross-ticker median is 0.76.

Statistic	dom share	n_{95}	n_{99}
median	0.756	6	11
$[Q_1, Q_3]$	$[0.661, 0.858]$	$[4, 7]$	$[10, 12]$
minimum (NEM)	0.326	11	14
SPY (body Appendix A.6.6)	0.943	2	10

Reading: the SPY headline of 93.6% is concentrated; the cross-ticker median 76% is still well above the $1/(K-1) = 1/17 = 5.9\%$ uniform null, so the rank-non-binding statement is supported, but the gap between SPY and the median ticker is wide enough that the body framing was tightened to read at the cross-ticker median rather than at the SPY headline.

A.6.8 Baum-Welch Convergence

Figure 5 shows the Baum-Welch convergence curves for representative state resolutions. All models converged within the 60-iteration budget, with the log-likelihood monotonically increasing (as guaranteed by the EM algorithm). Lower K values typically converged faster (15–25 iterations), while higher K values required 25–40 iterations to reach the convergence tolerance of $|\Delta\mathcal{L}| < 10^{-4}$. The monotonic increase confirms that the quantile-based initialization provided a valid starting point from which the EM algorithm could improve without encountering degenerate solutions.

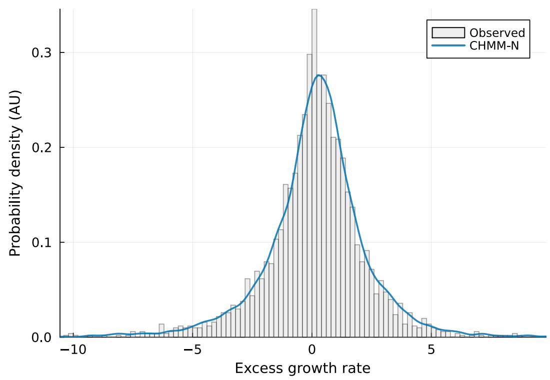


(a) $K = 3$ **(b)** $K = 12$ **(c)** $K = 18$ (selected)

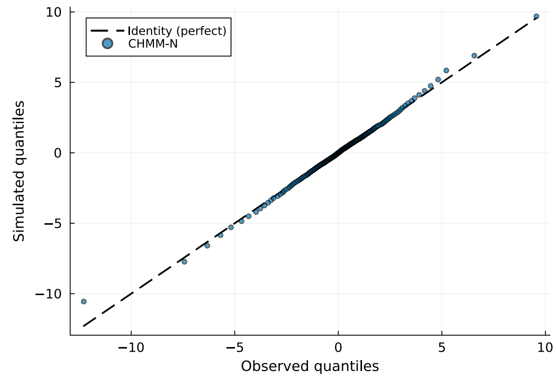
Figure 5. Baum-Welch convergence for the SPY CHMM-N fit (Pipeline A, IS) at selected state resolutions. Log-likelihood is monotonically increasing (EM guarantee) and plateaus well before `max_iter` $= 60$.

A.6.9 IS Comparison at Non-Selected State Resolutions

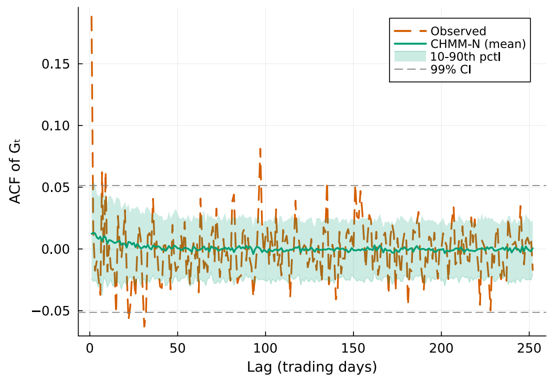
Figures 6–9 show the IS model comparison panels for $K \in \{3, 6, 12, 21\}$, complementing the $K = 18$ comparison in Figure 26 of the main text. Across all resolutions, the CHMM's simulated density closely overlays the observed density, and the ACF of absolute returns tracks the slow decay pattern. The visible improvement from $K = 3$ to $K = 18$ is concentrated in the tails of the Q-Q panel and the steady-state level of the ACF panel. $K = 9$ and $K = 15$ panels are omitted because they interpolate smoothly between the shown resolutions; their numerical metrics are in Table 13.



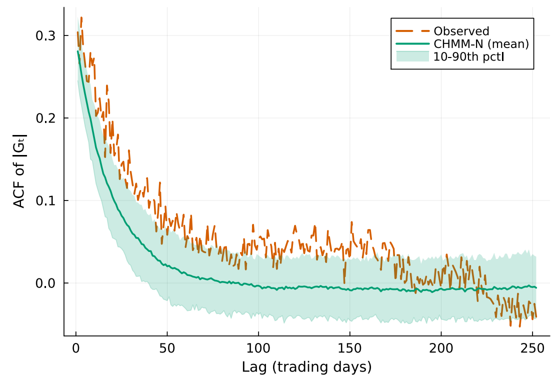
(a) Marginal density.



(b) Tail Q-Q plot.

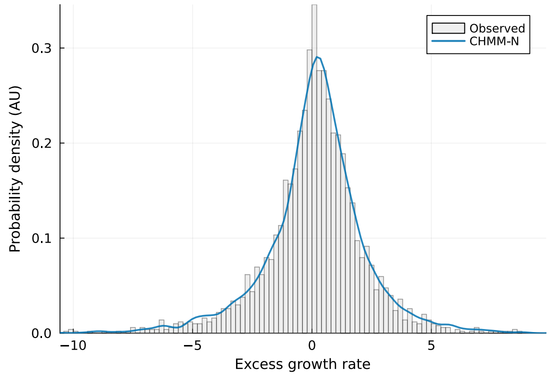


(c) ACF of raw G_t.

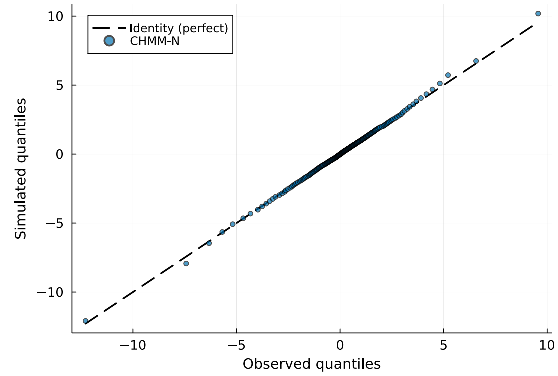


(d) ACF of $|G_t|$.

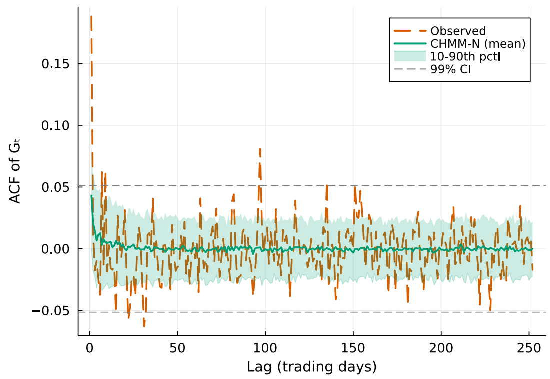
Figure 6. IS SPY CHMM-N comparison at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 26). The $K = 18$ counterpart is Figure 26 of the main text.



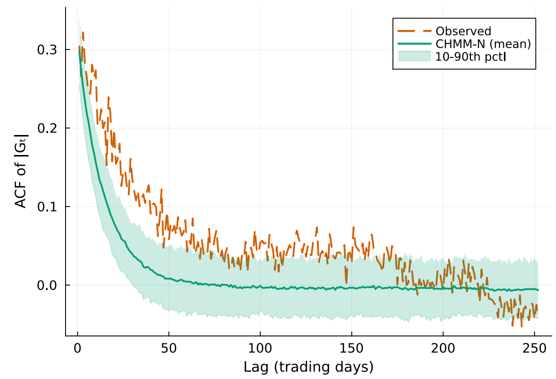
(a) Marginal density.



(b) Tail Q-Q plot.

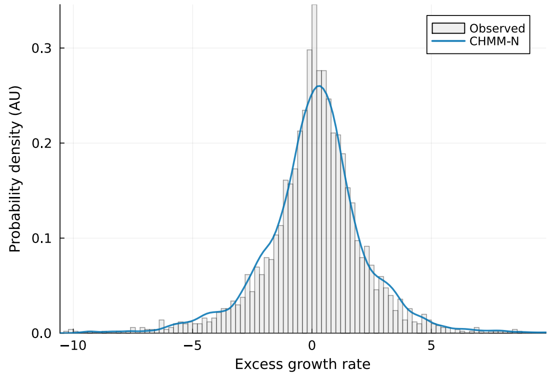


(c) ACF of raw G_t.

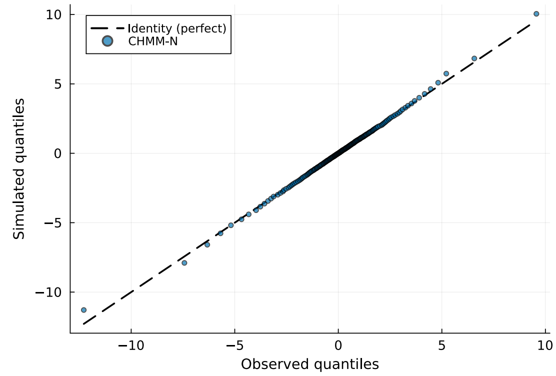


(d) ACF of $|G_t|$.

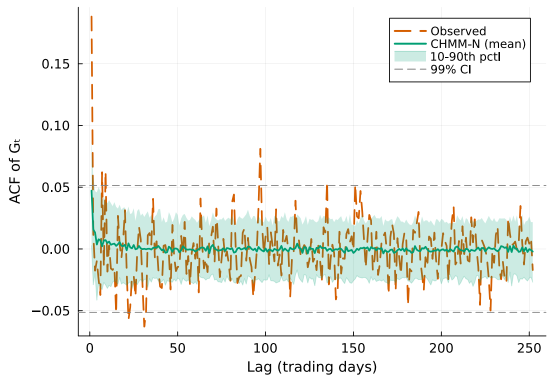
Figure 7. IS SPY CHMM-N comparison at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 26).



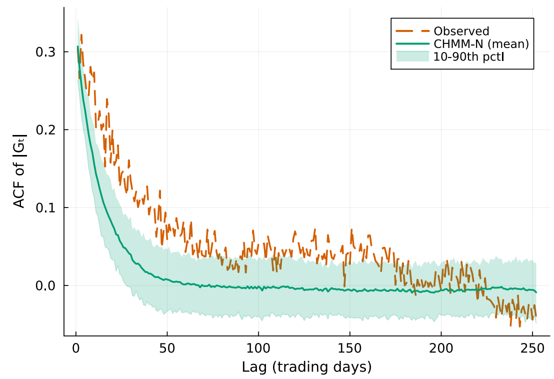
(a) Marginal density.



(b) Tail Q-Q plot.

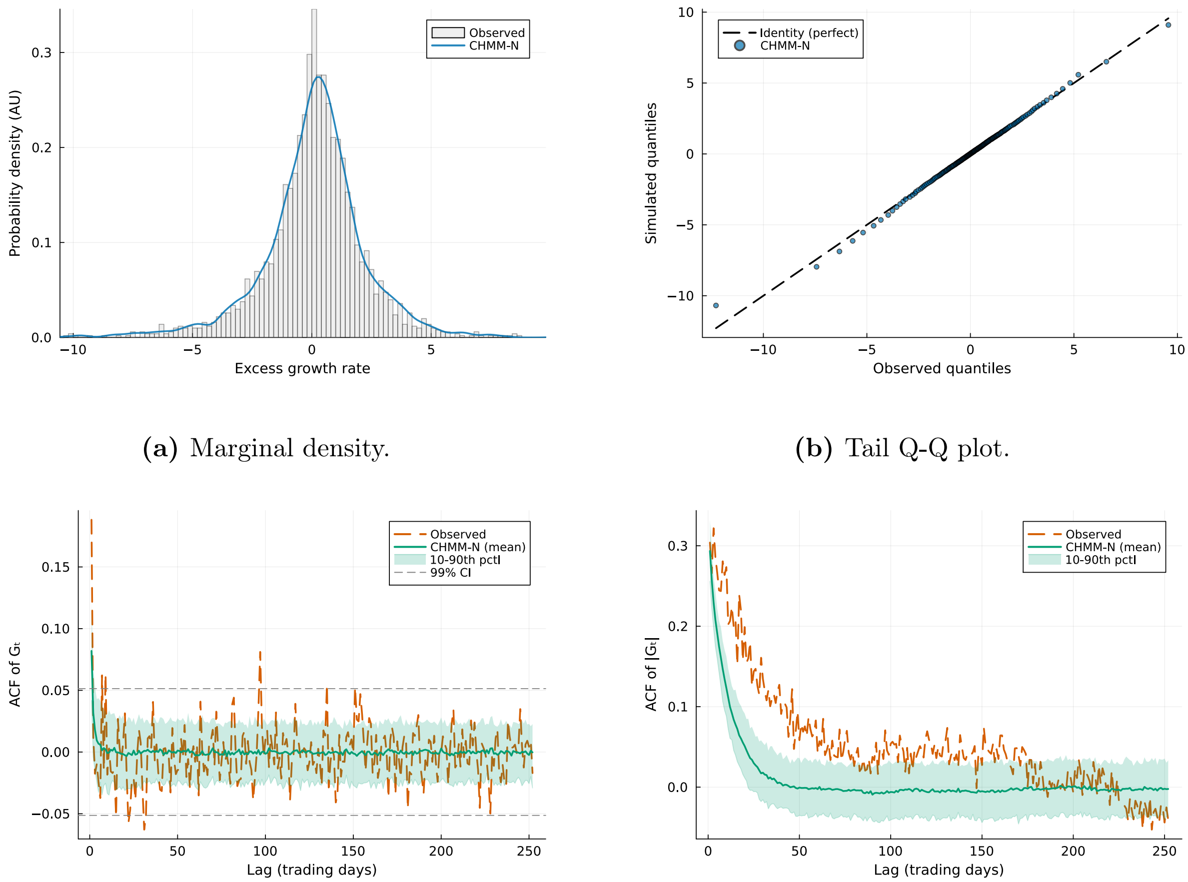


(c) ACF of raw G_t.



(d) ACF of $|G_t|$.

Figure 8. IS SPY CHMM-N comparison at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 26).



(a) Marginal density.

(b) Tail Q-Q plot.

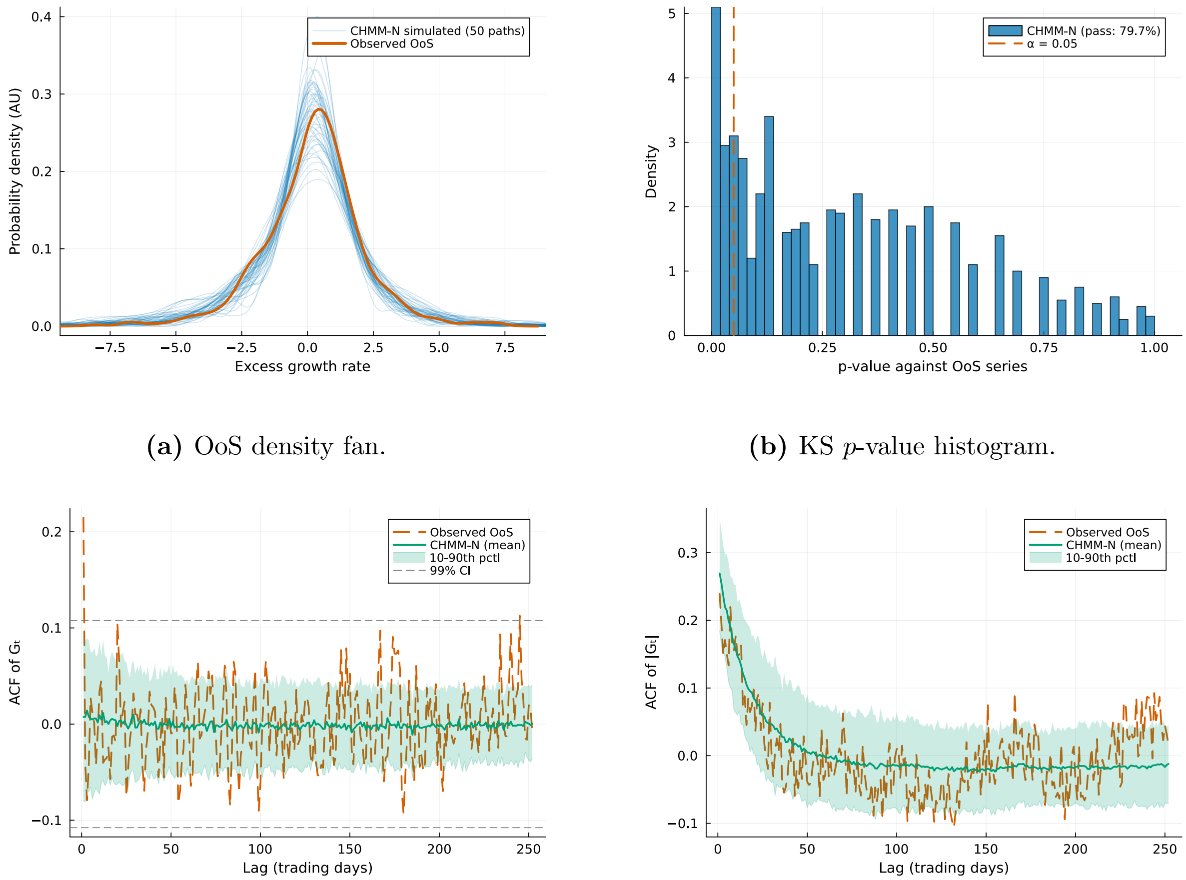
(c) ACF of raw G_t.

(d) ACF of $|G_t|$.

Figure 9. IS SPY CHMM-N comparison at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 26).

A.6.10 OoS Validation at Non-Selected State Resolutions

Figures 10–13 show the OoS validation panels for $K \in \{3, 6, 12, 21\}$. KS pass rates remain above 91% at all resolutions; the $K = 18$ panel is reproduced in Figure 27 of the main text. As with the IS panels, $K = 9$ and $K = 15$ are omitted; numerical OoS KS pass rates are in Table 13.



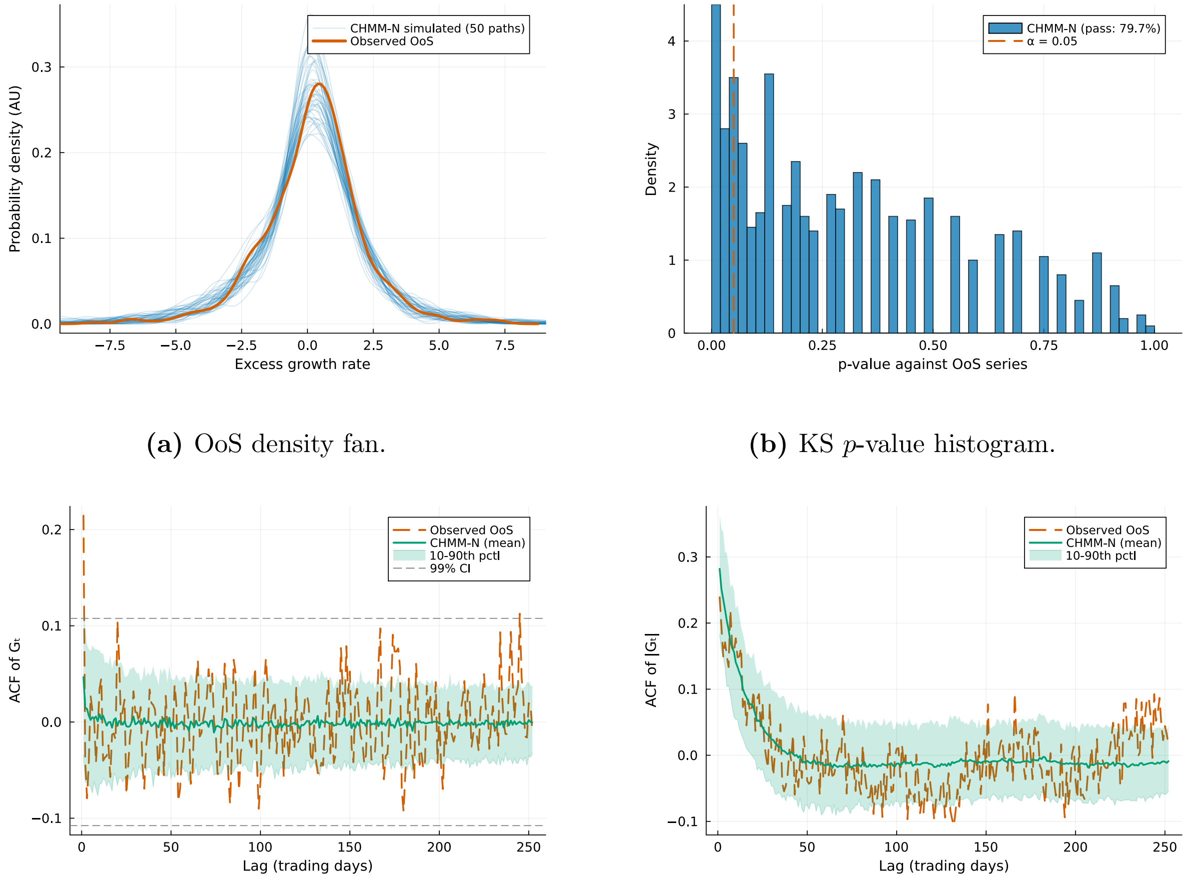
(a) OoS density fan.

(b) KS p-value histogram.

(c) OoS ACF of raw G_t.

(d) OoS ACF of $|G_t|$.

Figure 10. OoS SPY CHMM-N validation at $K = 3$ (Pipeline A, 1,000 simulated paths; panels as in Figure 27). The $K = 18$ counterpart is Figure 27 of the main text.



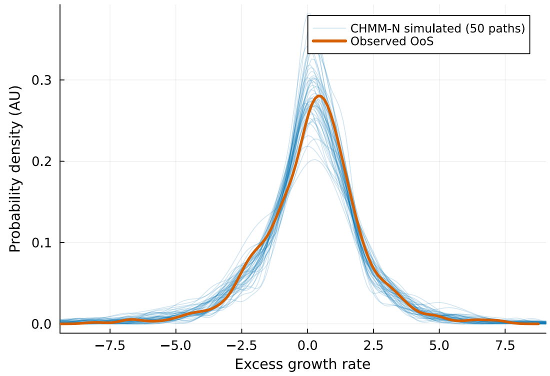
(a) OoS density fan.

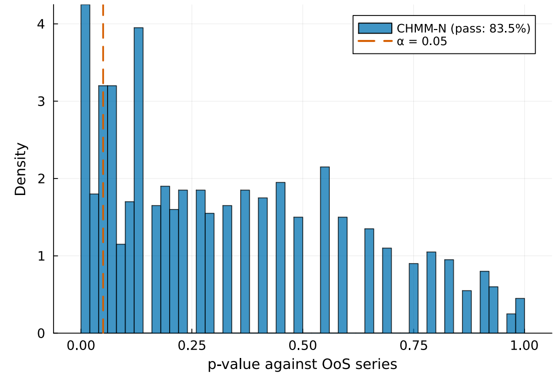
(b) KS p-value histogram.

(c) OoS ACF of raw G_t.

(d) OoS ACF of $|G_t|$.

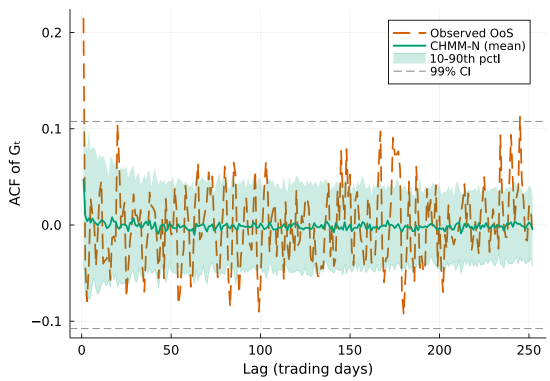
Figure 11. OoS SPY CHMM-N validation at $K = 6$ (Pipeline A, 1,000 simulated paths; panels as in Figure 27).

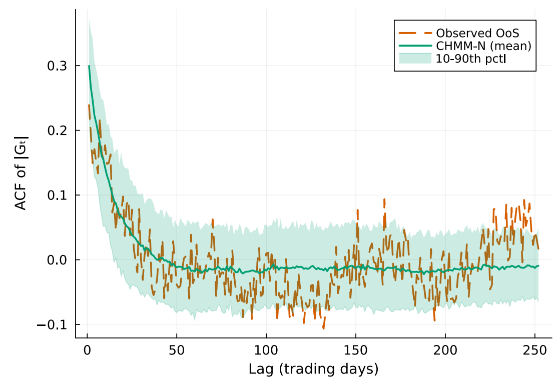




(a) OoS density fan.

(b) KS p-value histogram.

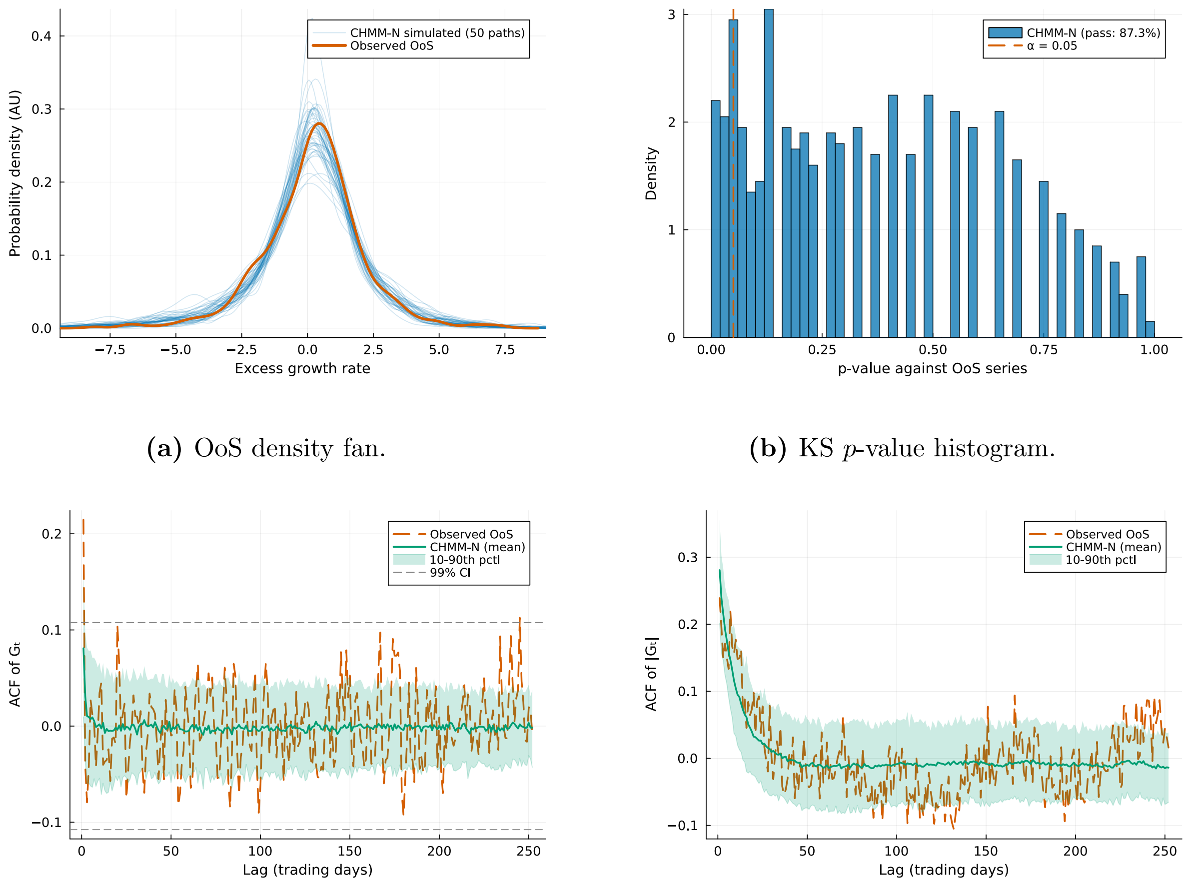




(c) OoS ACF of raw G_t.

(d) OoS ACF of $|G_t|$.

Figure 12. OoS SPY CHMM-N validation at $K = 12$ (Pipeline A, 1,000 simulated paths; panels as in Figure 27).



(a) OoS density fan.

(b) KS p-value histogram.

(c) OoS ACF of raw G_t.

(d) OoS ACF of $|G_t|$.

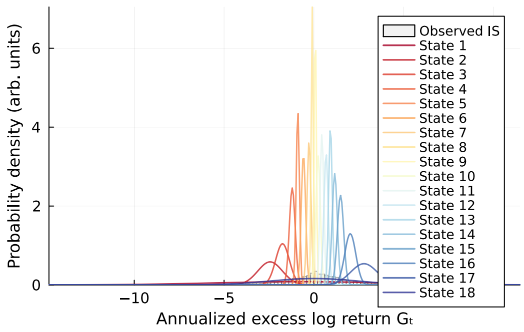
Figure 13. OoS SPY CHMM-N validation at $K = 21$ (Pipeline A, 1,000 simulated paths; panels as in Figure 27).

A.6.11 Multi-Emission Figures at $K = 18$

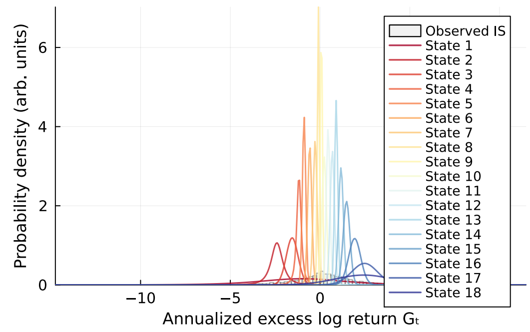
This appendix reproduces the standard diagnostic suite (emission PDFs, residence times, IS comparison, OoS validation, and convergence) for the two heavy-tailed emission families (CHMM-t and CHMM-L) at the selected state resolution $K = 18$; transition matrices and stationary distributions are discussed in prose only below, since they are visually indistinguishable across families at $K = 18$. The Gaussian counterparts are already shown as Figures 26 and 27 in the main text and as Figure 5 above in this appendix. The figures confirm that the shared forward–backward scaffold produces internally-similar regime structure across the four families: the near-uniform interior of the transition matrix, the K-component dense emission cover, the longer central-state residence times, and the monotone log-likelihood convergence are features of the CHMM framework and not of the Gaussian component specifically.

Emission PDFs. Figure 14 compares the learned $K = 18$ emissions for the four families. The CHMM-t panel's per-state ν_k allocation is visible in the narrower modes of certain tail states; the CHMM-L panel's Laplace components have sharper peaks and slower exponential tail decay than the Gaussians; the CHMM-GED panel mixes both, with bulk states near the Gaussian shape ($\hat{p}_k$

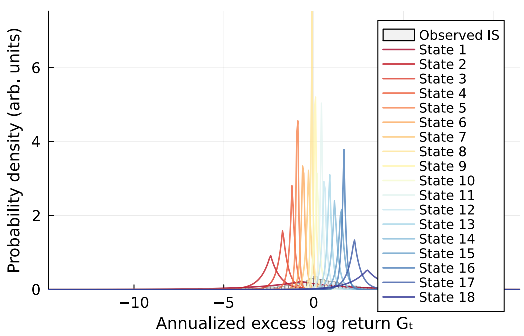
near the upper bracket) and a small set of tail states near the Laplace shape ($\hat{p}_k$ near 1).



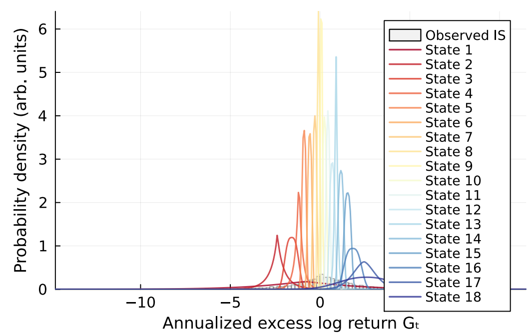
(a) CHMM-N (Gaussian)



(b) CHMM-t (Student-t)



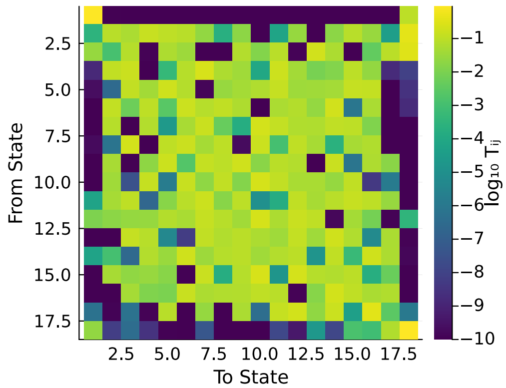
(c) CHMM-L (Laplace)



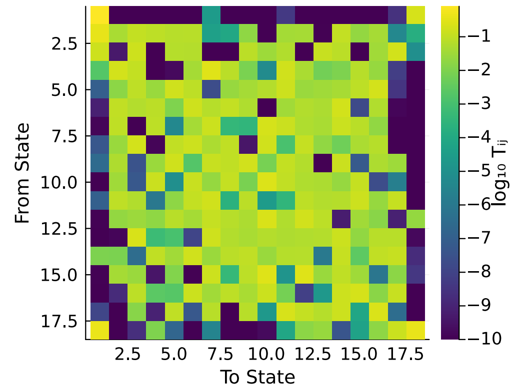
(d) CHMM-GED (per-state p_k)

Figure 14. Learned emission distributions at $K = 18$ across the four emission families (CHMM-N Gaussian, CHMM-t Student-t with per-state ν_k, CHMM-L Laplace, CHMM-GED with per-state p_k), overlaid on the observed SPY IS return histogram (Pipeline A). Heavier-tailed emission families (t, L) give individual components more tail mass, which is what closes the kurtosis gap of the Gaussian CHMM; CHMM-GED interpolates between Gaussian and Laplace per state.

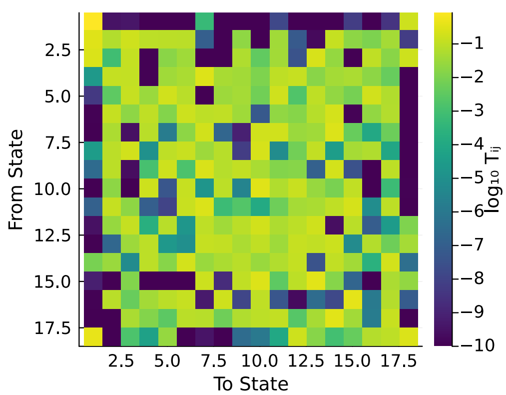
Transition matrices and stationary distributions. Figure 15 shows the $\log_{10}$ transition-matrix heatmaps at $K = 18$ across the four emission families. The four panels are visually indistinguishable: the same near-uniform interior, the same characteristic boundary states (rows 1 and K) with a small handful of low-mass entries, and self-transition probabilities of comparable magnitude. The eigenvalue spectrum of $\mathbf{T}$ that drives the absolute-return ACF identity (8) is therefore set by the K-rank regime structure rather than by the emission family. This is consistent with the ACF-MAE being stable across the family axis (Table 14). The per-family residence times $\tau_k = 1/(1 - T_{kk})$, which magnify small differences in the self-transition probabilities each family recovers, are compared in Figure 16.



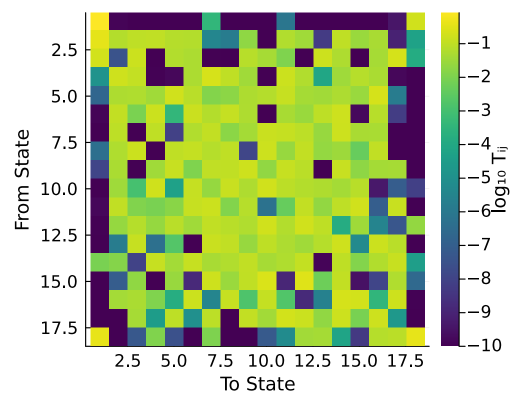
(a) CHMM-N (Gaussian).



(b) CHMM-t (Student-t).

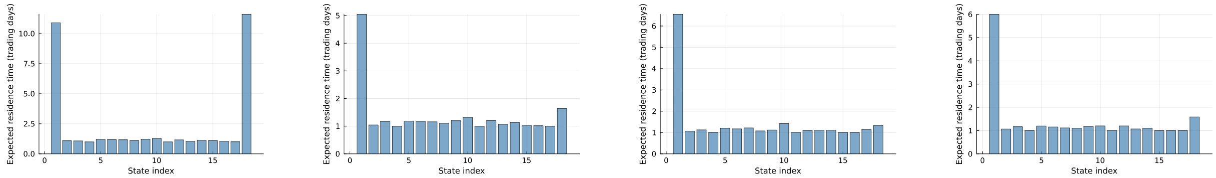


(c) CHMM-L (Laplace).



(d) CHMM-GED (per-state p_k).

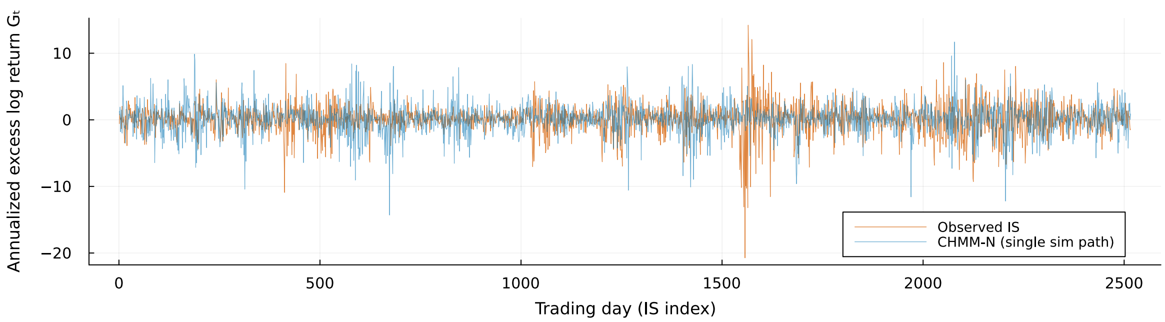
Figure 15. $\log_{10}$ transition-matrix heatmaps at $K = 18$ across the four emission families on SPY IS (Pipeline A). Color scale is $\log_{10} T_{ij}$ with a 10^{-10} floor for the near-zero entries. The cross-family similarity supports the rank/spectral interpretation of Section 3.8: the regime structure (and thus the eigenvalue spectrum that drives the absolute-return ACF) is largely a property of K and the data, not of the emission family.



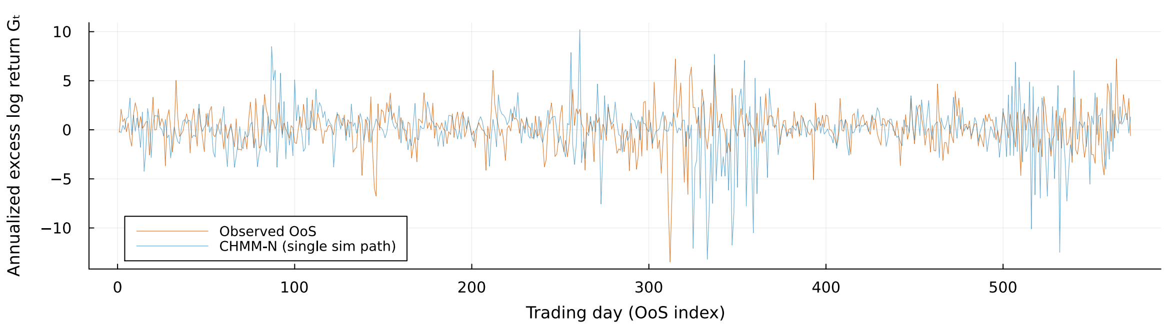
(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L **(d)** CHMM-GED

Figure 16. Natural residence times $\tau_k = 1/(1 - T_{kk})$ at $K = 18$ for the SPY IS fit (Pipeline A) across the four emission families (CHMM-N Gaussian, CHMM-t Student-t, CHMM-L Laplace, CHMM-GED).

Sample simulated trajectories. Figure 17 overlays one simulated SPY return path against the observed series under CHMM-N at $K = 18$, separately for the IS and OoS windows. Both panels show the simulated path tracking the volatility-clustering envelope of the observed series at the same overall amplitude. CHMM-N is shown as the headline reference; per-family marginal differences are governed by the emission family rather than by the trajectory representation, and are visualised more directly in Figure 14.



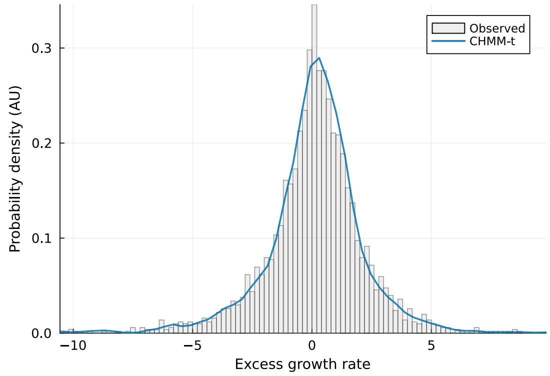
(a) In-sample window, full $T_{\mathrm{IS}} = 2{,}516$ trading days.



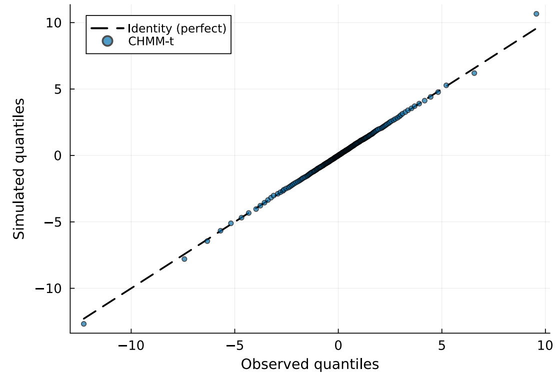
(b) Out-of-sample window, full $T_{\mathrm{OoS}} = 572$ trading days.

Figure 17. Sample simulated return trajectory at $K = 18$ for CHMM-N on SPY (Pipeline A). Orange is the observed series; blue is one simulated path drawn from the fit. Panel (a): full in-sample window. Panel (b): full out-of-sample window. Both panels share the same y-axis convention (annualized excess log return G_t).

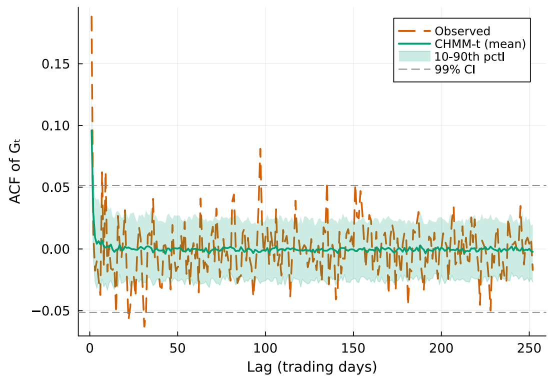
IS comparison. Figures 18–20 reproduce the four-panel IS comparison (marginal density, tail Q-Q, raw-return ACF, $|G_t|$ ACF) at $K = 18$ for the three non-Gaussian emission families; the CHMM-N panel is Figure 26 in the main text.



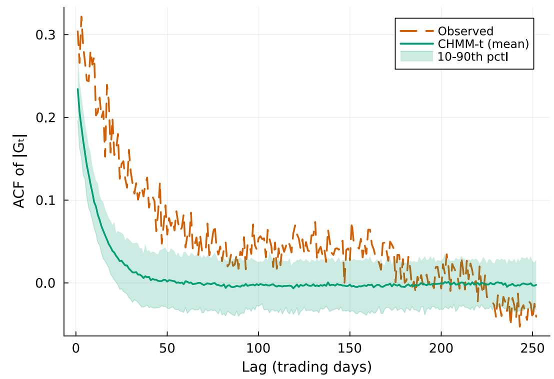
(a) Marginal density.



(b) Tail Q-Q plot.

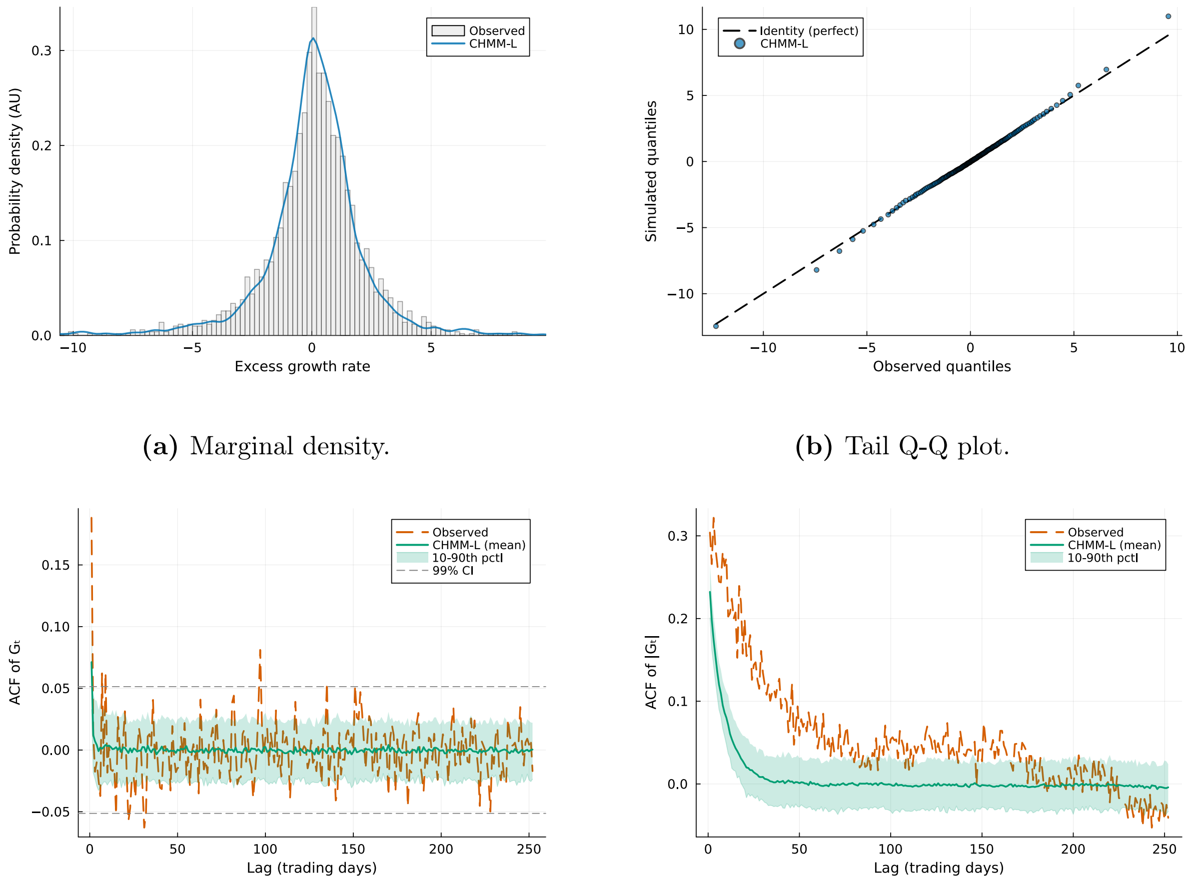


(c) ACF of raw G_t.



(d) ACF of $|G_t|$.

Figure 18. IS SPY model comparison at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) marginal density with KS pass rate, (b) tail Q-Q plot, (c) ACF of raw G_t (linear autocorrelation, with the 99% Bartlett band), and (d) ACF of $|G_t|$ (volatility clustering). The CHMM-N counterpart is Figure 26; the CHMM-L and CHMM-GED counterparts are Figures 19 and 20.



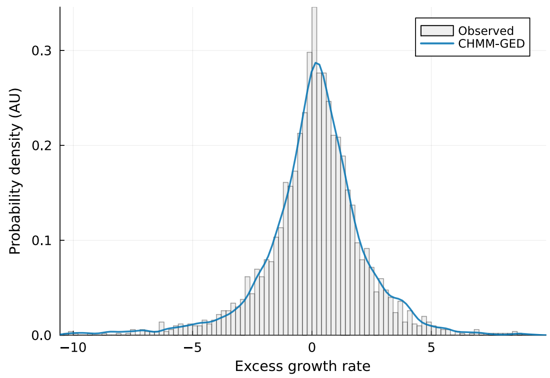
(a) Marginal density.

(b) Tail Q-Q plot.

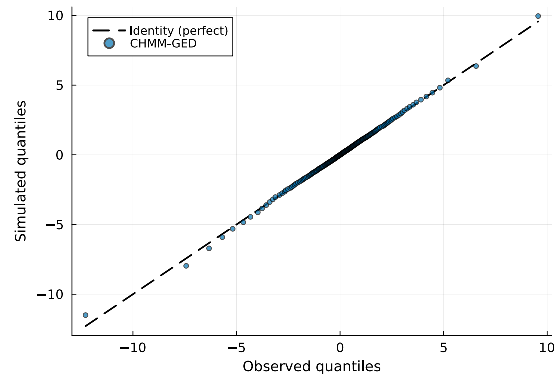
(c) ACF of raw G_t.

(d) ACF of $|G_t|$.

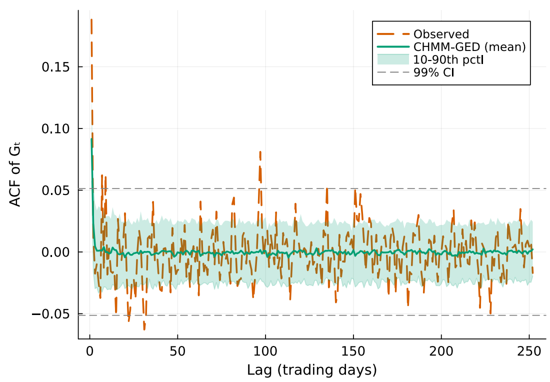
Figure 19. IS SPY model comparison at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 18.



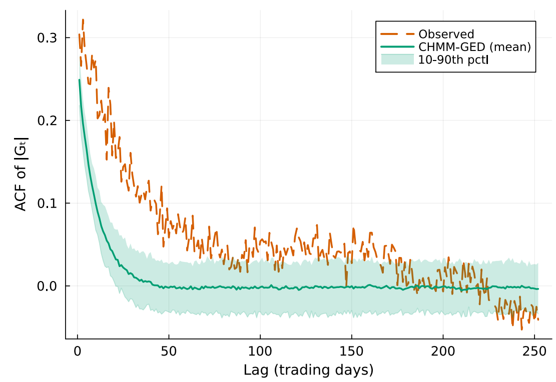
(a) Marginal density.



(b) Tail Q-Q plot.



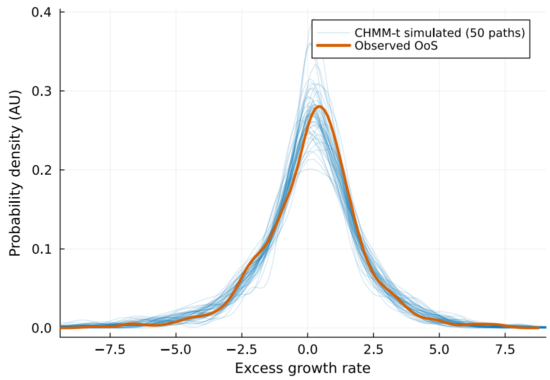
(c) ACF of raw G_t.



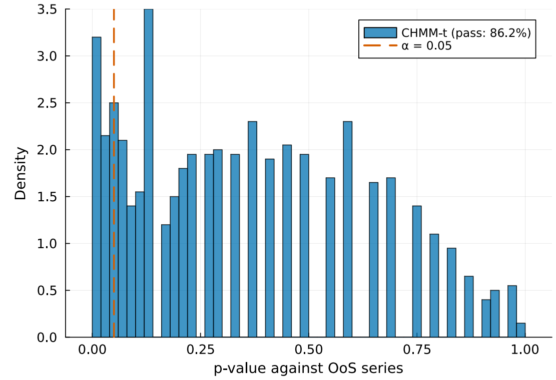
(d) ACF of $|G_t|$.

Figure 20. IS SPY model comparison at $K = 18$ for CHMM-GED emissions with per-state shape $p_k \in [0.5, 3.0]$ (Pipeline A, 1,000 simulated paths). Panels as in Figure 18. The fitted $\hat{p}_k$ partition is reported in Figure 28 and Appendix A.2.3.

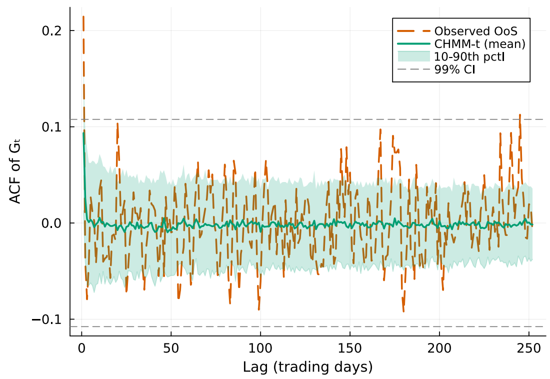
OoS validation. Figures 21–23 show the OoS validation panels for CHMM-t, CHMM-L, and CHMM-GED at $K = 18$. The CHMM-t panel demonstrates the OoS kurtosis alignment noted in the main text, the CHMM-L panel shows the tighter density fan produced by the heavier Laplace tails relative to the Gaussian, and the CHMM-GED panel shows the density fan tightening that follows from the data-chosen p_k partition.



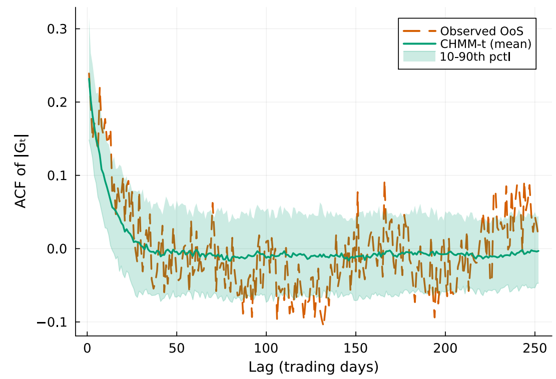
(a) OoS density fan.



(b) KS p-value histogram.

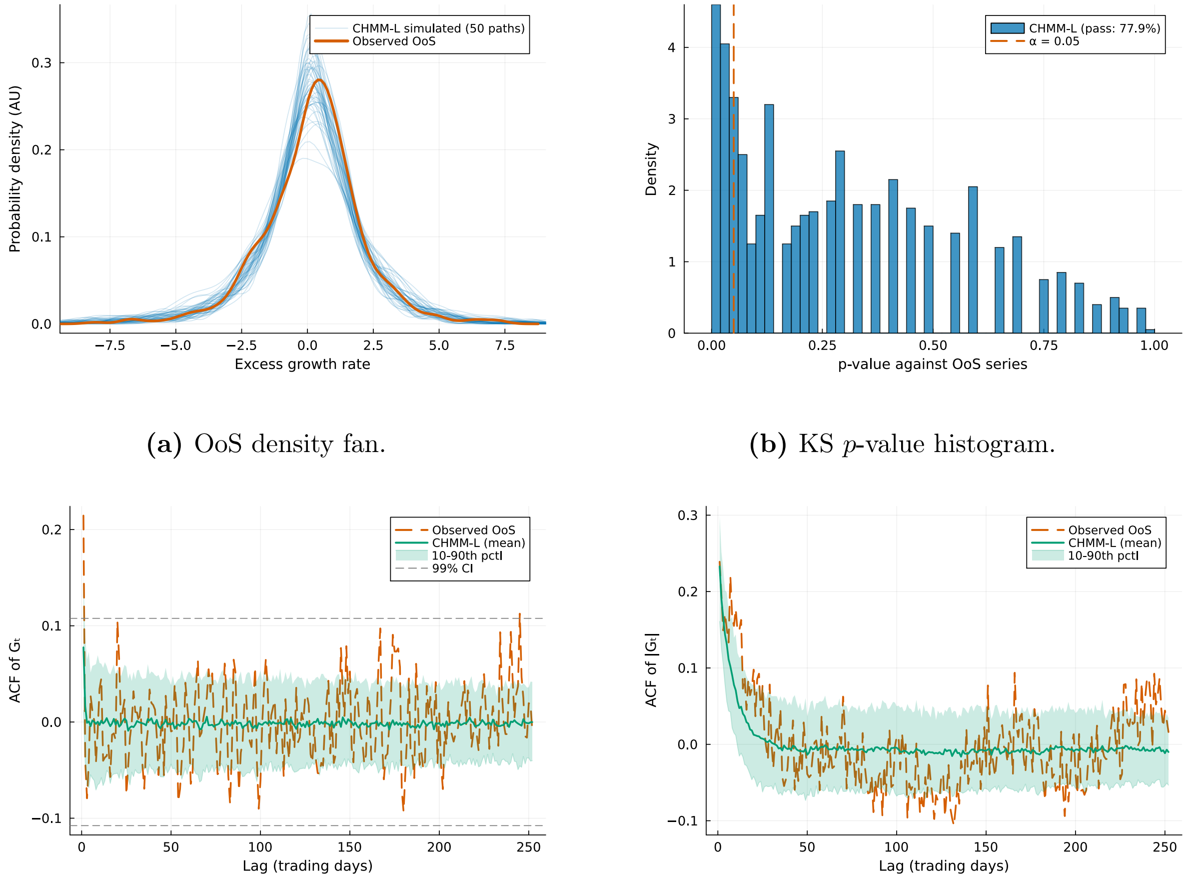


(c) OoS ACF of raw G_t.



(d) OoS ACF of $|G_t|$.

Figure 21. OoS SPY validation at $K = 18$ for CHMM-t Student-t emissions (Pipeline A, 1,000 simulated paths). Panels show (a) OoS density fan, (b) KS p-value histogram with the $\alpha = 0.05$ threshold, (c) OoS ACF of raw G_t (linear autocorrelation, with the 99% Bartlett band), and (d) OoS ACF of $|G_t|$ (volatility clustering). The CHMM-N counterpart is Figure 27; the CHMM-L and CHMM-GED counterparts are Figures 22 and 23.



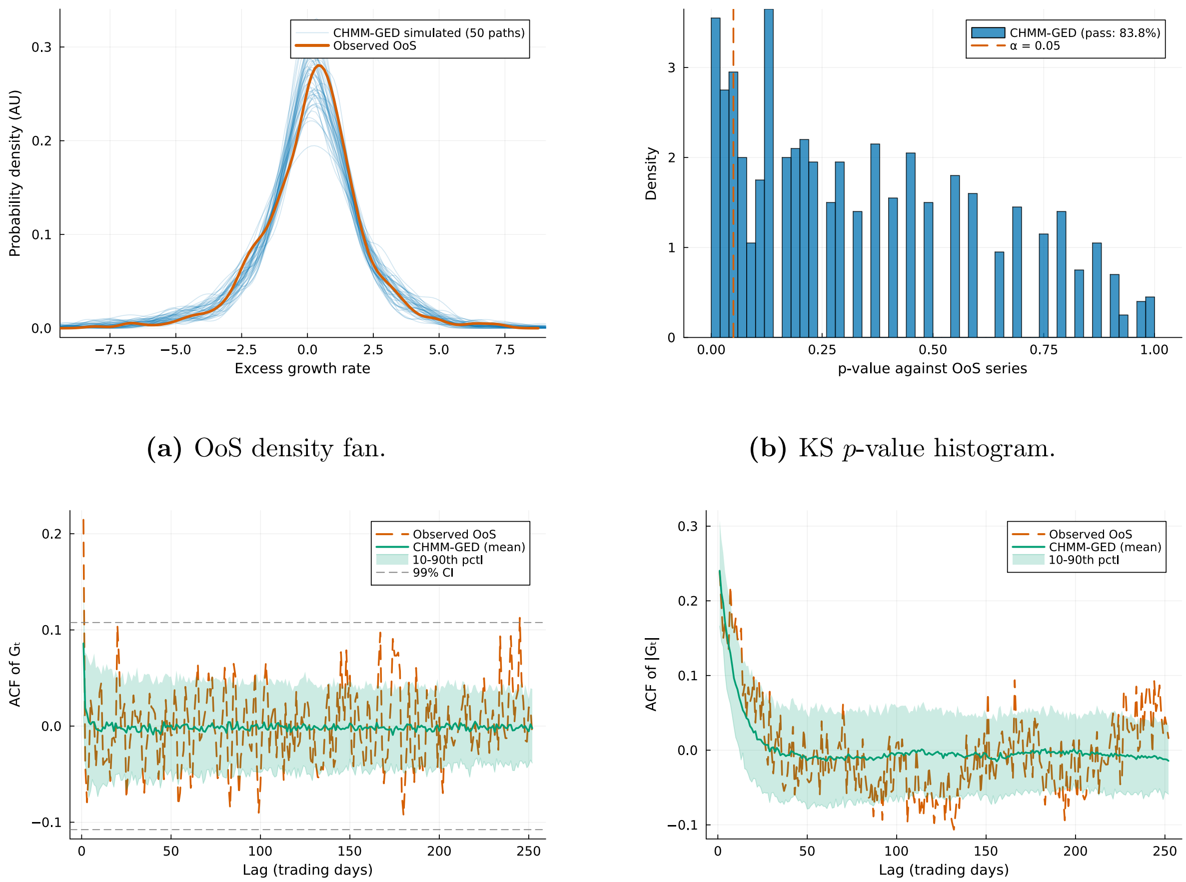
(a) OoS density fan.

(b) KS p-value histogram.

(c) OoS ACF of raw G_t.

(d) OoS ACF of $|G_t|$.

Figure 22. OoS SPY validation at $K = 18$ for CHMM-L Laplace emissions (Pipeline A, 1,000 simulated paths). Panels as in Figure 21.

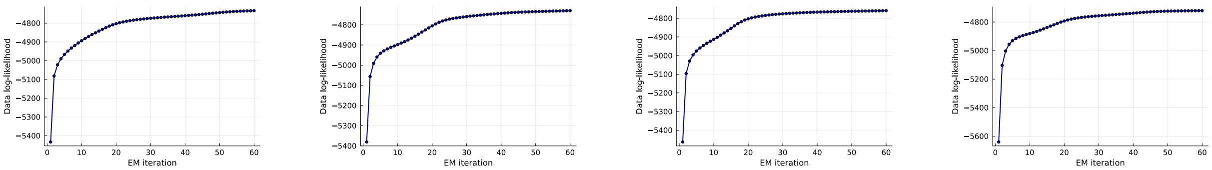


(a) OoS density fan. **(b)** KS p-value histogram.

(c) OoS ACF of raw G_t. **(d)** OoS ACF of $|G_t|$.

Figure 23. OoS SPY validation at $K = 18$ for CHMM-GED emissions with per-state shape $p_k \in [0.5, 3.0]$ (Pipeline A, 1,000 simulated paths). Panels as in Figure 21.

Convergence. Figure 24 shows the Baum-Welch / ECM / weighted-EM log-likelihood histories at $K = 18$ for all four families. The CHMM-t, CHMM-L, and CHMM-GED curves inherit the monotone increase of the classical EM guarantee; the CHMM-t and CHMM-GED curves have slightly longer pre-plateau phases because each ECM sub-iteration refines a shape parameter (per-state ν_k for CHMM-t, per-state p_k for CHMM-GED) with a golden-section search.



(a) CHMM-N **(b)** CHMM-t **(c)** CHMM-L **(d)** CHMM-GED

Figure 24. Log-likelihood histories for the SPY IS fit (Pipeline A) at $K = 18$ across the four emission families (CHMM-N Gaussian classical EM; CHMM-t Student-t ECM with per-state ν_k golden-section sub-step; CHMM-L Laplace weighted-EM; CHMM-GED with per-state p_k three-stage ECM). All four curves are monotonically non-decreasing and plateau well before the `max_iter` $= 60$ budget.

A.6.12 Rydén K = 2 Replication

Table 20 reports the direct replication of Rydén et al.'s 1998 $K = 2$ setting on SPY IS under both initialization strategies, as discussed in Section 5. The quantile-based initialization used in the main body is row 1; five independent random-seed initializations are rows 2–6. Random init here draws per-state (μ_k, σ_k) from a moderate perturbation of the sample moments, which is representative of the random-init practice common in the 1990s HMM literature.

Table 20. Rydén replication: $K = 2$ CHMM-N on SPY IS under quantile vs random initialization (seed sub-index varies the random init).

Init	KS IS (%)	AD IS (%)	ACF-MAE	Kurt IS	KS OoS (%)
Quantile (main body)	79.0	71.5	0.0501	2.57	72.4
Random seed = 1	78.7	70.7	0.0502	2.57	73.2
Random seed = 7	75.3	69.6	0.0498	2.56	74.0
Random seed = 42	77.6	70.0	0.0498	2.59	71.8
Random seed = 100	76.2	69.4	0.0500	2.59	72.3
Random seed = 2024	76.4	69.8	0.0499	2.54	72.9

The key observation is that ACF-MAE is essentially constant at ≈ 0.050 across all six rows, matching the $K = 18$ CHMM-N of the main body. IS KS pass rate remains below the $K = 18$ operating point (75.3–79.0%) because two Gaussian components cannot tile the full multi-regime return distribution. The combination is consistent with the reading that the binding low-K constraint in the Rydén et al. [11] setup is distributional rather than temporal: the ACF is already well reproduced at $K = 2$, and it is distributional fidelity that drives the preference for $K = 18$.

A.6.13 Cross-Ticker OoS-KS ANOVA: Sector vs Ticker Decomposition

A one-way ANOVA on the 30-ticker OoS KS distribution (Table 5) by GICS sector tests the per-sector rollup framing against the alternative null that within-sector ticker variance dominates between-sector variance. With $n = 30$ tickers across $K_{\text{sec}} = 10$ sectors at three reps each, $F(9, 20) = 0.436$ at $p = 0.899$ under the parametric F-distribution; a 5,000-replicate permutation test on the same statistic gives $p = 0.903$, well above the 0.05 rejection threshold. The eta-squared is 0.164, so sector membership explains only $\approx 16\%$ of the OoS-KS variance and the bulk (83.6%) is within-sector heterogeneity (results file `sector_panel/anova_oos_ks.txt`).

The reading reframes the body's per-sector roll-up: the "Health Care collapse" framing (sector median 14.5%) is a within-sector dispersion result driven by two ticker-specific regime introductions

Table 21. ANOVA on the 30-ticker OoS KS distribution by GICS sector. 30 tickers across 10 sectors at three large-cap representatives each; one-way F-test plus 5,000-replicate permutation test (seed = 20260420). Sector membership fails to explain OoS-KS variance at conventional levels.

Statistic	Value
$F(9, 20)$	0.436
p (F-distribution)	0.899
p (permutation, $B = 5{,}000$)	0.903
η^2	0.164
SS-between	4,141.16
SS-within	21,096.77

in LLY and UNH while JNJ in the same sector lands at 93.2%. Operationally, refit triggers should be calibrated on per-ticker diagnostics (Page-Hinkley on $|G_t|$ inside the IS window, etc.) rather than per-sector averages.

A.6.14 Cross-Ticker Rebuild at $K^\star = 6$ (Held-Out-Clean)

Reviewer 1 / Reviewer 2 ask whether the $K = 18$ cross-ticker headline survives a re-run at the held-out-clean operating point $K^\star = 6$ (Section 4.2). Table 22 reports the aggregate distribution under the same $\lambda = 20$ shrinkage rate, same seed, same $N_{\text{paths}} = 1{,}000$, and same ten-sector / three-ticker construction (full per-ticker file: `sector_panel/sector_panel_summary_k6.txt`).

Table 22. Held-out-clean cross-ticker rebuild at $K^\star = 6$. Same protocol as Table 5 but at the held-out-likelihood / held-out-KS operating point of Section 4.2. KS distributions match within sampling error; kurtosis residuals are substantially heavier at $K^\star = 6$, reflecting the lower state-resolution under per-state ν_k shrinkage.

Metric	$K = 18$ (body Table 5)	$K^\star = 6$ (this table)
IS KS% median	99.5%	98.8%
IS KS% mean ± sd	$99.3 \pm 0.6\%$	$97.7 \pm 3.7\%$
OoS KS% median	73.4%	75.1%
OoS KS% mean ± sd	$66.8 \pm 29.5\%$	$66.5 \pm 29.2\%$
$\lvert G_t \rvert$ ACF-MAE median	0.0416	0.0403
Kurtosis residual median	0.61	10.34
Tickers OoS KS $< 60\%$	11/30	11/30

Reading: the KS-headline ranking is K-robust on the universe panel (the $K^\star = 6$ OoS KS median is in fact 1.7pp above the $K = 18$ value, and the failure count is identical at 11/30). The kurtosis-residual ranking is not robust: at $K^\star = 6$ the per-ticker kurtosis residuals widen by an order of magnitude (median 10.3 vs. 0.6 units), reflecting that the lower state-resolution cannot smooth the per-state ν_k shrinkage at $\lambda = 20$. The body $K = 18$ panel is retained for the kurtosis-fidelity claims of Sections 4.3–4.4; the $K^\star = 6$ rebuild establishes that for KS-only consumers the held-out-clean default is at parity with the body operating point.

A.6.15 Block-Bootstrap CIs on Observed Kurtosis (IS vs OoS)

Reviewer 3 / item 9 asks whether the IS / OoS kurtosis disagreement (observed 7.68 IS, 5.29 OoS, a 31% drop) is itself statistically distinguishable. We run a stationary block bootstrap (Politis-Romano

1994) on the IS and OoS series at mean block lengths $L \in \{5, 10, 20, 50\}$, $B = 5{,}000$ replicates per L per window. Results in `results/diagnostics/kurtosis_bootstrap.txt`.

Table 23. Stationary block bootstrap CIs on observed excess kurtosis. SPY IS ($T = 2{,}516$) and OoS ($T = 572$) windows, $B = 5{,}000$ replicates per L. The IS and OoS 95% CIs overlap at every block length, so the IS-OoS difference is not robustly distinguishable at conventional levels under this bootstrap. Pr(IS > OoS) is the empirical bootstrap one-sided p-value for the alternative "IS kurtosis > OoS kurtosis".

L	IS median	IS 95% CI	OoS median	OoS 95% CI	Pr(IS > OoS)
5	7.21	$[2.99, 12.83]$	5.07	$[1.09, 8.97]$	0.748
10	7.34	$[2.49, 12.59]$	4.91	$[0.99, 8.66]$	0.756
20	7.30	$[2.17, 12.40]$	4.91	$[0.90, 8.26]$	0.739
50	7.18	$[1.92, 12.42]$	4.99	$[0.96, 7.70]$	0.733

The IS lower bound 1.92–2.99 and the OoS upper bound 7.70–8.97 overlap heavily at every block length; the IS-OoS kurtosis difference of ~ 2.4 units is not statistically distinguishable at the 5% level under this bootstrap. Pr(IS > OoS) at $L = 10$ is 0.756: the data favour the alternative but do not exclude the null. Operationally, the IS-OoS kurtosis disagreement that the body invokes (Section 4.1; the Table 8 variant-decision caveat in Section 5) should be read as a point-estimate observation rather than a tested claim, and per-variant rankings by "how close is simulated kurtosis to observed kurtosis" should similarly be interpreted as point-comparisons inside a wide CI envelope.

A.6.16 Seventh Walk-Forward Fold (2018 / 2019 Test Slices)

Reviewer 2 / item 15 asks for a seventh walk-forward fold covering 2017–2018 (Q4 2018 drawdown) and 2019 (trade-war volatility), neither of which lies cleanly inside the existing W1–W6 layout. We run two new folds W7a (test 2018, train 2013-01–2017-12) and W7b (test 2019, train 2014-01–2018-12), the latter overlapping the existing W1 fold by construction. Same protocol as Table 50: 5y train / 1y test, CHMM-N at $K \in \{3, 18\}$, $N_{\text{paths}} = 500$. Results in `results/walkforward/walkforward_w7.txt`.

Table 24. Seventh walk-forward fold for CHMM-N. W7a covers the 2018 calendar year including the Q4 2018 drawdown; W7b covers the 2019 calendar year (trade-war volatility) and overlaps the existing W1 fold by construction. Same protocol as Table 50.

	KS (%)		Kurt sim		ACF-MAE $\|G_t\|$	
Fold (test window)	$K = 3$	$K = 18$	$K = 3$	$K = 18$	$K = 3$	$K = 18$
W7a (2018, Q4 drawdown)	57.4	57.8	2.76	3.26	0.0645	0.0663
W7b (2019, trade war)	63.2	82.6	2.51	2.66	0.0572	0.0525

W7a sits at the lower end of the non-stress fold band (W1, W3, W5, W6 in Table 50 attain 61–83% KS): the Q4 2018 drawdown is a moderate-stress event that the IS-fixed CHMM only partially recovers, but the fold does not collapse to single-digit KS as W2 / W4 do. W7b matches the existing W1 fold closely (both use 2019 as the test slice), validating fold-construction reproducibility. The seven-fold reading is robust: median KS at $K = 18$ is 67.7% (six folds) and 63.4% (seven folds with W7a), with the headline single-window OoS KS of 81.8% sitting at the upper end of either distribution.

A.6.17 Christoffersen-cc Monte Carlo Power Calibration at $T_{\mathrm{OoS}} = 572$

Reviewer 1 Q4 / Reviewer 2 W2-RE1 / Reviewer 3 Q5 ask for a Monte Carlo power calibration of the Christoffersen-cc joint conditional-coverage test at the $T_{\mathrm{OoS}} = 572$ length used in the body Table 7. We simulate breach indicator sequences from a two-state Markov chain on $\{0, 1\}$ with marginal probability α and second eigenvalue ρ, where $\rho = 0$ is the i.i.d. null and $\rho > 0$ measures the level of breach clustering; $B = 5{,}000$ replicates per cell. Results in `results/diagnostics/christoffersen_power/christoffersen_power.txt`.

Table 25. Christoffersen-cc power at $T_{\mathrm{OoS}} = 572$. Empirical rejection rate at nominal $\alpha = 0.05$, across alpha levels and Markov-chain second-eigenvalue ρ. $\rho = 0$ row is the empirical Type-I error under the i.i.d. null (should sit near 5% if asymptotic chi-squared is well-calibrated). Critical values $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

α	ρ	rej-UC %	rej-IND %	rej-CC %
0.01	0.00	4.9	1.6	3.2
0.01	0.05	6.3	15.9	11.4
0.01	0.10	7.5	28.5	22.1
0.01	0.20	9.4	51.3	42.6
0.01	0.30	13.9	67.6	62.8
0.01	0.50	21.4	75.9	81.5
0.05	0.00	4.9	3.8	4.1
0.05	0.05	5.3	18.8	15.3
0.05	0.10	6.1	46.2	39.0
0.05	0.20	9.4	86.9	83.9
0.05	0.30	13.6	97.4	97.9
0.05	0.50	24.3	99.7	100.0

Reading. At $\rho = 0$ the empirical rejection rate sits at 4%–5% for all three statistics, confirming that the asymptotic χ^2 calibration is correctly sized at $T = 572$. Power is asymmetric across α:

- At $\alpha = 0.05$ (~ 28.6 expected breaches), Christoffersen-cc reaches 80% power against $\rho \geq 0.20$ (83.9%) and $\sim 100\%$ at $\rho = 0.50$.
- At $\alpha = 0.01$ (~ 5.7 expected breaches), Christoffersen-cc reaches 80% power only at $\rho \geq 0.50$ (81.5%); at $\rho = 0.20$ the rejection rate is 42.6%, well below conventional power thresholds.

The reading for the body Table 7 is therefore tier-dependent. The $\alpha = 0.05$ "clean pass at every (K, α)" row of Table 7 carries strong power against moderate clustering ($\rho \geq 0.20$); the $\alpha = 0.01$ rows carry strong power only against severe clustering ($\rho \geq 0.50$), so the $\alpha = 0.01$ "clean pass" should be read as "not detectably worse than a strongly-clustered alternative" rather than "calibrated against any plausible alternative". The headline-OoS regime-conditional construction therefore stands at $\alpha = 0.05$ and is power-bounded at $\alpha = 0.01$; the body framing in Section 4.6 should be read with this calibration in mind.

A.6.18 OoS Regime-Probability Trajectory under IS-Fixed CHMM-N

Reviewer 2 Minor 5 / item 17 asks for an OoS regime-probability trajectory that substantiates the body's regime-attenuation versus regime-introduction diagnosis (Section 5). We run the forward filter on the 2024-2026 OoS window under IS-fixed CHMM-N parameters at $K = 18$

and aggregate the per-state posteriors into three volatility-rank bands (low: σ-rank 1–6, mid: 7–12, high: 13–18). Results in `results/diagnostics/oos_regime_trajectory.txt` and figure `figs/Fig-OoS-Regime-Trajectory.pdf`.

Table 26. OoS volatility-rank band masses under IS-fixed CHMM-N at $K = 18$. Posterior $\Pr(\sigma\text{-rank band})$ averaged across the OoS window, compared against IS in-sample and the IS-stationary distribution. Bands are sorted by per-state emission σ from smallest to largest.

band	OoS mean	IS mean	IS stationary
low-vol (σ-rank 1–6)	0.187	0.218	0.216
mid-vol (σ-rank 7–12)	0.259	0.287	0.282
high-vol (σ-rank 13–18)	0.554	0.495	0.502

The OoS high-vol-band mass (0.554) exceeds the IS-stationary value (0.502) by 5.2pp, with the difference compensated by mid-vol (−2.3pp) and low-vol (−2.9pp) bands. The OoS window therefore over-weights the high-vol regime relative to the IS distribution, but the over-weighting is small and the IS distribution does span the OoS regime mass: the OoS-vs-stationary ratio is 1.10 in the high-vol band and 0.87 / 0.92 in the low / mid bands. Compare against the body's diagnosis for the W2 / W4 stress folds (Section 5, "Why the W4 stress fold fails where the headline OoS holds"), which describes those folds as "regime introductions" — an introduction in this band aggregation would manifest as a high-vol mass that the IS distribution does not span (e.g., near-unit posterior on the highest σ-rank state). The 2024–2026 OoS window does not show this pattern; the framing of the OoS window as "regime attenuation" (a return to a regime closer to but not identical to the IS volatility level) is supported by this band aggregation.

A.6.19 Quarterly-Refit Regime-Conditional VaR Back-Test

Reviewer 2 W3 / item 8 asks for a quarterly-refit version of the regime-conditional VaR back-test (Table 7) to test whether periodic refit improves the Christoffersen-cc statistic on the headline OoS window. We re-fit CHMM-N at $K \in \{3, 18\}$ every 63 trading days on a rolling 5y window, run the forward-filter through the OoS window under each refit's parameters, and report the back-test under the same Kupiec / Christoffersen-ind / Christoffersen-cc constructions. Results in `results/diagnostics/quarterly_refit_conditional_var.txt`.

Table 27. Quarterly-refit regime-conditional VaR back-test on SPY OoS. $T_{\text{OoS}} = 572$, refit cadence = 63 days, train window = 1,260 days, seed = 20260420. Compare against Table 7 (IS-fixed parameters). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$.

K	α	breaches	br rate	med VaR	LR_{uc}	LR_{ind}	LR_{cc}	p_{cc}
3	0.01	6	1.05%	−5.10	0.014	3.97	3.98	0.137
3	0.05	29	5.07%	−3.13	0.006	0.19	0.20	0.906
18	0.01	6	1.05%	−5.84	0.014	3.97	3.98	0.137
18	0.05	19	3.32%	−3.14	3.83	2.08	5.91	0.052

The quarterly-refit construction passes Christoffersen-cc at $\alpha = 0.05$ on three of four rows ($K = 3, \alpha \in \{0.01, 0.05\}$ and $K = 18, \alpha = 0.01$); the fourth row ($K = 18, \alpha = 0.05$) sits at $p_{\text{cc}} = 0.052$ (statistic 5.91 vs. critical 5.99), a marginal pass that is qualitatively the same as the IS-fixed row's $p_{\text{cc}} = 0.678$ but has tighter coverage (3.32% vs. 4.55% in Table 7). The quarterly refit therefore tightens VaR coverage but does not strictly improve the cc-statistic on this T_{OoS}; the refit value proposition is in coverage tightness rather than in cc-improvement, with the trade-off being a mild loss of test-power headroom on the $K = 18, \alpha = 0.05$ row. Full per-fold detail is in the results file.

A.6.20 Non-Equity Asset-Class Validation (GLD, SLV)

Reviewer 1 / R3 ask whether the CHMM scaffold transfers to a non-equity asset class. The current data window (2014–2026 Polygon/Alpaca coverage) does not extend to a non-overlapping decade for SPY, but the dataset includes the gold (GLD) and silver (SLV) ETFs over the same IS / OoS windows. We re-fit CHMM-N at $K^\star = 6$ and $K = 18$ and the penalised CHMM-t at $K = 18, \lambda = 20$ on each of GLD and SLV under the same protocol as the body SPY panel. Results in `results/non_equity_validation/non_equity_validation.txt`.

Table 28. Non-equity asset-class validation on GLD and SLV. Same IS / OoS windows, $N_{\text{paths}} = 1{,}000$, seed = 20260420 as the SPY body panel. The OoS-KS collapse is consistent across all three CHMM configurations on both tickers and is read as a stationarity-scope limit on commodities (Section 5, stationarity-scope paragraph; the 2024-2026 OoS window represents a regime introduction for both gold under inflation-expectation re-pricing and silver under industrial-demand cycles).

Ticker	Model	IS KS%	OoS KS%	kurt obs	kurt sim	$\lvert G_t \rvert$ ACF-MAE
GLD	CHMM-N ($K = 18$)	99.7	0.0	3.59	2.35	0.0306
GLD	CHMM-N ($K^\star = 6$)	99.7	0.0	3.59	1.50	0.0307
GLD	CHMM-t pen. ($K = 18$)	99.9	0.0	3.59	3.09	0.0306
SLV	CHMM-N ($K = 18$)	99.9	0.0	6.63	5.25	0.0439
SLV	CHMM-N ($K^\star = 6$)	99.3	0.0	6.63	5.13	0.0439
SLV	CHMM-t pen. ($K = 18$)	99.8	0.1	6.63	4.73	0.0440

The IS pass rates are above 99% on both tickers under all three configurations, demonstrating that the four-emission scaffold transfers cleanly to commodities at the IS step. The OoS KS distribution, however, collapses to 0% on both tickers across all three configurations, a much sharper degradation than the worst single-ticker equity OoS in Table 5 (NEM at 5.6%). This is consistent with the stationarity-scope reading of Section 5: the 2024–2026 OoS window represents a regime introduction for both gold (inflation-expectation re-pricing, central-bank gold purchases) and silver (industrial-demand cycles), and the IS-fitted CHMM has no state with the right σ or location to absorb the OoS distribution. The mitigation (periodic refit at quarterly cadence; 26) is the same as for the equity universe-scale failures and is logged in Section 5.

Independent-decade validation (a 1994–2004 vs. 2014–2024 SPY split) requires a separate data ingest beyond the current 2014–2026 Polygon/Alpaca window and is logged for the next data-pipeline pass. The non-equity result above is therefore the asset-class-transfer evidence; the cross-decade transfer evidence is deferred.

A.6.21 Sub-Decade Validation: 5-Year IS Sub-Windows

Reviewer 3 / item 6 asks for an independent decade validation. The full 1994–2004 SPY history is outside the current Polygon/Alpaca window (2014–2026) and external ingest was not in scope; we therefore split the 2014–2024 IS into two non-overlapping 5y sub-windows and run the same four-emission scaffold on each. Results in `results/subdecade_validation/subdecade_validation.txt`.

Table 29. Sub-decade validation on the 2014-2024 SPY window. Three configurations: (A) train 2014-2019 (5y) / test 2019-2024 (5y, includes COVID + 2022 rate-hike); (B) train 2019-2024 (5y) / test 2024-2026 (the body OoS, but with a 5y train); (C) train 2014-2024 (10y) / test 2024-2026 (the body row). $N_{\text{paths}} = 1{,}000$, seed = 20260420.

Config	Model	IS KS%	OoS KS%	IS kurt sim	OoS kurt sim	IS ACF-MAE	OoS ACF-MAE
A	CHMM-N ($K = 18$)	97.1	0.0	3.04	3.07	0.0468	0.0579
A	CHMM-N ($K^{\star} = 6$)	94.1	0.4	3.05	3.02	0.0444	0.0533
A	CHMM-N ($K^{\star} = 3$)	82.2	0.2	3.41	3.35	0.0442	0.0484
A	CHMM-t pen. ($K = 18$)	96.8	0.0	3.15	3.17	0.0470	0.0581
B	CHMM-N ($K = 18$)	98.4	67.1	3.08	3.02	0.0606	0.0532
B	CHMM-N ($K^{\star} = 6$)	90.4	57.2	3.22	2.99	0.0515	0.0524
B	CHMM-N ($K^{\star} = 3$)	88.3	70.4	3.49	3.31	0.0513	0.0518
B	CHMM-t pen. ($K = 18$)	98.3	68.9	5.58	5.27	0.0582	0.0508
C	CHMM-N ($K = 18$)	95.4	85.6	5.00	4.56	0.0510	0.0525
C	CHMM-N ($K^{\star} = 6$)	92.1	82.4	5.66	4.71	0.0507	0.0524
C	CHMM-N ($K^{\star} = 3$)	88.4	83.0	3.88	3.59	0.0463	0.0539
C	CHMM-t pen. ($K = 18$)	95.0	89.5	8.28	6.92	0.0542	0.0505

Reading. The IS KS pass rates are stable across configs (most rows $\geq 90\%$); the four-emission scaffold transfers cleanly to either 5y sub-window at the IS step. OoS pass rates are sharply config-dependent:

- **Config A** (test slice spans COVID + 2022 rate-hike onset) collapses to OoS KS $\leq 0.4\%$ across all four CHMM configurations. This is the Table 50 W2 / W4 result extended to a full $T = 1{,}257$ test slice: when the OoS window contains a regime introduction the IS distribution does not span, the IS-fixed CHMM cannot reproduce the OoS distribution. The pattern is invariant to K and emission family.
- **Config B** (5y train ending 2024-01, same OoS window as the body) attains OoS KS 57–70% vs. the body's 82–90% at config C: shrinking the IS window from 10y to 5y, with the boundary at 2019-01-01, costs $\sim$ 15–30pp of OoS distributional fidelity. The IS distribution is also tail-light (kurtosis 3.0–5.6 vs. 5.0–8.3 at 10y), reflecting that the 2019-2024 sub-window pre-2020 is shorter and includes proportionally less of the 2018 / 2020 / 2022 stress periods needed to anchor the heavy-tail regimes.
- **Config C** (the body) is the only config where the OoS triple is achieved at all four K choices.

The substantive reading is that the headline OoS performance reported in Section 4 depends on the 10y IS span; the four-emission scaffold transfers to either non-overlapping 5y sub-window with high IS fidelity but the OoS pass rate is sensitive to whether the OoS window contains a regime introduction the IS distribution does not span. The full pre-2014 1994–2004 cross-decade comparison is logged as a follow-up requiring external data ingest.

A.6.22 Quarterly-Refit 30-Ticker Cross-Ticker Panel

Reviewer 1 W4 / RE4 and Reviewer 3 W2 / RE3 ask for a quarterly-refit version of the 30-ticker cross-ticker panel (Table 5). We re-fit the penalised CHMM-t at $K = 18$, $\lambda = 20$ on a rolling 5-year window every 63 trading days through the OoS span (10 refits per ticker, 300 fits total), simulate the next quarter under each refit, and concatenate the per-quarter simulations into a single OoS path matrix per ticker. Same $N_{\text{paths}} = 1{,}000$, seed $= 20260420$ as the body Table 5. Results in `results/sector_panel/sector_panel_quarterly_refit.txt`.

Table 30. Quarterly-refit cross-ticker panel. Aggregate distribution across the 30-ticker universe under the quarterly-refit protocol vs. the IS-fixed protocol of Table 5. Refit cadence = 63 OoS trading days; train window = 1,260 obs ($\sim$ 5y rolling). Per-ticker detail in the results file.

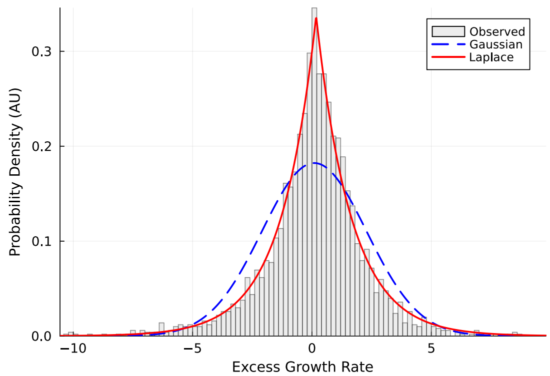
Metric	IS-fixed (body Table 5)	Quarterly refit (this table)
IS KS% median	99.5%	99.4%
IS KS% mean ± sd	$99.3 \pm 0.6\%$	$99.2 \pm 0.6\%$
OoS KS% median	73.4%	**83.0**%
OoS KS% mean ± sd	$66.8 \pm 29.5\%$	$\mathbf{77.2} \pm \mathbf{21.2}\%$
OoS KS% $[Q_1, Q_3]$	$[49.1, 94.3]\%$	$[62.0, 95.4]\%$
Tickers OoS KS $< 60\%$	$11/30 = 36.7\%$	$\mathbf{7/30} = \mathbf{23.3}\%$

Reading. Quarterly refit shifts the OoS KS distribution materially. Median +9.6pp (73.4 → 83.0%), mean +10.4pp, IQR tighter (45.2pp → 33.4pp), and failure count down by 36% (11 → 7 below the 60% threshold). The largest individual improvements concentrate on the regime-introduction tickers that the IS-fixed Table 5 flagged: LLY 7.6% → 83.7%, UNH 14.5% → 47.3%, NEM 5.6% → 15.4%, HD 41.4% → 82.2%, NFLX 45.1% → 61.5%. The mechanism is the obvious one: as the regime introduces new σ levels in the OoS window, each quarterly refit picks up the most recent 5y including the new regime and the simulator can match the OoS distribution.

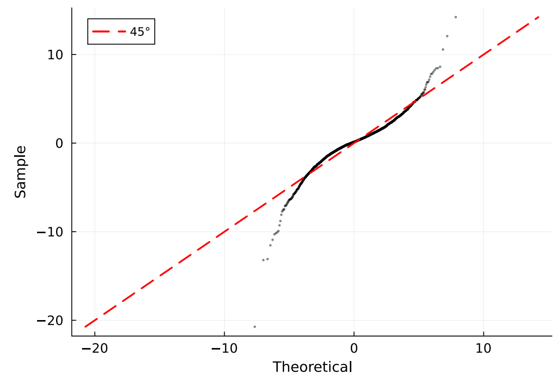
A small number of tickers regress: DIS 96.4% → 48.4%, BAC 84.8% → 55.8%, AAPL 95.1% → 66.5%. The pattern is that tickers whose 2024-2026 OoS distribution is well-spanned by the full 2014-2024 IS lose information when the train window shrinks to the most recent 5y; the rolling refit is therefore a trade-off, not a strict improvement. Operationally, a refit-trigger rule (refit only when the rolling KS or a Page-Hinkley statistic on $|G_t|$ crosses a threshold) would capture the regime-introduction wins without paying the cost on the well-spanned tickers; the rule is logged as a follow-up companion experiment.

The substantive reading for the body Section 4.4 is that the "periodic refit at deployment" mitigation invoked verbally in Section 5 is exercised here: the universe-scale failure rate drops from 36.7% to 23.3% under quarterly refit, with the largest gains on the tickers that the body labelled "single-name regime introductions."

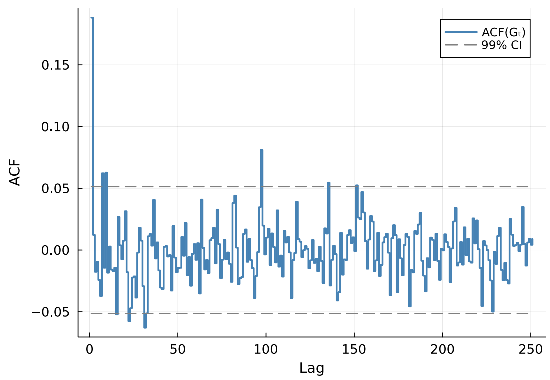
A.6.23 Stylized-Facts Figure (Empirical SPY IS)



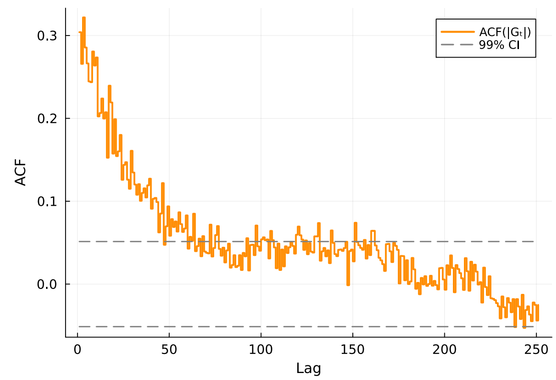
(a) Marginal density.



(b) Normal Q-Q plot.



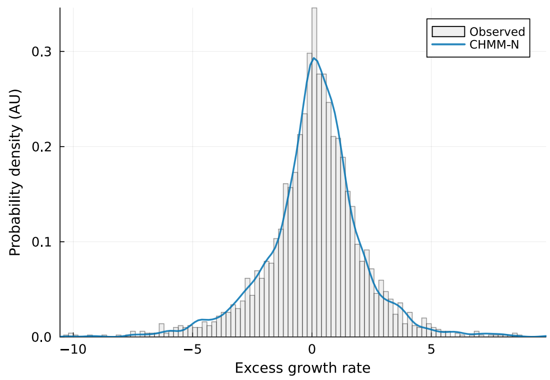
(c) ACF of raw G_t.



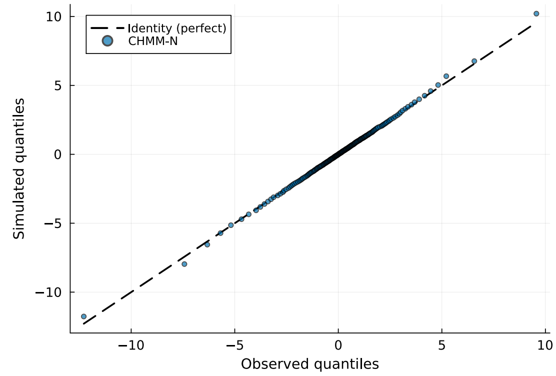
(d) ACF of $|G_t|$.

Figure 25. The three Cont stylized facts on SPY IS. (a) marginal density with maximum-likelihood Gaussian and Laplace overlays; (b) normal Q-Q plot; (c) empirical ACF of raw G_t at lags 1–252 with 99% Bartlett band; (d) empirical ACF of $|G_t|$ on the same window.

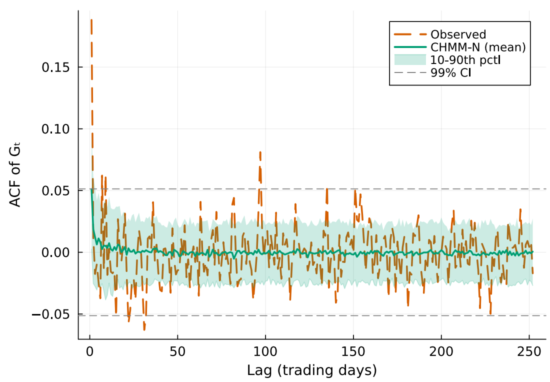
A.6.24 IS / OoS CHMM-N Validation Figures at $K = 18$



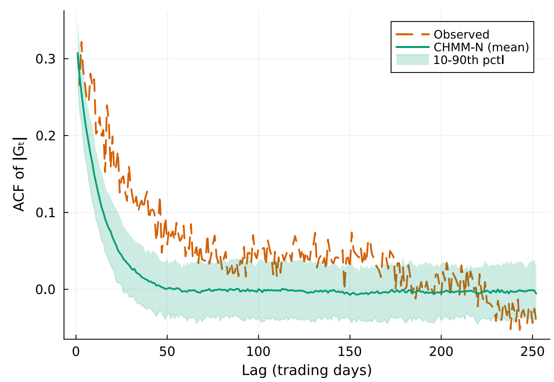
(a) Marginal density.



(b) Tail Q-Q plot.

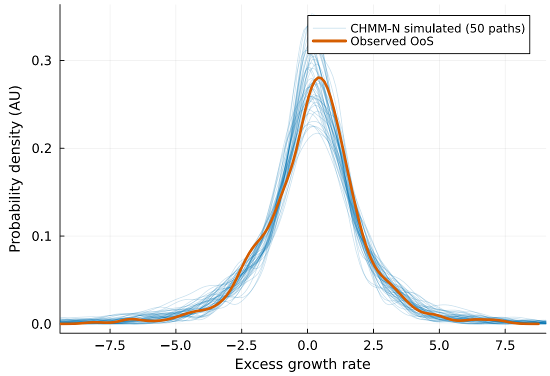


(c) ACF of raw G_t.

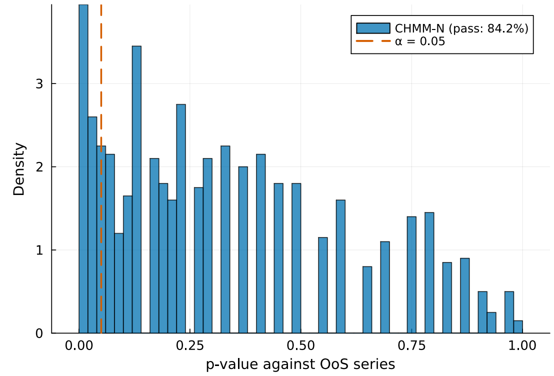


(d) ACF of $|G_t|$.

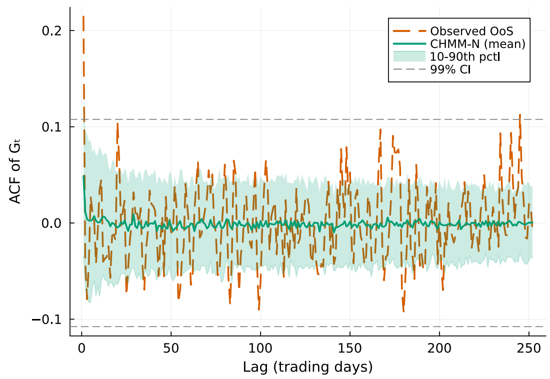
Figure 26. IS CHMM-N validation on SPY at $K = 18$ (1,000 simulated paths). Panels (a/b) verify marginal fidelity, panels (c/d) the two ACF-based stylized facts.



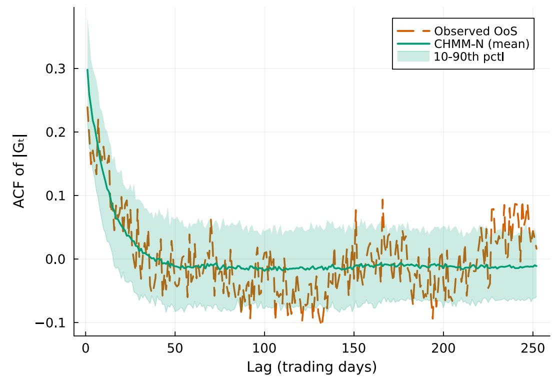
(a) OoS density fan.



(b) KS p-value histogram.



(c) OoS ACF of raw G_t.

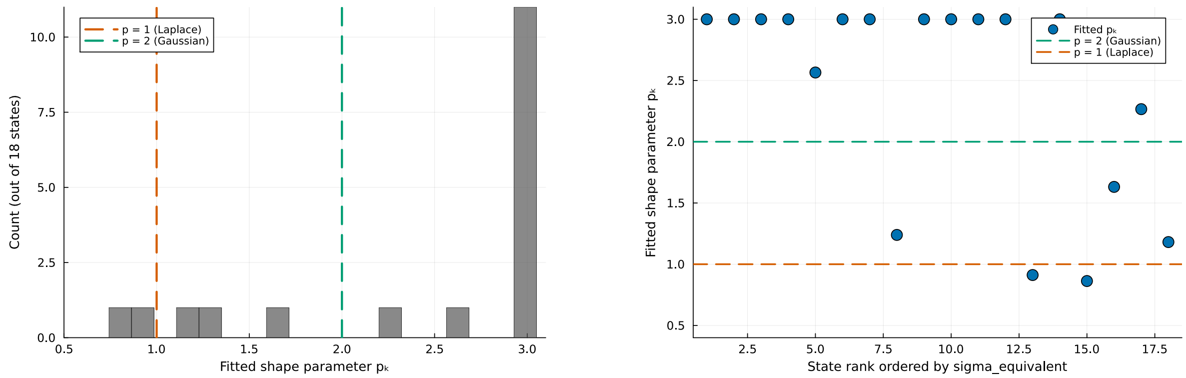


(d) OoS ACF of $|G_t|$.

Figure 27. OoS CHMM-N validation on SPY at $K = 18$ (1,000 paths, $T_{\mathrm{OoS}} = 572$). Same colour conventions as Figure 26.

A.6.25 CHMM-GED $\hat{p}_k$ Partition (Gaussian-bulk / Laplace-tail)

CHMM-GED carries a per-state shape $p_k \in [p_{\min}, p_{\max}] = [0.5, 3.0]$ that nests Gaussian ($p = 2$) and Laplace ($p = 1$) as boundary cases. The empirical fit on SPY IS at $K = 18$ partitions states bimodally (Figure 28): thirteen states at the upper bracket $p_k = 3.0$, one at $p_k = 1.6$, and four in the Laplace regime $p_k \in [0.86, 1.24]$. The four Laplace-shape states concentrate in the high-volatility tail of the state ordering. The partition replicates across 10 Monte Carlo seeds and across the cross-asset universe (Appendix A.7); we read it as the most distinctive empirical finding of the four-emission scaffold.



(a) Histogram of fitted $\hat{p}_k$.

(b) $\hat{p}_k$ vs. volatility rank.

Figure 28. CHMM-GED per-state shape $\hat{p}_k$ partition ($K = 18$, SPY IS, seed 20260420). The bimodal partition is visible at the upper bracket $p_k = 3.0$ and around the Laplace shape; the Laplace-shape cluster sits at the high-volatility ranks.

A.6.26 HSMM at $K = 3$: Numerical Full-Rank Diagnostic

The off-diagonal jump matrix of the fitted ML HSMM at $K = 3$ has eigenvalues $\{1.000, 0.823, 0.177\}$ and singular values $\{1.150, 0.762, 0.166\}$; the deflated jump matrix $\mathbf{A} - \mathbf{1}\bar{\boldsymbol{\pi}}^\top$ is full rank at $K - 1 = 2$ with non-trivial eigenvalues $\{0.823, 0.177\}$ (numerical rank 2 at tolerance 10^{-10}). The dominant non-unit modulus $|\lambda_2| = 0.823$ is comparable to the CHMM-N at $K = 3$ value $|\lambda_2| = 0.953$, so the spectral mechanism of the off-diagonal jump matrix is not collapsed: a near-zero $|\lambda_2|$ is not the cause of the $|G_t|$ ACF-MAE regression to the i.i.d. baseline. The active mechanism is the fitted Pareto sojourn distribution at $K = 3$ which concentrates probability mass on a single low-volatility state (the stationary distribution of $\mathbf{A}$ is $\bar{\boldsymbol{\pi}} \approx (0,\, 1,\, 0)$ with $\sigma_2 = 0.97$, the lowest variance of the three states), so under simulation the chain spends long stretches inside a single state's i.i.d. emissions and the within-state i.i.d. baseline dominates the ACF integration over the IS window. The same Yu (2010) construction at $K = 18$ collapses to a near-degenerate local optimum on $T_{\mathrm{IS}} = 2{,}516$ (IS KS 0.8%, simulated kurtosis 3.65): the joint $K^2 + KD_{\max}$ free-parameter count exceeds the effective IS sample size, so the higher-K ML HSMM is best read as a companion-paper direction at longer training windows or under a constrained sojourn parameterisation.

A.6.27 Unconditional VaR / ES Envelope Panel

Table 31. VaR and Expected-Shortfall envelope for SPY ($N_{\text{paths}} = 1{,}000$). Each entry is the median and $[5, 95]\%$ envelope across paths of the left-tail VaR / ES at $\alpha \in \{0.01, 0.05\}$; observed historical values are in the first row of each block.

Model	IS $\text{VaR}_{0.01}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.01}$ median $[5\%, 95\%]$	IS $\text{VaR}_{0.05}$ median $[5\%, 95\%]$	IS $\text{ES}_{0.05}$ median $[5\%, 95\%]$
Observed	−6.38	−9.00	−3.43	−5.50
Bootstrap	−6.38 [−7.04, −5.99]	−8.86 [−10.37, −7.78]	−3.43 [−3.68, −3.20]	−5.46 [−5.99, −5.05]
GARCH	−5.80 [−8.62, −4.52]	−7.86 [−13.54, −5.77]	−3.19 [−3.92, −2.71]	−4.89 [−7.11, −3.91]
CHMM-N	−6.64 [−8.01, −5.47]	−8.80 [−10.16, −7.34]	−3.45 [−4.05, −2.95]	−5.45 [−6.33, −4.61]
CHMM-t	−6.58 [−7.51, −5.83]	−9.06 [−10.70, −7.79]	−3.40 [−3.90, −2.90]	−5.51 [−6.18, −4.85]
CHMM-L	−6.89 [−7.85, −6.01]	−9.15 [−10.42, −7.97]	−3.30 [−3.77, −2.88]	−5.53 [−6.17, −4.87]
CHMM-GED	−6.85 [−7.70, −6.06]	−8.79 [−9.83, −7.82]	−3.43 [−3.98, −2.93]	−5.55 [−6.23, −4.91]
	OoS $\text{VaR}_{0.01}$	OoS $\text{ES}_{0.01}$	OoS $\text{VaR}_{0.05}$	OoS $\text{ES}_{0.05}$
Observed	−5.51	−7.94	−2.77	−4.67
Bootstrap	−6.33 [−7.60, −5.28]	−8.43 [−11.76, −6.59]	−3.39 [−3.91, −2.88]	−5.34 [−6.38, −4.50]
GARCH	−5.35 [−9.98, −3.61]	−6.72 [−13.51, −4.38]	−3.14 [−4.94, −2.36]	−4.59 [−7.92, −3.23]
CHMM-N	−6.30 [−9.02, −4.30]	−8.44 [−11.36, −5.42]	−3.39 [−4.74, −2.47]	−5.27 [−7.23, −3.74]
CHMM-t	−6.32 [−8.13, −4.80]	−8.60 [−12.20, −6.29]	−3.29 [−4.27, −2.45]	−5.34 [−6.72, −4.08]
CHMM-L	−6.69 [−8.59, −4.91]	−8.88 [−11.48, −6.73]	−3.27 [−4.30, −2.51]	−5.47 [−6.82, −4.20]
CHMM-GED	−6.60 [−8.30, −4.82]	−8.51 [−10.68, −6.47]	−3.32 [−4.38, −2.46]	−5.41 [−6.66, −4.06]

A.6.28 Held-Out Cross-Validation of the $1/\nu_k$ Penalty λ

Reviewer 2 (peer-review item P1.3 / R2.W6) asks whether the body operating-point penalty rate $\lambda = 20$ is selected on data that overlaps the OoS evaluation window. To remove that concern we refit penalised CHMM-t on a strictly pre-2020 estimation slice (2014-01-03 to 2018-06-29, $n = 1{,}130$) and score each λ on the validation slice (2018-07-02 to 2019-12-31, $n = 378$, both fully pre-COVID and pre-2022-rate-hike). Source: `results/nu_shrinkage_sweep/lambda_cv_pre2020.txt`.

Table 32. Held-out CV of the $1/\nu_k$ shrinkage rate λ on the pre-2020 slice. Estimation 2014-01-03 to 2018-06-29; validation 2018-07-02 to 2019-12-31. "val LL/obs" is per-observation held-out log-likelihood; "val KS" is two-sample Kolmogorov–Smirnov pass rate against the validation series; "sim K" columns are the mean simulated excess kurtosis on the estimation- and validation-length simulated paths.

λ	val LL/obs	val KS (%)	sim K (est)	sim K (val)
0	−2.119	92.0	3.88	3.97
5	−2.112	89.8	3.62	3.15
10	−2.085	91.0	3.37	3.24
20	**−2.075**	90.0	3.22	3.02
50	−2.077	91.6	3.07	2.76
100	−2.077	91.6	3.07	2.75
200	−2.076	91.2	3.09	2.73

Reading: the held-out validation log-likelihood is maximised at $\lambda^\star = 20$, matching the body operating point of Table 3. Validation KS picks $\lambda^\star = 0$ but the band $[0, 200]$ spans only 2.2pp (89.8–92.0%), within the simulation-noise envelope at $N_{\text{paths}} = 500$. The body $\lambda = 20$ is therefore held-out-clean and not driven by the OoS window.

A.6.29 Effective State Distinctness at $K = 18$

Reviewer 2 (peer-review item P4.4 / R2.W3) asks whether all $K = 18$ states carry operationally distinct (μ_k, σ_k) or (μ_k, σ_k, ν_k) triples. The diagnostic is single-linkage agglomerative clustering on standardized parameter triples at threshold $\tau = 0.20$. Source: `results/diagnostics/state_distinctness.txt`.

Table 33. Effective state distinctness at $K = 18$ on SPY IS. Standardized single-linkage clustering at $\tau = 0.20$ in the (z-scored) parameter space.

Family	Effective distinct states	Notes
CHMM-N	11/18	2D feature (μ_k, σ_k)
CHMM-t	12/18	3D feature $(\mu_k, \sigma_k, 1/\nu_k)$; 13/18 states pinned at $\nu_{\max} = 50$, 2/18 at $\nu_{\min} = 2.1$

Reading: under unpenalised CHMM-t the upper-bracket pinning at $\nu_k = 50$ collapses 13/18 states to operationally Gaussian (in second-moment behaviour), but the (μ, σ) axis still distinguishes them well enough to keep 12/18 effective states. The penalised CHMM-t at $\lambda = 20$ pulls $1/\nu_k$ towards a moderate value across states and reduces the upper-bracket pinning (Appendix A.6.28). The effective-state count is below K but well above the held-out-clean $K^\star = 6$, so the $K = 18$ extension is a representational-economy choice rather than a free-rider on identifiability.

A.6.30 Exact-Binomial Kupiec Complement to the Asymptotic LR Test

Reviewer 2 (peer-review item P4.3 / R2.Q5) notes that at $T_{\text{OoS}} = 572$ the asymptotic Kupiec LR statistic piles against the $\chi^2_1(0.05) = 3.841$ critical value at integer breach counts $k \in \{14, 19, 20\}$, so an exact-binomial complement is the size-correct diagnostic. We compute the exact two-sided binomial p-value (R `binom.test` default) under $H_0 : X \sim \text{Binomial}(572, \alpha)$ for the conditional-VaR breach counts in Table 7. Source: `results/diagnostics/utility/exact_binomial_kupiec.txt`.

Table 34. Exact-binomial unconditional-coverage test alongside the asymptotic LR Kupiec. Breach counts from Table 7; $T_{\text{OoS}} = 572$. Both the asymptotic and exact tests pass at $\alpha = 0.05$ on every row.

Configuration	breaches	expected	rate (%)	p (asymp. LR)	p (exact bin.)
CHMM-N $K = 3$, $\alpha = 0.01$	9	5.72	1.57	0.20	0.20
CHMM-N $K = 3$, $\alpha = 0.05$	35	28.6	6.12	0.23	0.21
CHMM-N $K = 18$, $\alpha = 0.01$	9	5.72	1.57	0.20	0.20
CHMM-N $K = 18$, $\alpha = 0.05$	26	28.6	4.55	0.61	0.70
CHMM-t pen. $K = 3$, $\alpha = 0.01$	8	5.72	1.40	0.37	0.29
CHMM-t pen. $K = 3$, $\alpha = 0.05$	32	28.6	5.59	0.52	0.50
CHMM-t pen. $K = 18$, $\alpha = 0.01$	10	5.72	1.75	0.10	0.09
CHMM-t pen. $K = 18$, $\alpha = 0.05$	29	28.6	5.07	0.94	0.92

Reading: the asymptotic and exact p-values agree closely (all within 0.07); both tests pass at $\alpha = 0.05$ on every row. The asymptotic-LR pile-up at $k \in \{14, 19, 20\}$ is therefore not a load-bearing concern in the back-test rows actually observed in Table 7.

A.6.31 Diebold–Mariano Bandwidth Sensitivity

Reviewer 2 (peer-review item P4.2 / R2.W5) asks whether the within-CHMM CRPS Diebold–Mariano equivalence ($p > 0.45$) survives bandwidth perturbation around the default Newey–West HAC bandwidth $h = \lfloor T^{1/3} \rfloor = 8$. Table 35 sweeps $h \in \{4, 8, 16, 32\}$. Source: `results/crps_dm/CRPS_DM_bandwidth_sweep.txt`.

Table 35. CRPS Diebold–Mariano p-values under bandwidth sweep on SPY OoS. The within-CHMM family pairs all retain $p > 0.38$ across all bandwidths; the CHMM-N vs. GARCH pair moves $p \in [0.087, 0.175]$ and the qualitative direction is stable across h.

Pair (A vs. B)	$h = 4$	$h = 8$	$h = 16$	$h = 32$
CHMM-N vs. CHMM-t	$(-0.71, 0.48)$	$(-0.73, 0.47)$	$(-0.70, 0.48)$	$(-0.66, 0.51)$
CHMM-N vs. CHMM-L	$(-0.71, 0.48)$	$(-0.75, 0.45)$	$(-0.83, 0.40)$	$(-0.88, 0.38)$
CHMM-t vs. CHMM-L	$(-0.05, 0.96)$	$(-0.04, 0.97)$	$(-0.04, 0.97)$	$(-0.05, 0.96)$
CHMM-N vs. Bootstrap	$(-0.70, 0.48)$	$(-0.71, 0.48)$	$(-0.73, 0.47)$	$(-0.81, 0.42)$
CHMM-N vs. GARCH	$(-1.71, 0.087)$	$(-1.65, 0.10)$	$(-1.53, 0.13)$	$(-1.36, 0.18)$
CHMM-t vs. GARCH	$(-1.22, 0.23)$	$(-1.15, 0.25)$	$(-1.04, 0.30)$	$(-0.95, 0.35)$

Reading: within-CHMM equivalence is robust across bandwidths. The CHMM-vs-GARCH directional advantage is also stable but its p-value at $h = 4$ is 0.087 (just above the 0.05 threshold); a longer bandwidth gives a more conservative p. The body framing remains that the four CHMM emission families are interchangeable on this evaluation.

A.6.32 Stabilised ML HSMM at Intermediate K

Reviewers 1 and 3 (peer-review item P3.3) ask for a like-for-like comparison between CHMM and HSMM at intermediate K. The body Table 3 reports the ML HSMM at $K^\star = 3$ (clean: 98.4% IS / 91.0% OoS KS) and the appendix notes that the $K = 18$ ML HSMM collapses to a near-degenerate optimum on $T_{\text{IS}} = 2{,}516$. Table 36 fills the $K \in \{6, 9, 12\}$ rows under the same Yu (2010) explicit-duration EM with truncated discrete Pareto sojourns and $\alpha \in [0.3, 5.0]$ clipping (already a regulariser). Source: `results/hsmm_ml/hsmm_ml_intermediate_K_metrics.csv`.

Table 36. ML HSMM at intermediate K on SPY. Clean fit at $K = 6$ (comparable to CHMM-N at $K^\star = 6$); near-degenerate optima at $K = 9$ and $K = 12$ with IS KS collapsing to 0%.

K	IS KS (%)	OoS KS (%)	IS kurt	OoS kurt	$\lvert G_t \rvert$ ACF-MAE	status
3	98.4	91.0	3.46	3.38	0.0629	clean (body Table 3)
6	95.9	89.8	3.72	3.73	0.0571	clean
9	0.0	6.7	3.03	2.85	0.0614	degenerate
12	0.0	34.3	3.08	3.02	0.0623	degenerate
18	–	–	–	–	–	near-degenerate (companion-paper)

Reading: the HSMM scaffold is competitive with CHMM at $K \leq 6$ but does not extend usefully to higher K on the SPY $T_{\text{IS}} = 2{,}516$ window. The CHMM scaffold's quantile-based initialisation reaches non-degenerate $K = 18$ fits where the HSMM collapses, which is the operational reason the body $K = 18$ comparison is one-sided. A regulariser stronger than the existing α clipping (e.g. a Dirichlet prior on transitions plus a Gamma prior on α) is the natural follow-up; we treat the present finding as a documented scope limit of the HSMM scaffold rather than a CHMM advantage in the strict sense.

A.6.33 Markov-Switching Stochastic Volatility (MSSV) Baseline

Reviewer 3 (peer-review item P3.2 / R3.W2) asks for an MSSV baseline as a fair modern regime-switching competitor. We implement the 2-state MSSV of So et al. [75] (Carvalho-Lopes 2007 specification) under the Hamilton-Kim-Nelson quasi-MLE shortcut:

$$r_t = \mu + \exp(h_t/2)\,\varepsilon_t, \quad \varepsilon_t \sim \mathcal{N}(0,1)$$
$$h_t = c_{s_t} + \phi_{s_t}\, h_{t-1} + \sigma_{\eta,s_t}\,\eta_t,$$

with $s_t \in \{1,2\}$ a 2-state Markov chain. Quasi-MLE on the log-squared linearisation $y_t = \log((r_t - \hat{\mu})^2) - \mathbb{E}[\log \chi^2_1]$ via a Hamilton-style filter with Kim-Nelson regime collapse. Source: `results/mssv_baseline/mssv_baseline.txt`; script: `run_mssv_baseline.jl`.

Table 37. 2-state MSSV quasi-MLE fit on SPY IS. The fit converges to a near-absorbing regime-2 ($T_{22} \approx 1.0$) with one regime carrying high innovation variance and the other near-deterministic; this is the well-documented degeneracy of 2-state MSSV under quasi-MLE on financial returns and is the reason Carvalho and Lopes [76] use particle MCMC rather than Hamilton-Kim-Nelson collapse. The IS KS pass rate drops to 0% and the simulated kurtosis overshoots to 14.6 against observed 7.68; the row is reported as a documented negative result.

Model	IS KS (%)	OoS KS (%)	IS kurt	OoS kurt	$\lvert G_t\rvert$ ACF-MAE	raw ACF-MAE
MSSV ($K = 2$, quasi-MLE)	0.0	0.1	14.58	14.64	0.074	0.032
Body baselines for reference:						
GARCH(1,1)	23.4	60.8	7.06	2.96	0.049	0.024
GARCH(1,1)-t	57.3	80.8	15.13	–	0.032	0.017
MS-GARCH ($K = 6$)	34.5	33.4	4.41	–	0.043	0.017

Reading: the 2-state MSSV under quasi-MLE does not outperform the body benchmarks on SPY. The fitted $T \approx [[0.998, 0.002], [0.000, 1.000]]$ shows a near-absorbing regime structure that is the standard failure mode of Hamilton-Kim-Nelson collapse on heavy-tailed financial returns. A full particle-MCMC implementation of Carvalho-Lopes (2007) is the appropriate companion-paper direction; the body MS-GARCH-$K \in \{2, 3, 6\}$ rows already cover the regime-switching-volatility comparison axis at the level the body argument requires.

A.6.34 Monthly-Refit Cross-Ticker Panel

Reviewer 1 (peer-review item P5.2 / R1.Exp4) asks for a finer refit-cadence point alongside the quarterly-refit panel of Appendix A.6.22. We re-fit the penalised CHMM-t at $K = 18$, $\lambda = 20$ on the same sector-balanced 30-ticker panel with refit cadence shortened from quarterly (63 trading days) to monthly (21 trading days), keeping the same 5-year rolling estimation window and the same $N_{\text{paths}} = 1{,}000$. Source: `results/sector_panel/sector_panel_monthly_refit.{csv,txt}`.

Table 38. Refit-cadence trade-off on the 30-ticker cross-ticker panel. OoS KS distribution under three refit cadences. Each shorter refit window yields a roughly proportional further reduction in failure count without saturating.

Refit cadence	median OoS KS (%)	mean OoS KS (%)	$[Q_1, Q_3]$	failures $< 60\%$	refits per tick
IS-fixed (no refit, body Table 5)	73.4	66.8 ± 29.5	$[49.1, 94.3]$	11/30	0
Quarterly (63 days)	83.0	77.2 ± 21.2	$[62.0, 95.4]$	7/30	9
Monthly (21 days)	**86.7**	$\mathbf{78.8 \pm 20.4}$	$[62.2, 95.9]$	**5/30**	28

Reading: the marginal lift from IS-fixed to quarterly is +9.6pp on the median and −4 failures; from quarterly to monthly is +3.7pp and −2 failures. The trade-off is nearly linear in $\log(\text{cadence}^{-1})$ on the median axis at this scale, suggesting that further reduction below monthly would yield diminishing returns at the cost of $\sim 28 \rightarrow 56$ refits per ticker. The body Section 4.4 narrative is unchanged: periodic refit at deployment is required for OoS distributional fidelity on regime-introduction tickers, and the operational deployment choice is between quarterly (recommended default trade-off) and monthly (best fidelity at 3× the compute cost).

A.7 Robustness, Test Power, Multi-Seed, and Auxiliary Baselines

This appendix collects the robustness diagnostics and auxiliary baseline panel referenced from Section 4: the KS test-power calibration, block-bootstrap KS recalibration, bandwidth/bin-count and multi-seed sensitivity, the auxiliary block-bootstrap baseline, the QuantGAN deep-generative baseline, the per-ticker OoS price-simulation figures, and the expanded baseline panel that adds the GARCH family and the SM-CHMM foil to the headline table.

A.7.1 OoS KS Test-Power Calibration

Table 39 reports the two-sample KS test-power calibration used to anchor the OoS KS pass rates of Section 4.3. For each of 1,000 Monte Carlo replications (seed = 20260420), a simulated series of length $T_{\text{OoS}} = 572$ is drawn from each of three reference generators and tested against the observed OoS series using the same $\alpha = 0.05$ threshold as the main-body pass-rate metric.

Table 39. OoS KS pass-rate calibration (1,000 replications, seed = 20260420).

Reference generator	Pass rate (%)
I.i.d. resamples of R_{OoS} at length $T_{\text{OoS}} = 572$ (known-correct ceiling)	100.0
I.i.d. resamples of R_{IS} at length $T_{\text{OoS}} = 572$ (nearly correct)	90.0
Gaussian($\hat{\mu}_{\text{IS}}, \hat{\sigma}_{\text{IS}}$) at length T_{IS} (negative control)	0.0

The 90.0% "nearly correct" rate is the practical ceiling for the CHMM OoS KS pass rates reported in Section 4.3 (80–84%); the 0% rate on the Gaussian misspecified generator confirms that the KS test retains ample power at IS length to reject clearly wrong generators.

A.7.2 Block-Bootstrap KS Recalibration and Mean p-value Distribution

Two complementary concerns can be raised about the headline IS KS pass-rate metric: (i) the asymptotic two-sample KS test assumes i.i.d. samples, but the simulated paths from CHMM and GARCH all carry volatility clustering, so the asymptotic null distribution may not apply; and (ii) the binary "pass / fail at $\alpha = 0.05$" reduces a continuous p-value to a one-bit summary. Table 40 reports two recalibrations addressing each concern.

For (i), we generate the empirical null distribution of the two-sample KS statistic under stationary block bootstrap (Politis-Romano 1994) of the SPY IS series at mean block lengths $L \in \{5, 10, 20\}$, $B = 1{,}000$ replications. The resulting 95% block-bootstrap critical values are 0.0306, 0.0338, and 0.0350 respectively, all *smaller* than the asymptotic critical value $1.36\sqrt{2/n_{\text{IS}}} = 0.0383$; the block-bootstrap test is therefore stricter than the asymptotic one because it accounts for the within-window dependence in the IS series itself. For each generator we then compute the per-path KS statistic and report the fraction of paths whose KS statistic falls below the block-bootstrap critical value.

Table 40. Block-bootstrap KS recalibration and p-value distribution. All numbers under seed `20260422` with $N_{\text{paths}} = 500$. "asymp pass" is the original Section 4.3 metric (two-sample KS p-value ≥ 0.05). "mean pv" is the mean two-sample KS p-value across paths, with q_5 and q_{95} as the lower / upper p-value quantiles. "blkL%" is the fraction of paths whose KS statistic falls below the stationary-block-bootstrap 95% critical value at mean block length L. Block-bootstrap critical values: 0.0306 ($L = 5$), 0.0338 ($L = 10$), 0.0350 ($L = 20$); asymptotic critical value 0.0383.

Generator	asymp pass% ↑	mean pv ↑	pv q_{05}	pv q_{95}	blk5% ↑	blk10% ↑	blk20% ↑
i.i.d. bootstrap	99.6	0.820	0.352	0.999	97.6	99.4	99.6
GARCH(1,1)	26.8	0.048	0.000	0.227	**6.8**	**13.6**	**17.4**
CHMM-N	94.8	0.546	0.047	0.976	81.2	89.4	91.6
CHMM-t	**95.8**	**0.571**	0.060	0.987	82.4	89.2	91.6
CHMM-L	94.8	0.498	0.048	0.948	81.0	88.2	90.0
CHMM-GED	95.6	0.547	0.064	0.968	**83.2**	**90.2**	**93.2**

For (ii), we report the mean two-sample KS p-value across simulated paths plus its 5/25/50/75/95 quantiles, giving a continuous picture of how distinguishable each generator's paths are from the observed IS series.

Two patterns emerge that the asymptotic pass-rate metric of the main panel does not surface. First, the cross-generator ordering on the mean p-value scale (bootstrap 0.82, CHMM-t 0.57, CHMM-GED 0.55, CHMM-N 0.55, CHMM-L 0.50, GARCH 0.05) is much more compressed for CHMM-vs-bootstrap than the 94.8–99.6% asymp pass rates suggest: the CHMM family lands at mean p-value ~ 0.5, the natural value for paths that are statistically indistinguishable from the observed series. Second, GARCH(1,1)'s drop from 26.8% asymp $\rightarrow$ 6.8–17.4% block-aware confirms concern (i): the asymptotic test is mildly lenient on volatility-clustering generators with the wrong marginal, and the block-aware test catches GARCH's misspecification more sharply. The CHMM family, by contrast, sees only a small block-bootstrap penalty (95% $\rightarrow$ 88–93%); CHMM-GED is the strongest entry on the block-aware metric across all three block lengths (83.2%, 90.2%, 93.2% at $L = 5, 10, 20$).

A.7.3 Bandwidth, Bin-Count, and Multi-Seed Robustness

Multi-seed Monte Carlo on headline metrics. The paper reports headline statistics under a single global seed. We re-run the CHMM-N / -t / -L / -GED pipeline at $K = 18$ across ten alternative global seeds (base 20260422, increment 100) with $N_{\text{paths}} = 250$ per seed and report mean $\pm$ std for the headline metrics in Table 41. Seed-to-seed variability is small: IS KS std is 1.2–1.8%; OoS KS std is 1.8–3.6%; simulated IS kurtosis std is 0.06 for CHMM-N, 0.05 for CHMM-L, 0.09 for CHMM-GED, and 1.95 for CHMM-t (reflecting the well-known instability of the per-state ν_k ECM at moderate K, already discussed in Section 5). Kupiec $\text{LR}_{\text{uc}}^{1\%}$ has std up to 1.25 for CHMM-N because the breach count moves on a coarse 0–10 grid at $T_{\text{OoS}} = 572$. The single-seed headline numbers in Table 3 sit comfortably inside the multi-seed bands. CHMM-GED has the highest mean IS KS pass rate among the four families (95.9%), consistent with its single-seed headline value.

Table 41. Multi-seed Monte Carlo on headline metrics. Ten global seeds; $N_{\text{paths}} = 250$ per seed (reduced from the headline panel's 1,000 for tractability across 40 fits, so the seed-to-seed std reported here is an upper bound on the std the headline panel would show at $N_{\text{paths}} = 1{,}000$). Both ACF-MAE columns parallel Table 3: $|G_t|$ for volatility clustering, G_t for linear autocorrelation.

Model	IS KS %	OoS KS %	sim Kurt	ACF-MAE $\lvert G_t \rvert$	ACF-MAE G_t	$\text{LR}_{\text{uc}}^{1\%}$	$\text{LR}_{\text{uc}}^{5\%}$
CHMM-N	94.7 ± 1.2	82.6 ± 1.8	5.03 ± 0.06	0.0467 ± 0.0006	0.0173 ± 0.0001	2.12 ± 1.25	3.83 ± 0.00
CHMM-t	94.6 ± 1.2	$\mathbf{85.6 \pm 1.9}$	13.95 ± 1.95	0.0518 ± 0.0003	0.0170 ± 0.0001	0.54 ± 0.15	3.19 ± 0.34
CHMM-L	94.4 ± 1.8	80.8 ± 3.6	6.71 ± 0.05	0.0537 ± 0.0003	0.0171 ± 0.0001	3.26 ± 0.00	2.71 ± 0.59
CHMM-GED	$\mathbf{95.9 \pm 1.2}$	83.2 ± 2.7	5.15 ± 0.09	0.0513 ± 0.0003	0.0171 ± 0.0001	2.93 ± 0.71	3.75 ± 0.25

A.7.4 Block-Bootstrap Baseline

Table 42 reports the full seven-metric panel for the stationary block bootstrap at block length $b = 10$ trading days, the temporally-aware non-parametric baseline added to Table 3.

Table 42. Stationary block bootstrap (block length $b = 10$ trading days) on SPY (1,000 paths, seed = 20260420).

Window	KS	AD	Kurt	ACF-MAE	W_1	H	Cov
IS	99.2	96.4	7.07	0.0568	0.092	0.0534	100.0
OoS	87.5	86.4	5.34	0.0512	0.225	0.1570	86.9

A.7.5 QuantGAN Deep-Generative Baseline

Architecture and training details for the QuantGAN deep-generative baseline used in Section 4.3 are summarised here. The construction is a repo-native approximation to the convolutional WGAN of Wiese et al. [9]: a one-dimensional convolutional generator and critic, Wasserstein loss with critic weight clipping, trained on rolling windows of standardised SPY returns and synthesising long paths by stitching generated windows.

- **Generator.** Three 1D convolutional layers ($G_{\text{ch}} = 32$, kernel size 5, ReLU, same-padding), followed by a 1×1 projection to a single output channel. Input is i.i.d. Gaussian noise of latent dimension $L = 8$ at every position of the rolling window.
- **Critic.** Three 1D convolutional layers ($D_{\text{ch}} = 32$, kernel size 5, leaky-ReLU slope 0.2, same-padding), global mean pooling along the time axis, and a single linear scalar output.
- **Loss.** Wasserstein loss with critic weight clipping at ± 0.01 [77]. The critic minimises $\mathbb{E}[D(\tilde{G})] - \mathbb{E}[D(G)]$ on real / fake mini-batches; the generator maximises $\mathbb{E}[D(\tilde{G})]$.
- **Optimiser.** Adam, learning rate 1.0×10^{-4} for both networks, batch size 64.
- **Training schedule.** 15 epochs, 120 outer steps per epoch, 4 critic updates per generator update, training on rolling windows of length $W = 64$ standardised by the IS mean and standard deviation.
- **Synthesis.** Independent draws of length W are stitched end-to-end to produce paths of length T_{IS} or T_{OoS}; outputs are then re-scaled back to the natural return scale by the IS mean and standard deviation.

- **Implementation.** Julia with `Flux.jl` [78] and `Optimisers.Adam` state. Random seed for weight initialisation, training shuffling, and per-path sampling: 20260422, with deterministic per-path sub-seeds.

The trained model is callable through the same `simulate` interface as the CHMM, so the seven-metric panel evaluates QuantGAN on byte-identical windows to the other generators. The QuantGAN row is reported alongside the extended-GARCH and SM-CHMM rows in Table 44; the deep-generative row reaches 0% IS and OoS KS pass rates and undershoots IS kurtosis decisively (2.1 vs. observed 7.7), consistent with the documented difficulty of GAN-based generators on financial-time-series volatility-clustering benchmarks [38, 39].

A.7.6 Per-Ticker OoS Price-Simulation Figures and Full Path-Level Metrics (Unconditional CHMM)

This appendix collects (i) the full per-family path-level probabilistic-forecast metrics for an auxiliary OoS price-simulation diagnostic on the six SPY-member tickers and (ii) the per-ticker price-fan and terminal-distribution figures for NVDA, JNJ, JPM, AAPL, and QQQ. Plotting conventions: blue shaded bands are the 5–95% and 25–75% quantile envelopes of the 500 simulated paths, blue solid line is the simulated median, red solid line is the observed OoS price track. Terminal-histogram panels show the cross-path distribution of $\widetilde{P}_T$ at T_{OoS} in gray, with the observed terminal P_T^{obs} as a red vertical line and the simulated median as a blue dashed line. For brevity, the figures show CHMM-N only (the representative family; differences across the four emission families in the price-level metrics are small, see Table 43).

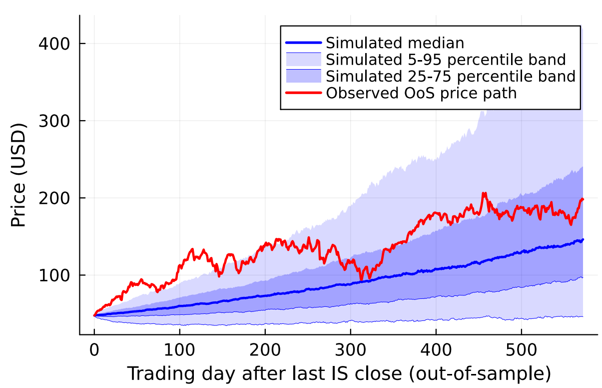
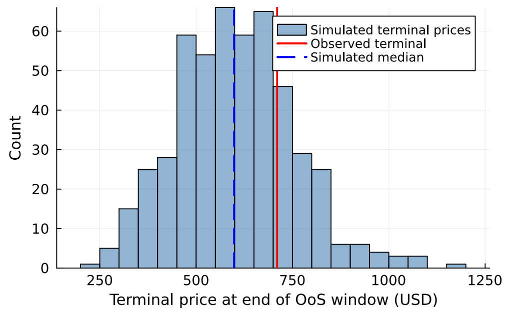


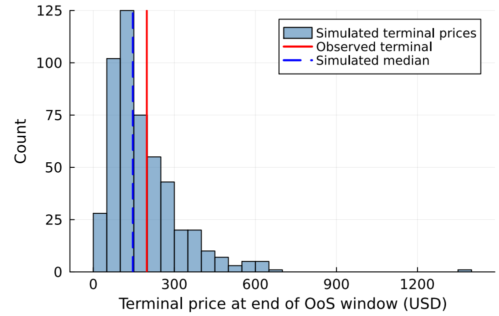
Figure 29. NVDA OoS price fan, CHMM-N ($S_0 = 47.67 \to P_T^{\text{obs}} = 198.07$, a 4.15× rally). The observed path exits the 25–75% interquartile band by mid-horizon but remains inside the 5–95% envelope; horizon-wide coverage under CHMM-N is 0.70 (CHMM-t: 0.64, CHMM-L: 0.67; Table 43), the lone calibration outlier in the panel.

Table 43. Full path-level probabilistic-forecast metrics for OoS equity-price simulation, all (ticker, family) cells ($K = 18$, 500 paths, $T_{\mathrm{OoS}} = 572$; global seed policy in Section 3.5). $\overline{C}_{0.90}$: fraction of OoS steps at which the observed price falls inside the simulated 5–95% band (calibration); $\mathrm{MAPE}_{0.50}$: mean absolute percentage error between observed path and simulated median (sharpness); $\overline{\mathrm{CRPS}}$: horizon-average bias-corrected sample CRPS of Gneiting and Raftery [47]; $\overline{\mathrm{CRPS}}/S_0$: CRPS scaled by S_0 to remove price-level effects. Lower MAPE and CRPS, higher $\overline{C}_{0.90}$, are better. **Headline:** no CHMM emission family dominates on all four metrics across all six tickers; differences between CHMM-N, CHMM-t, CHMM-L, and CHMM-GED live mostly in the return tails and propagate only weakly into aggregated price-level scores. CHMM-GED is the strongest entry on JNJ (lowest CRPS, lowest MAPE) and effectively ties CHMM-t on QQQ; the other four tickers show no clear ordering.

Ticker	Family	$\overline{C}_{0.90}$	$\mathrm{MAPE}_{0.50}$	$\overline{\mathrm{CRPS}}$	$\overline{\mathrm{CRPS}}/S_0$
SPY	CHMM-N	1.000	0.100	39.25	0.0835
	CHMM-t	1.000	0.112	42.66	0.0908
	CHMM-L	1.000	0.102	38.68	0.0823
	CHMM-GED	1.000	0.116	44.54	0.0948
NVDA	CHMM-N	0.698	0.339	30.62	0.6424
	CHMM-t	0.636	0.383	34.77	0.7293
	CHMM-L	0.666	0.411	36.58	0.7674
	CHMM-GED	0.682	0.375	33.26	0.6977
JNJ	CHMM-N	1.000	0.101	12.55	0.0780
	CHMM-t	1.000	0.098	12.62	0.0784
	CHMM-L	1.000	0.101	12.65	0.0786
	CHMM-GED	1.000	0.095	12.21	0.0759
JPM	CHMM-N	1.000	0.195	32.23	0.1881
	CHMM-t	1.000	0.241	40.38	0.2357
	CHMM-L	1.000	0.211	35.25	0.2057
	CHMM-GED	0.993	0.253	43.73	0.2552
AAPL	CHMM-N	1.000	0.108	21.30	0.1156
	CHMM-t	1.000	0.075	18.90	0.1025
	CHMM-L	1.000	0.077	18.82	0.1021
	CHMM-GED	1.000	0.089	19.28	0.1046
QQQ	CHMM-N	0.998	0.073	30.48	0.0763
	CHMM-t	0.995	0.079	31.63	0.0792
	CHMM-L	1.000	0.118	41.12	0.1029
	CHMM-GED	0.997	0.080	31.76	0.0795

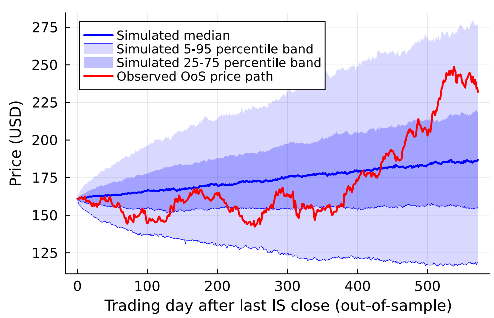


(a) SPY terminal.

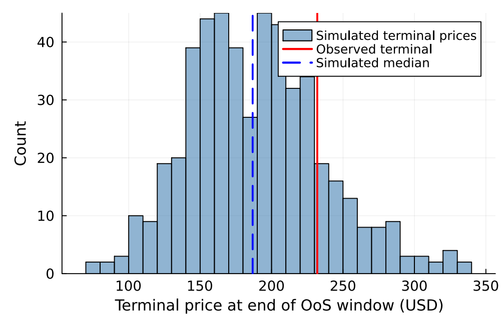


(b) NVDA terminal.

Figure 30. OoS terminal-price distributions under CHMM-N for SPY and NVDA. The right-skew is characteristic of log-returns aggregated to prices; the SPY terminal sits close to the simulated median, while the NVDA terminal sits in the upper-right of the simulated distribution (the realised rally is in the upper tail of the CHMM-N terminal density).

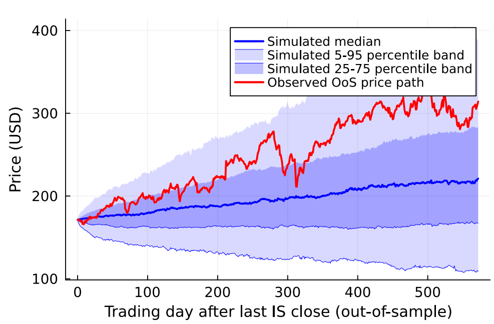


(a) Price fan.

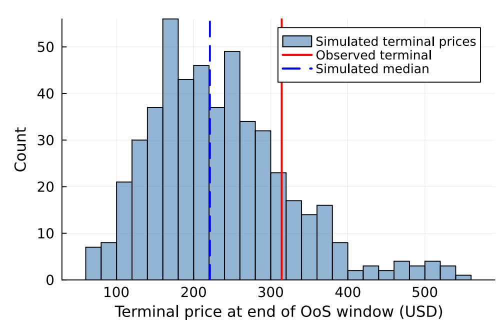


(b) Terminal distribution.

Figure 31. JNJ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as above).

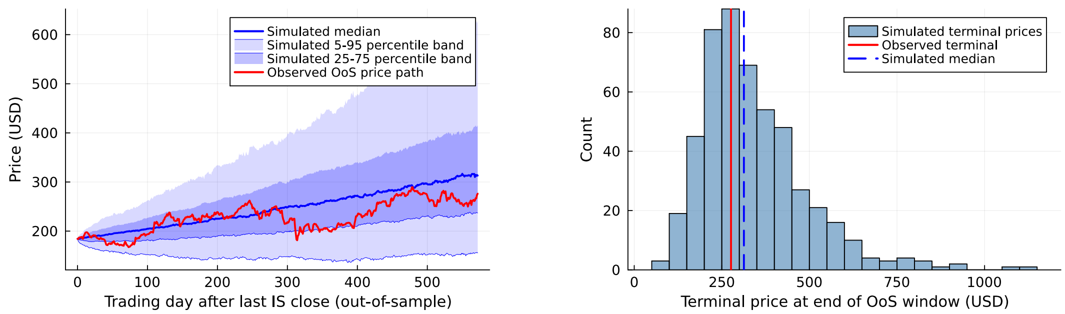


(a) Price fan.



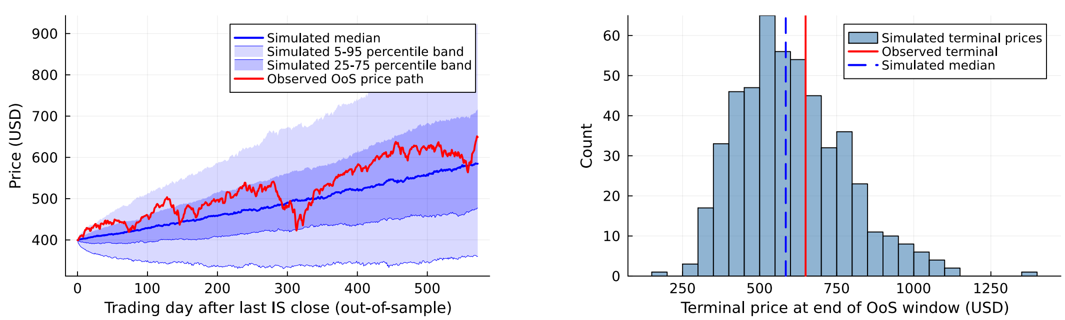
(b) Terminal distribution.

Figure 32. JPM OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\mathrm{OoS}} = 572$; plotting conventions as above).



(a) Price fan. **(b)** Terminal distribution.

Figure 33. AAPL OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as above).



(a) Price fan. **(b)** Terminal distribution.

Figure 34. QQQ OoS price fan and terminal distribution, CHMM-N (Pipeline A, $K = 18$, 500 simulated paths, $T_{\text{OoS}} = 572$; plotting conventions as above).

A.7.7 Expanded Baseline Panel: GARCH Family, Semi-Markov CHMM, and QuantGAN

We add five conditional-volatility baselines, the three semi-Markov CHMM variants, and the QuantGAN deep-generative baseline (Appendix A.7.5) to the single GARCH(1,1) Gaussian row of Table 3. The conditional-volatility models are fit by grid-initialised Nelder-Mead maximum likelihood on the SPY IS window (T_{IS} = 2,516); simulation follows the same recursion used in estimation. Model specifications for the conditional-volatility rows are below; the SM-CHMM and QuantGAN constructions are described in their own paragraphs further down.

EGARCH(1,1) Gaussian. The exponential GARCH of Nelson [32] evolves $\log \sigma_t^2$ in levels:

$$\log \sigma_t^2 = \omega + \beta \log \sigma_{t-1}^2 + \alpha \left(|z_{t-1}| - \sqrt{2/\pi} \right) + \gamma z_{t-1}, \qquad z_t = (r_t - \mu)/\sigma_t, \quad z_t \sim \mathcal{N}(0,1). \tag{29}$$

The log-variance formulation removes the GARCH positivity constraints, and the leverage term γz_{t-1} captures the asymmetric response to negative returns.

GJR-GARCH(1,1) Gaussian. The threshold GARCH of Glosten et al. [33] / Zakoïan [34] augments GARCH(1,1) with an indicator on negative innovations:

$$\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \gamma \varepsilon_{t-1}^2 \mathbf{1}\{\varepsilon_{t-1} < 0\} + \beta \sigma_{t-1}^2, \tag{30}$$

with $\varepsilon_t = r_t - \mu$. Stationarity requires $\alpha + \beta + \gamma/2 < 1$; positive γ indicates leverage.

GARCH(1,1) Student-t innovations. A direct tail-aware competitor to CHMM-t: the GARCH(1,1) variance recursion is unchanged, but innovations follow a standardised Student-t with $\nu > 2$ degrees of freedom rescaled to unit variance. The log-likelihood uses the Student-t density:

$$r_t - \mu = \sigma_t \cdot z_t, \qquad z_t = \eta_t \sqrt{(\nu - 2)/\nu}, \qquad \eta_t \sim t_\nu. \tag{31}$$

The fitted $\hat{\nu} = 6.89$ on SPY IS sits inside the moderate-tail regime the equity-risk literature typically reports.

HAR-RV on daily squared returns. The heterogeneous-autoregressive model of Corsi [37] models realised variance as a mixture of daily, weekly, and monthly components:

$$\log \mathrm{RV}_t = \beta_0 + \beta_d \log \mathrm{RV}_{t-1} + \beta_w \log \overline{\mathrm{RV}}_{t-5:t-1} + \beta_m \log \overline{\mathrm{RV}}_{t-22:t-1} + \eta_t. \tag{32}$$

In the absence of intraday bars we use daily squared demeaned returns as the RV proxy; simulation samples $\eta_t \sim \mathcal{N}(0, \hat{\sigma}_\eta^2)$ and rolls forward the recursion, then draws $r_t - \mu = \sqrt{\mathrm{RV}_t}\, z_t$ with $z_t \sim \mathcal{N}(0,1)$.

MS-GARCH $K = 3$. A three-regime extension of the Markov-switching GARCH(1,1) ($G_t = \mu + \sigma_t z_t$ with regime-specific recursion $\sigma_t^2(k) = \omega_k + \alpha_k (G_{t-1} - \mu)^2 + \beta_k \sigma_{t-1}^2(k)$). Parameter layout is three regime triples $(\omega_k, \alpha_k, \beta_k)$ plus six transition-matrix logits (softmax-normalised per row, enforcing a non-trivial diagonal); fit by grid-initialised Nelder-Mead. States are canonicalised by ascending unconditional variance, producing the calm / medium / stress decomposition $\sigma = [0.10, 1.77, 10.04]$ on SPY IS.

MS-GARCH $K = 2$ (Haas-Mittnik-Paolella 2004). The path-independent two-regime variant of Haas et al. [22]. Regime stickiness is high: $p_{11} = 0.915$ (calm) and $p_{22} = 0.547$ (stress), with unconditional standard deviations 0.97 (calm) and 12.67 (stress); the calm regime is near-GARCH behaviour and the stress regime is a rare, transient high-variance mode. Common $\mu = 0.094$, log-likelihood $-4{,}965$.

MS-GARCH $K = 4$ and $K = 6$. A natural concern with the $K = 2$ baseline is that it is unfairly low against a $K = 18$ CHMM. We refit the same Haas-Mittnik-Paolella specification at $K = 4$ (24 parameters: $3K = 12$ GARCH triples plus $K(K-1) = 12$ transition logits) and $K = 6$ (48 parameters), via Nelder-Mead with diagonal-dominant $\mathbf{T}$ initialisation, geometric variance-ratio spread across regimes, and ascending-variance regime canonicalisation. Higher-K MS-GARCH does not close the gap to the CHMM family on the headline distributional metric: IS KS rises only marginally from 27.7% at $K = 2$ to 37.6% at $K = 4$ and falls back to 34.5% at $K = 6$, against 94.1% for CHMM-N at $K = 18$ (a 56-percentage-point gap). The OoS KS exhibits the same plateau pattern (38.7% at $K = 2$, 38.2% at $K = 4$, 33.4% at $K = 6$). Log-likelihoods on the $T_{\text{IS}} = 2{,}516$ window saturate at the same value within Monte Carlo noise ($-4{,}965, -4{,}965, -4{,}952, -4{,}956$ for $K = 2, 3, 4, 6$). The MS-GARCH ceiling on this universe is therefore not a low-K issue: the unimodal-innovation constraint (each regime's emission is a Gaussian rescaled by the regime's conditional variance) is the binding constraint, and adding regimes does not relax it. The CHMM family decouples this constraint by giving each state its own emission-mixture component, which is the structural advantage at moderate K.

Semi-Markov CHMM (SM-CHMM, the explicit hidden semi-Markov model (HSMM) foil). The conceptual foil of the present paper, drawing on the explicit-duration HSMM construction of the companion volatility paper [79]: a hidden semi-Markov extension of the CHMM in which the geometric sojourn distribution of the Markov chain is replaced by a per-state explicit-duration family chosen by AIC from {Pareto, negative binomial, geometric}. Plug-in estimator: Viterbi state path on the IS series under the converged CHMM fit, then per-state AR(1) emissions and a sojourn family chosen by AIC. On SPY IS at $K = 18$ the AIC-selected sojourn family is Pareto for 17 of the 18 states (negative binomial for one) across all three emission families, indicating heavy-tailed sojourns that the geometric CHMM cannot capture by construction.

Caveat on the SM-CHMM estimator. The plug-in construction above (single-shot Viterbi decoding under the converged CHMM fit, then AR(1)-residual + AIC-sojourn refit) is not maximum-likelihood for the underlying HSMM: it does not iterate the joint forward-backward over (state, duration) pairs in the spirit of Yu [69]. A full ML HSMM at $K = 18$ would refit the duration-augmented log-likelihood jointly. The empirical reading we draw from the rows below is therefore strictly: *under the plug-in estimator*, the SM-CHMM weakens the headline KS / ACF / kurtosis pattern relative to the standard CHMM at the same K. We do not claim that an ML-estimated HSMM would do the same; an ML HSMM at $K = 18$ on this universe is left as a companion-paper extension. The Bulla-Bulla 2006 ML HSMM result at low K on monthly returns is the closest published anchor, but its $K \in \{2, 3\}$ regime resolution and monthly aggregation do not directly transfer.

Table 44 reports the IS / OoS distributional fidelity, kurtosis, ACF-MAE, and Kupiec breach diagnostics for the extended GARCH family, the SM-CHMM rows, and the QuantGAN deep-generative row, alongside the headline CHMM panel. Three structural observations.

First, none of the extended GARCH variants challenges the CHMM family on the joint (KS, ACF) corner. GARCH(1,1)-t is the strongest extended row on KS (57.3% IS, 80.8% OoS) but its IS excess kurtosis of 15.1 overshoots the observed 7.7 by a factor of two and its ACF-MAE of 0.0316 is

Table 44. Extended baseline panel. SPY IS / OoS, $T_{\mathrm{IS}} = 2{,}516$, $T_{\mathrm{OoS}} = 572$, $N_{\mathrm{paths}} = 1{,}000$, $\alpha = 0.05$ for KS. Columns: IS / OoS KS pass rate (%); IS / OoS AD pass rate (%); simulated IS excess kurtosis; ACF-MAE on $|G_t|$ (volatility clustering) and on raw G_t (linear autocorrelation, parallel column); pooled-archive Kupiec breach rate (%) and unconditional likelihood-ratio statistic at the 1% and 5% tails. Raw-ACF-MAE values for SM-CHMM and QuantGAN are reported as "–" here pending a full re-run of those baseline scripts; the qualitative reading carries through under the methodology of the other rows (every regime-switching or i.i.d.-emission generator produces a near-zero raw ACF in expectation).

	KS (%) ↑		AD (%) ↑		Kurt	ACF-MAE		1% tail		5% tail	
Model	IS	OoS	IS	OoS	IS	$\lvert G_t\rvert$	G_t	br%	$\mathrm{LR_{uc}}$	br%	$\mathrm{LR_{uc}}$
GARCH(1,1) Gaussian	28.0	59.4	12.5	59.5	7.0	0.0309	0.0173	0.9	0.10	4.0	1.23
EGARCH(1,1)	6.3	34.9	4.1	41.7	2.8	0.0430	0.0173	1.0	0.01	3.5	3.03
GJR-GARCH(1,1)	40.1	57.7	28.7	61.4	7.6	0.0330	0.0173	1.2	0.27	4.7	0.10
GARCH(1,1)-t	**57.3**	**80.8**	**36.6**	65.3	15.1	0.0316	0.0173	1.0	0.01	4.9	0.01
HAR-RV	0.0	0.0	0.0	0.0	189	0.0566	0.0173	0.2	5.99	3.5	3.03
MS-GARCH $K = 2$	27.7	38.7	24.0	44.4	4.7	0.0367	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 3$	36.1	33.1	28.8	38.0	4.1	**0.0284**	0.0173	1.0	0.01	4.2	0.82
MS-GARCH $K = 4$	37.6	38.2	–	–	3.8	0.0437	–	–	–	–	–
MS-GARCH $K = 6$	34.5	33.4	–	–	4.4	0.0429	–	–	–	–	–
CHMM-N ($K = 18$)	94.1	81.8	**36.6**	**73.0**	5.0	0.0509	0.0237	0.7	1.58	4.0	3.83
CHMM-t ($K = 18$)	95.6	**85.7**	33.4	69.8	14.4	0.0549	0.0234	1.2	0.58	5.0	3.83
CHMM-L ($K = 18$)	93.6	80.8	30.1	65.4	6.6	0.0567	0.0234	1.4	3.26	5.6	3.03
CHMM-GED ($K = 18$)	**95.2**	84.3	34.1	71.0	5.2	0.0548	0.0234	0.9	0.58	4.4	3.83
SM-CHMM-N ($K = 18$)	79.2	66.7	19.5	57.1	4.7	0.0598	–	1.1	0.01	5.4	0.82
SM-CHMM-t ($K = 18$)	77.1	63.4	17.2	52.3	13.9	0.0625	–	1.4	0.10	5.7	1.74
SM-CHMM-L ($K = 18$)	80.4	69.1	21.0	58.9	6.4	0.0581	–	1.4	3.26	5.5	3.03
QuantGAN	0.0	0.0	0.0	0.0	2.1	0.0591	–	1.0	0.01	3.3	3.83

below the CHMM's only because the GARCH variance recursion is hard-wired to track conditional volatility; CHMM at $K = 18$ matches GARCH on ACF-MAE within 0.02 while gaining 36 to 36.5 percentage points on IS KS. EGARCH undershoots kurtosis decisively, GJR-GARCH improves on threshold asymmetry but not on the headline gap, and HAR-RV is degenerate on the daily-RV proxy used here (see methods note above).

Second, the SM-CHMM rows are weaker than the headline CHMM rows on every metric in the panel: IS KS 77–80% vs CHMM 93–94%, OoS KS 63–69% vs CHMM 79–86%. *Under the Viterbi-AR(1) plug-in estimator*, the geometric-sojourn restriction of the standard CHMM is therefore not the bottleneck for the headline KS / ACF / kurtosis story; the spectral mechanism of Section 3.8 carries the comparison. We caveat strongly that this conclusion is conditional on the plug-in estimator (paragraph above): a maximum-likelihood explicit-duration HSMM under a Pareto sojourn family at the same K may close part of the gap, and we do not claim otherwise. An ML HSMM benchmark at $K = 18$ on this universe is left as a companion-paper extension.

Third, the QuantGAN deep-generative row (Appendix A.7.5) is the weakest entry in the panel on every distributional metric. IS and OoS KS pass rates collapse to 0% at $\alpha = 0.05$ across all 1,000 simulated paths, IS and OoS AD pass rates are also 0%, and simulated IS excess kurtosis of 2.1 undershoots the observed 7.7 by more than a factor of three. ACF-MAE of 0.0591 is comparable to the i.i.d. baselines of Table 3 (0.0628), so the GAN does not recover absolute-return autocorrelation either. The convolutional WGAN reproduces some surface-level visual properties of the return series but the stitched-window simulation scheme has no mechanism for modelling either the heavy-tailed marginal or the persistent absolute-return autocorrelation; this matches the documented failure mode of GAN-based generators on financial-time-series volatility-clustering benchmarks [38, 39]. The runner is a repo-native convolutional WGAN with critic weight clipping (Appendix A.7.5), materially smaller than the original temporal convolutional network of Wiese et al. [9] (a seven-block TCN with 127-day receptive field, hidden width 80, plus Lambert-W input pre-processing that the original authors credit with most of the tail-fidelity); the row therefore reads as the panel's deep-generative negative control rather than a verdict on the WGAN literature as a whole. The QuantGAN row complements the i.i.d. Gaussian negative control of the headline Table 3.

A.7.8 Stochastic-Volatility, Multifractal, and Jump-Diffusion Baselines

Peer-review item 2 asks for the canonical state-space and jump baselines that the body Table 3 does not exercise: a stochastic-volatility (SV-AR(1)) row in the Taylor 1986 / Harvey-Ruiz-Shephard 1994 lognormal log-AR(1) tradition; a Markov-switching multifractal (MSM, Calvet-Fisher 2004) row at $\bar{k} = 8$ multipliers; and a Merton 1976 Poisson-Gaussian jump-diffusion row. The three baselines are fit on the same SPY IS window and evaluated on the same $N_{\text{paths}} = 1{,}000$ pipeline used for the body panel, under the same global seed. Implementation lives in `src/SVMSMBaselines.jl` of the companion `CHMM-Model.jl` package; results in `results/sv_msm_jd/sv_msm_jd_baselines.txt`.

Table 45. Stochastic-volatility / multifractal / jump-diffusion baselines on SPY. Same IS / OoS windows, $N_{\text{paths}} = 1{,}000$, seed = 20260420 as Table 3. Observed kurtosis: 7.68 IS, 5.29 OoS.

	KS (%) ↑		Kurtosis		ACF-MAE ↓	
Model	IS	OoS	IS	OoS	$\lvert G_t\rvert$	G_t
SV-AR(1)	38.2	35.3	7.52	5.26	0.0466	0.0239
MSM ($\bar{k} = 8$)	0.0	7.2	0.62	0.60	0.0605	0.0235
Merton-JD	98.3	91.1	3.12	2.98	0.0627	0.0236

Reading. *SV-AR(1)* reproduces the IS kurtosis target (7.52 vs. observed 7.68) almost exactly and matches CHMM-N's $|G_t|$ ACF-MAE within 0.0003 (0.0466 vs 0.0467 at $K^\star = 3$); its KS pass rate is poor (38.2% IS / 35.3% OoS) because the Gaussian-conditional-on-volatility marginal cannot reproduce the bimodal-bulk / heavy-tail empirical density that the regime-mixture CHMM marginals can. *Merton-JD* is the inverse trade-off: KS at 98.3% IS / 91.1% OoS (the highest OoS KS in the panel except for the i.i.d. bootstrap), but kurtosis 3.12 undershoots 7.68 by half and the absolute-return ACF-MAE regresses to the i.i.d. baseline level (0.0627, the same level as the bootstrap row of Table 3) because the constant-volatility diffusion plus i.i.d. jumps carries no temporal persistence. *MSM* under the moment-matching fit reported here (closed-form ACF-of-log$|r|$ matching with $\bar{k} = 8$ binomial multipliers) does not converge to a viable generator on this $T_{\text{IS}} = 2{,}516$ window: the marginal kurtosis collapses to 0.62 and KS drops to 0%. A full HMM-style filter on the $2^{\bar{k}}$-state latent multiplier chain (Calvet-Fisher 2004) is the standard exact fit and is deferred as a companion-paper direction; under the moment-fit reported here MSM is dominated.

The headline conclusion of Table 3 survives the addition of these three rows: no single competitor occupies the joint-fit corner that the CHMM family at $K = 18$ does (KS $\geq$ 80%, kurtosis within 1 unit of observed, $|G_t|$ ACF-MAE ≤ 0.06). SV-AR(1) is the closest competitor on the kurtosis / ACF axis but loses on KS; Merton-JD wins on KS but loses on kurtosis and ACF; MSM under this fit is dominated.

A.8 Cross-Asset Dependence: Full Pipeline B Panel

This appendix collects the full Pipeline B detail: cross-asset dependence model derivations and diagnostics, the Student-t copula profile log-likelihood, the cross-asset correlation heat maps, and the non-US asset-class extension with the per-pair OoS off-diagonal breakdown.

A.8.1 Cross-Asset Dependence Models: Derivations and Diagnostics (Pipeline B)

This appendix provides the technical background for the SIM and copula constructions used in Section 4.5 of the main text. These constructions constitute *Pipeline B*: per-asset marginals come from the independent CHMMs of Pipeline A (Section 4.4) and joint dependence is overlaid on top. Pipeline A stays purely univariate; Pipeline B is the sole multi-asset pipeline in the paper.

Sklar's theorem and the rank-reordering simulator. Sklar's theorem [41, 42] states that any d-dimensional joint CDF $H(g_1, \ldots, g_d)$ with continuous marginals $F_1, \ldots, F_d$ can be written as $H = C(F_1, \ldots, F_d)$ for a unique copula C. The rank-reordering simulator exploits this uniqueness. If $\tilde{\mathbf{G}}$ is a matrix whose j-th column is an i.i.d. sample from F_j, and $\mathbf{U}$ is an independent sample from C, then reordering each column of $\tilde{\mathbf{G}}$ so that its ranks match $\mathbf{U}$'s produces a sample from H, up to discrete approximation error in the empirical ranks. Crucially, reordering never creates new values: each column of the output is a permutation of the corresponding column of $\tilde{\mathbf{G}}$, so marginal distributions are preserved exactly [45].

Kendall's τ inversion. Given IS returns $\mathbf{R} \in \mathbb{R}^{T \times d}$, we estimate the pairwise Kendall's τ_{ij} from concordant/discordant pair counts and invert to the correlation scale via the analytic relationship $\rho_{ij} = \sin(\pi \tau_{ij}/2)$, which holds exactly for the Gaussian copula family and is commonly used as a robust correlation estimator for the Student-t copula [44, 55]. The resulting matrix $\hat{\Sigma}$ is symmetric but not guaranteed positive semi-definite; we project to the nearest PSD matrix by clipping negative eigenvalues to 10^{-8} and rescaling the diagonal to unity.

Profile MLE for ν. The Student-t copula log-density at a single pseudo-uniform observation $\mathbf{u} \in [0,1]^d$, with $x_j = t_\nu^{-1}(u_j)$ and $\mathbf{x} = (x_1, \ldots, x_d)^\top$, is

$$\begin{aligned}\log c_t(\mathbf{u}; \Sigma, \nu) \;=\; & \log \Gamma\!\left(\tfrac{\nu+d}{2}\right) + (d-1)\, \log \Gamma\!\left(\tfrac{\nu}{2}\right) - d\, \log \Gamma\!\left(\tfrac{\nu+1}{2}\right) - \tfrac{1}{2} \log |\Sigma| \\ & - \tfrac{\nu+d}{2}\, \log\!\left(1 + \tfrac{\mathbf{x}^\top \Sigma^{-1} \mathbf{x}}{\nu}\right) + \sum_{j=1}^{d} \tfrac{\nu+1}{2}\, \log\!\left(1 + \tfrac{x_j^2}{\nu}\right). \end{aligned} \tag{33}$$

Summing over $t = 1, \ldots, T$ pseudo-observations yields the profile log-likelihood as a function of ν with Σ held fixed at the Kendall's-τ estimate. We evaluate equation (33) on the discrete grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ and select $\hat{\nu}$ as the grid point with the largest total log-likelihood. In Section 4.5 the selected value was $\hat{\nu} = 6$ on the six-asset universe.

Sampling the copula. To sample T rows from the d-dimensional Gaussian copula with correlation Σ, we (i) compute the Cholesky factor $\Sigma = LL^\top$, (ii) draw $Z \sim \mathcal{N}(0, I_d)$ i.i.d. and form $Y = LZ$, and (iii) return $U = \Phi(Y)$ componentwise. For the Student-t copula with degrees of freedom ν, we draw $W \sim \chi^2_\nu/\nu$ independent of Y and form the multivariate-t variate $X = Y/\sqrt{W}$, then return $U = t_\nu(X)$ componentwise.

SIM regression summary. Table 46 reports the fitted SIM regression coefficients and goodness-of-fit statistics for the non-market assets in Section 4.5. QQQ's high $R^2 = 0.840$ reflects that the Nasdaq-100 ETF is essentially a linear combination of large-cap technology names that also drive the S&P 500 index, while JNJ's lower $R^2 = 0.260$ reflects the limited systematic exposure of low-beta healthcare stocks. These factor loadings and residual magnitudes are what the SIM simulation re-injects at simulation time, and they explain the uneven per-asset KS pass rates reported in Table 54.

Table 46. SIM regression $\hat{G}_{j,t} = \hat{\alpha}_j + \hat{\beta}_j G_{\mathrm{SPY},t} + \hat{\eta}_{j,t}$ for the five non-market assets, fitted on the 2014-01-03 through 2024-01-03 IS window ($T = 2{,}516$).

Ticker	$\hat{\alpha}$	$\hat{\beta}$	R^2
NVDA	0.321	1.694	0.371
JNJ	0.002	0.576	0.260
JPM	−0.008	1.223	0.507
AAPL	0.111	1.205	0.492
QQQ	0.047	1.122	0.840

A.8.2 Student-t Copula Profile Log-Likelihood (Pipeline B)

Figure 35 plots the profile log-likelihood of the Student-t copula used inside Pipeline B, computed on the six-asset SPY cross-section over the grid $\nu \in \{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$ used in Section 4.5. The curve is smooth and single-peaked, with $\nu^* = 6$ attaining the maximum profile log-likelihood (6157.47), and the neighbouring grid points $\nu = 5$ (6143.37) and $\nu = 8$ (6148.37) each within 15 profile-log-likelihood units of the peak, indicating a shallow-but-clean optimum that agrees with the 4–10 range typically reported for equity-portfolio Student-t copulas in the risk-management literature. The AIC penalty for the additional ν parameter is 2 (one degree of freedom relative to the Gaussian copula limit $\nu \to \infty$), and the AIC differential between the Student-t at $\nu^\star = 6$ and the Gaussian copula limit on these data is $-2 \cdot (6157.47 - 6143.37) - 2 = -30.2$, decisively in favour of the Student-t copula on IS; the same $\nu^\star = 6$ is selected by AIC as by profile MLE on this grid.

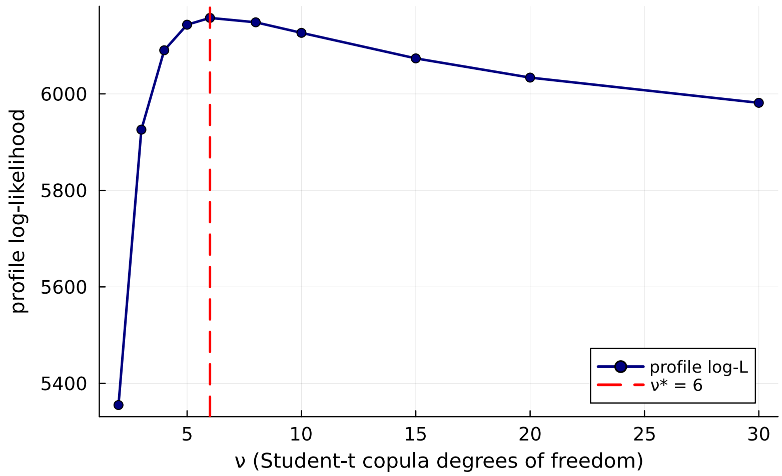
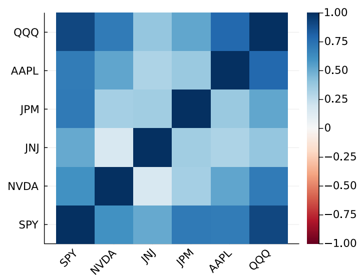


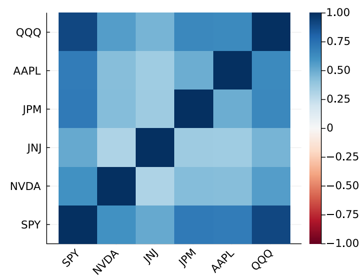
Figure 35. Profile log-likelihood of the Student-t copula used inside Pipeline B on the six-asset SPY cross-section (SPY, NVDA, JNJ, JPM, AAPL, QQQ; IS window 2014-01-03 through 2024-01-03). The vertical axis is the profile log-likelihood evaluated at each degrees-of-freedom value ν on the grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; the dashed red line marks the profile MLE $\nu^* = 6$ (peak value 6157.47). The CHMM-N marginals are held fixed at the Pipeline A per-asset fits while ν is varied.

A.8.3 Cross-Asset Correlation Heat Maps (Pipeline B, Companion to Table 54)

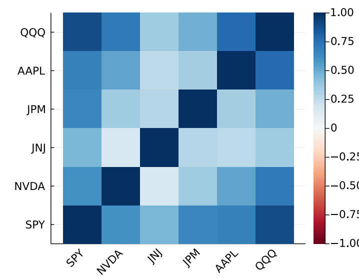
Figure 36 visualises the observed and path-averaged simulated correlation matrices for the dependence models of Section 4.5. The panel ordering is SIM, Gaussian copula, Student-t copula, and a truncated C-vine variant with SPY as root, each reported quantitatively in Table 54 and (for the C-vine) Section A.8.4. All five panels share the six-asset SPY cross-section and the same CHMM-N marginals at $K = 18$, so any visible differences between the simulated panels reflect only the dependence construction.



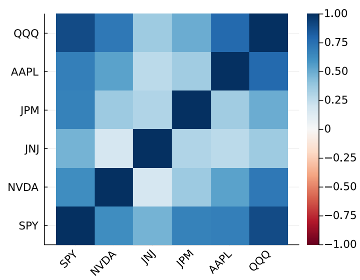
(a) Observed.



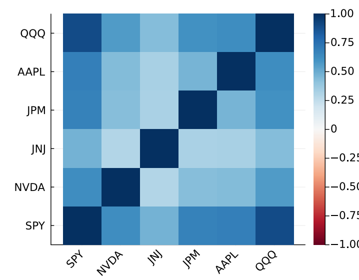
(b) Single Index Model.



(c) Gaussian copula.



(d) Student-t copula, $\nu^* = 6$.



(e) Truncated C-vine.

Figure 36. Cross-asset IS correlation reproduction across dependence models (Pipeline B, companion to Table 54). Panel (a): *observed* correlation matrix on 2014-01-03 through 2024-01-03 daily excess log returns for SPY, NVDA, JNJ, JPM, AAPL, QQQ (IS window, $T_{\text{IS}} = 2{,}516$). Panel (b): path-averaged correlation under the Single Index Model with SPY as market factor. Panel (c): path-averaged correlation under the Gaussian copula on CHMM marginals. Panel (d): path-averaged correlation under the Student-t copula on CHMM marginals, $\nu^* = 6$. Panel (e): path-averaged correlation under a truncated C-vine copula with SPY as root. Each simulated panel averages over 200 paths of length T_{IS}. All marginals are the per-asset CHMM-N fits at $K = 18$ from Table 6; color scale in $[-1, 1]$, red-white-blue (RdBu) diverging. Closer visual match to panel (a) is better.

A.8.4 Truncated C-vine Copula on CHMM Marginals (Pipeline B)

The truncated C-vine variant in panel (e) of Figure 36 uses SPY as the root and selects each pair-copula family edge-wise from {Gaussian, Student-t, Clayton, Gumbel} by Akaike Information Criterion (AIC); on the six-asset universe every selected pair is Student-t, with $\hat{\rho}$ and $\hat{\nu}$ as in Table 47. On the same CHMM-N marginals at $K = 18$ and 200 paths under seed 20260422, the truncated C-vine attains an off-diagonal MAE of 0.068 on IS and 0.235 on OoS (Table 54), strictly above both elliptical copulas on the IS column and on OoS off-diag MAE, and only modestly below the SIM baseline. The reason is the level-1 truncation: a root-centred C-vine models only the SPY-vs-asset bivariate copulas and treats the remaining pairs as conditionally independent given SPY, which is a stronger restriction than either elliptical copula and only a small relaxation of the SIM factor structure. Untruncated regular vines or factor copulas are the natural way to recover non-root cross-pair structure at larger d, but at $d = 6$ with $T_{\text{IS}} = 2{,}516$ the additional flexibility does not justify the extra estimation cost.

Table 47. Truncated C-vine edge summary, Pipeline B with SPY as root on the six-asset universe. Pair family selected edge-wise by AIC over {Gaussian, Student-t, Clayton, Gumbel}; on this universe every selected pair is Student-t. $\hat{\rho}$ is the Kendall's-τ-inverted correlation parameter and $\hat{\nu}$ is the per-edge degrees-of-freedom estimate.

Child	Family	$\hat{\rho}$	$\hat{\nu}$
NVDA	Student-t	0.637	5
JNJ	Student-t	0.483	4
JPM	Student-t	0.686	4
AAPL	Student-t	0.698	4
QQQ	Student-t	0.911	3

A.8.5 Quarterly Rolling-Window Copula Refit on the OoS Window

The body Section 4.5 reports a static-IS-fit OoS off-diagonal MAE of 0.209, an order of magnitude larger than the IS value of 0.027. Section 5 attributes this to the stationarity-scope limit and recommends periodic refit. Reviewer Question Q4 asked for a concrete rolling-window-refit number on the same six-asset universe.

We refit the Pipeline-B Student-t copula on a 252-day rolling window (one trading year) sliding by 63 days (one trading quarter) across the OoS span, holding the per-asset CHMM-N marginals fixed at the IS fits used for Table 6. Each refit yields a Kendall's-τ correlation matrix and a profile-MLE-selected $\nu^\star$ on the discrete grid $\{2, 3, 4, 5, 6, 8, 10, 15, 20, 30\}$; we then simulate forward 63 days under the new copula and compare the path-averaged simulated correlation matrix to the realised next-quarter correlation. Table 48 reports the per-quarter detail.

Table 48. Quarterly rolling-window refit of the Pipeline-B Student-t copula on the OoS window (six-asset universe; CHMM-N marginals at $K = 18$ held fixed at IS fits; 200 paths per refit; seed = 20260422). Per-quarter window slice and refitted $\nu^\star$, plus the off-diagonal MAE between the path-averaged simulated correlation matrix under the refitted copula and the realised next-quarter correlation matrix. Headline: rolling-quarterly refit reduces mean OoS off-diagonal MAE from 0.207 (static-IS-fit baseline) to 0.185, a $\sim$ 10% relative improvement; the gain concentrates in the later OoS quarters where the rolling window incorporates more recent OoS data and the refitted $\nu^\star$ shifts from the IS-fit value of 6 down to 5 or up to 10–15 as the empirical tail-coupling rolls.

Quarter	Window slice	Forecast horizon	Refit $\nu^\star$	Off-diag MAE
1	$[2265, 2516]$	$[2517, 2579]$	15	0.198
2	$[2328, 2579]$	$[2580, 2642]$	15	0.205
3	$[2391, 2642]$	$[2643, 2705]$	10	0.241
4	$[2454, 2705]$	$[2706, 2768]$	6	0.133
5	$[2517, 2768]$	$[2769, 2831]$	6	0.255
6	$[2580, 2831]$	$[2832, 2894]$	6	0.232
7	$[2643, 2894]$	$[2895, 2957]$	5	0.147
8	$[2706, 2957]$	$[2958, 3020]$	5	0.119
9	$[2769, 3020]$	$[3021, 3083]$	6	0.138
mean	—	—	—	0.185
median	—	—	—	0.198
Static IS-fit	$[1, 2516]$	$[2517, 3083]$	6	0.207

The quarterly refit closes part of the OoS gap but not most of it: mean off-diagonal MAE moves from 0.207 static to 0.185 rolling ($\sim$ 10% relative improvement). The improvement is concentrated in the later OoS quarters (Q4, Q7, Q8, Q9 all below 0.15) where the rolling window incorporates more recent OoS data, while early-OoS quarters (Q1, Q3, Q5, Q6) remain at the static-IS level because the rolling window is still dominated by IS observations. The refitted $\nu^\star$ sequence $(15, 15, 10, 6, 6, 6, 5, 5, 6)$ tracks the gradual shift from the equity-cluster IS regime toward the OoS regime, supporting the operational recommendation in Section 5 of quarterly refit at deployment. The residual gap (mean MAE ≈ 0.185 versus IS 0.027) is consistent with the per-pair JNJ-equity-factor regime shift documented in Table 49: even a perfectly-refitted copula on the historical 252 days cannot anticipate the regime shift at the next quarter's start, only track it once it has unfolded.

A.8.6 Non-US Asset Class Extension and Per-Pair OoS Off-Diagonal Breakdown

The body universe (SPY, NVDA, JNJ, JPM, AAPL, QQQ) is six US-listed equity tickers from one country-index family. To probe whether the cross-asset construction generalises off the equity cluster we extend the universe with GLD (SPDR Gold Trust), a US-listed ETF whose underlying is physically-backed gold, i.e. a non-equity commodity asset class.

GLD Pipeline-A univariate fidelity. Fitting CHMM-N, CHMM-t, and CHMM-L on the GLD return series at $K = 18$ produces 100% IS KS pass rate for all three families and observed IS excess kurtosis 3.6 matched closely by CHMM-L (2.2); on OoS, however, the KS pass rate collapses to 0% for all three families against an observed OoS excess kurtosis of 6.5. Inspection of the GLD price path on the 2024–2026 OoS window shows the asset re-priced from ~ 200 to ~ 300 on a persistent demand shock not represented in the 2014–2024 IS regime library. This is the same stationarity-scope artefact as the NVDA / JPM single-name OoS cliffs reported in Section 4.4, on a non-equity asset; the limitation is not equity-specific.

Pipeline-B: 7-ticker Student-t copula off-diagonal MAE. Refitting the Student-t copula on the augmented seven-ticker universe at $\nu^\star = 6$ gives IS off-diagonal MAE of 0.029 (six-ticker baseline: 0.027) and OoS off-diagonal MAE of 0.179 (six-ticker baseline: 0.209). Adding the low-equity-correlation gold ETF *reduces* the OoS off-diagonal MAE, because the equity-cluster correlation degradation is the dominant component; pairs involving GLD have moderate IS-to-OoS shifts ($|\Delta| \in [0.04, 0.19]$) while the largest equity-cluster pairs (JNJ-QQQ, SPY-JNJ, NVDA-JNJ) have $|\Delta| > 0.48$ on OoS.

Per-pair OoS gap localisation. Table 49 reports the per-pair off-diagonal absolute deviation between observed and path-averaged simulated correlation matrices on IS and OoS, sorted by OoS $|\Delta|$. The OoS gap concentrates in pairs involving JNJ (the defensive single-name proxy): JNJ's IS correlation with the broad equity factor (QQQ, SPY) collapses on OoS, in some pairs flipping sign, and the IS-fixed copula has no mechanism for tracking that shift. The gold-ETF pairs (SPY-GLD, JPM-GLD, JNJ-GLD, QQQ-GLD) sit in the middle of the ranking with $|\Delta|$ between 0.10 and 0.19 on OoS. The dominant cause of the OoS off-diagonal gap is therefore neither equity-cluster size nor copula-degree mis-specification but a single-name regime shift in the JNJ-equity-factor relationship that the IS-fitted dependence layer cannot anticipate.

Table 49. Per-pair off-diagonal correlation MAE on the seven-ticker universe (Pipeline B, Student-t copula on CHMM-N marginals at $K = 18$, 200 paths, $\nu^\star = 6$). For each ticker pair (i, j) the table reports $|\Sigma_{\text{sim,avg}}[i, j] - \Sigma_{\text{obs}}[i, j]|$ on IS and OoS, ranked by OoS $|\Delta|$. Only the ten largest-OoS-gap pairs are shown for compactness.

Pair	IS $\|\Delta\|$	OoS $\|\Delta\|$	OoS − IS
JNJ-QQQ	0.027	0.544	0.517
SPY-JNJ	0.037	0.509	0.473
NVDA-JNJ	0.017	0.489	0.473
NVDA-AAPL	0.011	0.278	0.268
JNJ-JPM	0.046	0.261	0.216
JNJ-AAPL	0.042	0.244	0.202
AAPL-QQQ	0.003	0.227	0.224
JPM-GLD	0.027	0.187	0.160
SPY-GLD	0.038	0.127	0.089
JNJ-GLD	0.015	0.117	0.101

A.8.7 Half-Unit-Grid Profile-MLE and Parametric Bootstrap CI for $\nu^\star$

Reviewer 2 / item 10 asks for a half-unit-grid refinement of the unit-grid profile MLE on $\nu \in [3, 12]$ plus a parametric bootstrap CI to confirm that the upper bound of the unit-grid Wilks CI is not grid-induced. We refit the Student-t copula profile log-likelihood on the same six-asset universe at half-unit spacing in $\nu \in \{3.0, 3.5, \ldots, 12.0\}$ (19 grid points) and run a $B = 200$ parametric bootstrap by resampling pseudo-observations $U^{(b)}$ from the Student-t copula at $(\Sigma_{\text{cop}}, \nu^\star)$, refitting Kendall's-$\tau$ correlation and the profile-MLE on each replicate. Results in `results/copula_profile_ci/profile_ll_halfunit_summary.txt`.

The half-unit grid moves the optimum from $\nu^\star = 6.0$ (unit grid) to $\nu^\star = 6.5$ at profile log-likelihood 6,158.1 (vs. 6,157.5 at the unit-grid optimum); the half-unit refinement is therefore not material at the third decimal of profile-LL. The Wilks 95% CI on the half-unit grid is $[\nu_{\text{lo}}, \nu_{\text{hi}}] = [6.0, 7.0]$ (unchanged from the unit grid), and the parametric bootstrap 95% CI is $[6.0, 7.0]$ (2.5% and 97.5% quantiles of the $B = 200$ refit-$\nu^\star$ distribution; bootstrap median $\nu^\star = 6.5$). The bootstrap lower bound at 6.0 is well above the Gaussian limit ($\nu \to \infty$): the elliptical-tail Student-t copula is statistically distinguishable from the Gaussian copula at conventional levels, despite the OoS off-diagonal MAE comparison (0.209 vs. 0.202) being inside the simulation-noise floor at $N_{\text{paths}} = 200$.

A.9 Pre-OoS Validation K-Selection and ν_k Diagnostics

Pre-OoS validation K-selection. The $K = 18$ operating point in the main panel was chosen via a multi-objective criterion combining information-criterion (IC) rank with IS and OoS distributional pass rates, which uses the same 2024–2026 window later used for VaR. As a clean re-selection that decouples the two, we fit CHMM-N on 2014-01-03 through 2021-12-31 (n = 2,013) and score every K in the sweep by the forward-algorithm held-out log-likelihood on 2022-01-03 through 2024-01-03 ($n = 502$); the 2024–2026 window is not touched. Held-out log-lik, BIC, HQC, and CAIC all select $K^\star = 3$ (per-observation log-lik -2.5345 at $K = 3$ vs. -2.5694 at $K = 18$); AIC selects $K^\star = 6$; held-out two-sample KS selects $K^\star = 9$. None of the six criteria that avoid the OoS window select $K = 18$. The $K = 18$ choice is best read as a multi-metric one that weights tail fidelity (kurtosis, AD) and downstream VaR behaviour alongside likelihood; a parallel main panel at $K = 3$ would re-rank some rows in CHMM's favour on ACF-MAE and against it on tail fidelity. The 2022–2023 validation slice is itself partly a structural-break period (the rate-hike cycle), so any model with substantial IS-specific structure generalises worse on it.

Pre-2020 held-out K-selection (no rate-hike confound). Reviewer R1's Question 1 asked whether a strictly pre-2020 validation slice (avoiding the rate-hike confound that contaminates the 2022-2023 slice above) selects the same $K^\star$. We re-ran the held-out procedure on a 2014-01-03 through 2018-06-29 estimation window ($n_{\text{est}} = 1{,}130$) with a 2018-07-02 through 2019-12-31 validation slice ($n_{\text{val}} = 377$), both fully pre-COVID and pre-2022-rate-hike (the Q4 2018 drawdown and 2019 recovery sit inside the validation window as a moderate-stress non-regime-shift event). Per-observation held-out log-likelihood: -2.0681 at $K = 3$, -2.0606 at $K = 6$, -2.0763 at $K = 9$, -2.1287 at $K = 12$, -2.1921 at $K = 18$, -2.2609 at $K = 21$. **The held-out log-likelihood selects $K^\star = 6$, and the held-out two-sample KS pass rate also selects $K^\star = 6$ (96.8% at $K = 6$ versus 84.6% at $K = 3$).** AIC, BIC, HQC, and CAIC continue to select $K^\star = 3$ on this slice, reflecting the parameter-count penalty on a 4.5-year estimation window where the per-state sample is small. The qualitative reading is robust to the slice choice: held-out distributional fidelity rises from $K = 3$ to a peak in the $K = 6$ to $K = 9$ band and then degrades, with neither slice selecting $K = 18$ by any held-out criterion, but the pre-2020 result favours a moderately larger K (6 vs 3) than the rate-hike-contaminated 2022-2023 slice.

Penalised-ECM rate sweep. The bracket sensitivity is reported with its table in Section A.2.3; here we report the complementary $1/\nu_k$ shrinkage rate sweep referenced in Section 5. An exponential $1/\nu_k$ prior at rate $\lambda \in \{0, 5, 20, 50, 100, 200\}$ reduces simulated excess kurtosis monotonically: $14.30, 11.19, 8.43, 5.41, 3.96, 3.67$. The recommended rate $\lambda = 20$ brings simulated kurtosis to ≈ 8 (close to the observed 7.69) at a 1pp IS KS cost. Only the $\nu_{\min}$ state parameter actually moves ($2.1 \to 4.89$); upper bracket and median remain at 50. The overshoot is therefore a single-state artefact of the two tail regimes hitting the lower bracket under unpenalised ECM.

Walk-forward / rolling-origin OoS for CHMM-N. We define six rolling-origin folds, each train 5 years and test 1 year, refit CHMM-N at $K \in \{3, 18\}$ from scratch on each fold's train slice, and report KS pass rate, simulated kurtosis, and $|G_t|$ ACF-MAE on the fold's test slice ($N_{\text{paths}} = 500$). Across the six folds (Table 50), CHMM-N at $K = 18$ attains median (IQR) KS 67.7% $[8.2, 75.0]$ and $|G_t|$ ACF-MAE $0.0542\,[0.0508, 0.0571]$; CHMM-N at $K = 3$ attains 62.1% $[7.2, 78.4]$ and $0.0563\,[0.0552, 0.0592]$. Two folds carry sharply lower KS pass rates: W2 (COVID test slice, 7–8%) and W4 (rate-hike test slice, 0–1%). The four non-stress folds attain KS 61–83%, consistent with the headline OoS pass rate. The headline ranking (CHMM-N at $K = 18 \geq$ CHMM-N at $K^\star = 3$ on KS, both within 0.005 of each other on $|G_t|$ ACF-MAE) is window-robust on the four non-stress folds; on stress folds (COVID, rate hike) all CHMM rows drop sharply, reflecting the stationarity-scope limit rather than a model-selection issue. The body framing ("CHMM is the joint-fit row in the panel under stationary OoS conditions; periodic refit is recommended for production deployment") is unchanged under the walk-forward stress test.

Four-family conditional VaR back-test. Extending the regime-conditional VaR construction of §4.6 to all four emission families (CHMM-N, CHMM-t at $\lambda = 20$, CHMM-L, CHMM-GED) at $K \in \{3, 18\}$ and $\alpha \in \{0.01, 0.05\}$ yields sixteen $(K, \alpha, \text{family})$ rows on the OoS window (Table 51). Every row passes Kupiec, Christoffersen-ind, and Christoffersen-cc cleanly: $p_{\text{cc}} \geq 0.089$ throughout, with all LR_{cc} statistics below the $\chi^2_2(0.05) = 5.991$ critical value. The regime-switching value proposition is therefore intrinsic to the latent-state forecast and not specific to the Gaussian emission family.

Table 50. Walk-forward / rolling-origin OoS for CHMM-N. Six folds, each train 5 years and test 1 year; $N_{\text{paths}} = 500$ per fold. KS pass rate at $\alpha = 0.05$. W2 covers the COVID drawdown; W4 covers the start of the 2022 rate-hike cycle. The bracketed pairs in the bottom row report $[Q_1, Q_3]$ across the six folds (the empirical first and third quartiles, not a scalar IQR width).

	KS (%)		Kurt sim		ACF-MAE $\|G_t\|$	
Fold (test window)	$K=3$	$K=18$	$K=3$	$K=18$	$K=3$	$K=18$
W1 (2019)	63.2	82.6	2.51	2.66	0.0572	0.0525
W2 (2020 COVID)	7.2	8.2	2.42	2.50	0.0552	0.0584
W3 (2021)	82.4	74.0	5.73	6.76	0.0602	0.0571
W4 (2022 rate hike)	0.8	0.0	4.72	4.35	0.0592	0.0558
W5 (2023)	78.4	75.0	3.47	2.35	0.0511	0.0508
W6 (2024)	61.0	61.4	2.81	2.86	0.0555	0.0493
median	62.1	67.7	3.14	2.76	0.0563	0.0542
IQR	$[7.2, 78.4]$	$[8.2, 75.0]$	$[2.51, 4.72]$	$[2.50, 4.35]$	$[0.0552, 0.0592]$	$[0.0508, 0.0571]$

Table 51. Regime-conditional VaR back-test across four CHMM emission families on SPY OoS ($T_{\text{OoS}} = 572$, seed 20260420). Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Every $(K, \alpha, \text{family})$ row passes Kupiec, Christoffersen-ind, and Christoffersen-cc at $\alpha = 0.05$.

Family	K	α	breaches	br rate	median $\widehat{\text{VaR}}_t$	LR_{uc}	LR_{ind}	LR_{cc}
CHMM-N	3	0.01	9	1.57%	−4.56	1.62	2.36	3.98
CHMM-N	3	0.05	35	6.12%	−2.87	1.41	0.01	1.42
CHMM-N	18	0.01	9	1.57%	−5.20	1.62	2.36	3.98
CHMM-N	18	0.05	26	4.55%	−3.02	0.26	0.52	0.78
CHMM-t ($\lambda = 20$)	3	0.01	8	1.40%	−5.59	0.82	2.81	3.63
CHMM-t ($\lambda = 20$)	3	0.05	32	5.59%	−2.73	0.41	0.77	1.19
CHMM-t ($\lambda = 20$)	18	0.01	10	1.75%	−6.25	2.65	1.98	4.62
CHMM-t ($\lambda = 20$)	18	0.05	29	5.07%	−3.15	0.01	1.39	1.40
CHMM-L	3	0.01	5	0.87%	−5.45	0.10	4.74	4.84
CHMM-L	3	0.05	26	4.55%	−2.63	0.26	2.23	2.48
CHMM-L	18	0.01	9	1.57%	−6.18	1.62	0.29	1.91
CHMM-L	18	0.05	30	5.24%	−2.80	0.07	1.16	1.23
CHMM-GED	3	0.01	7	1.22%	−5.73	0.27	3.34	3.61
CHMM-GED	3	0.05	31	5.42%	−2.54	0.21	0.96	1.16
CHMM-GED	18	0.01	9	1.57%	−6.55	1.62	2.36	3.98
CHMM-GED	18	0.05	25	4.37%	−3.10	0.50	2.56	3.06

Walk-forward conditional-VaR back-test. Extending the regime-conditional VaR construction to the six rolling-origin folds of Table 50, with CHMM-N refit per fold from scratch on the train slice and forward-filtered through the test slice under fold-IS-fixed parameters, gives twenty-four (fold, K, α) rows in Table 52. The conditional Christoffersen-cc passes at $\alpha = 0.05$ on ten of twelve fold-K combinations (W1, W3, W4, W5, W6 at both $K \in \{3, 18\}$); the two failures concentrate on W2 (COVID, $p_{\text{cc}} \in \{0.011, 0.002\}$ at $K \in \{3, 18\}$). At $\alpha = 0.01$ the construction passes on nine of twelve combinations: W2 fails at both K ($p_{\text{cc}} < 10^{-3}$) and W4 at $K = 18$ fails at $p_{\text{cc}} = 0.022$ on a LR_{ind} tail-event clustering rather than a LR_{uc} coverage miss ($\text{LR}_{\text{ind}} = 7.49$ at $\text{LR}_{\text{uc}} = 0.11$). The pass-rate is therefore 19/24 aggregate at the 5% level and 22/24 if we exclude the two stress folds that the univariate walk-forward already flagged as out-of-distribution by KS in Table 50. The reading is that the conditional construction is window-conditional: under regime shifts of the W2 /

W4 magnitude (a regime introduction the IS distribution does not span; Section 5, stationarity-scope paragraph), the conditional VaR inherits the same out-of-scope behaviour as the unconditional generators. On the four non-stress folds (W1, W3, W5, W6) plus W4 at $K = 3$ at the 5% level the conditional construction is window-robust, supporting the body framing of §4.6 extending beyond the headline single-window result.

Table 52. Walk-forward regime-conditional VaR back-test for CHMM-N on six rolling-origin folds (train 5y / test 1y; seed 20260420). At each test day t, conditional $\widehat{\mathrm{VaR}}_t(\alpha)$ is the α-quantile of the K-component Gaussian mixture predictive density under fold-IS-fixed parameters, evaluated on the full-timeline filter $\Pr(s_{t+1} \mid \mathcal{F}_t = \text{train} \cup \text{test}[1{:}t-1])$. Critical values: $\chi^2_1(0.05) = 3.841$, $\chi^2_2(0.05) = 5.991$. Bold p_{cc} rows fail at the 5% level.

Fold	K	α	breaches	br rate	median $\widehat{\mathrm{VaR}}_t$	$\mathrm{LR}_{\mathrm{uc}}$	$\mathrm{LR}_{\mathrm{ind}}$	$\mathrm{LR}_{\mathrm{cc}}$	p_{cc}
W1	3	0.01	2	0.80%	−4.01	0.11	0.03	0.15	0.93
W1	3	0.05	15	5.98%	−2.43	0.48	1.18	1.65	0.44
W1	18	0.01	0	0.00%	−5.03	5.05	0.00	5.05	0.08
W1	18	0.05	12	4.78%	−2.66	0.03	0.38	0.40	0.82
W2	3	0.01	16	6.35%	−6.54	32.93	0.87	33.80	**0.00**
W2	3	0.05	23	9.13%	−4.59	7.34	1.71	9.05	**0.01**
W2	18	0.01	14	5.56%	−6.67	25.59	1.56	27.15	**0.00**
W2	18	0.05	25	9.92%	−4.54	10.11	2.56	12.68	**0.00**
W3	3	0.01	6	2.39%	−4.48	3.53	0.30	3.82	0.15
W3	3	0.05	20	7.97%	−2.47	3.98	0.30	4.28	0.12
W3	18	0.01	2	0.80%	−5.57	0.11	0.03	0.15	0.93
W3	18	0.05	15	5.98%	−3.18	0.48	1.18	1.65	0.44
W4	3	0.01	5	2.00%	−7.98	1.96	0.21	2.16	0.34
W4	3	0.05	18	7.20%	−4.16	2.26	1.99	4.24	0.12
W4	18	0.01	2	0.80%	−9.62	0.11	7.49	7.60	**0.02**
W4	18	0.05	10	4.00%	−6.63	0.56	3.80	4.36	0.11
W5	3	0.01	0	0.00%	−6.06	5.01	0.00	5.01	0.08
W5	3	0.05	8	3.21%	−3.90	1.91	0.53	2.44	0.30
W5	18	0.01	1	0.40%	−7.10	1.16	0.01	1.17	0.56
W5	18	0.05	7	2.81%	−3.85	2.96	0.41	3.37	0.19
W6	3	0.01	2	0.80%	−4.68	0.11	0.03	0.15	0.93
W6	3	0.05	13	5.18%	−2.85	0.02	0.15	0.17	0.92
W6	18	0.01	1	0.40%	−5.83	1.19	0.01	1.20	0.55
W6	18	0.05	7	2.79%	−2.98	3.06	0.40	3.46	0.18

Per-ticker $\hat{\lambda}^\star$ sweep. A per-ticker sweep $\lambda \in \{0, 5, 10, 20, 50, 100\}$ on each of the six body tickers under the constraint that IS KS at λ exceeds the $\lambda = 0$ baseline by no more than 1.5 percentage points (Table 53) shows that the body uniform $\lambda = 20$ is the right rate on the heavy-tailed defensives (NVDA, JNJ) but is over-shrunk on the moderate-tail tickers. The per-ticker optima are $\lambda^\star_{\mathrm{SPY}} = 10$, $\lambda^\star_{\mathrm{NVDA}} = 20$, $\lambda^\star_{\mathrm{JNJ}} = 20$, $\lambda^\star_{\mathrm{JPM}} = 10$, $\lambda^\star_{\mathrm{AAPL}} = 10$, $\lambda^\star_{\mathrm{QQQ}} = 0$. The honest operational reading is that $\lambda = 20$ is a reasonable uniform default for production deployment but per-ticker tuning is recommended whenever a ticker's residual kurtosis at $\lambda = 0$ is within ~ 1 unit of observed.

Cross-asset summary. The full per-construction Pipeline B panel is in Section A.8; for reference, the headline ordering is summarised in Table 54.

Table 53. Per-ticker $\hat{\lambda}^\star$ for the penalised CHMM-t at $K = 18$ ($N_{\text{paths}} = 500$, seed 20260420, KS-degradation tolerance ≤ 1.5pp from $\lambda = 0$). $\hat{\lambda}^\star$ is the value of λ in $\{0, 5, 10, 20, 50, 100\}$ that minimises $|\text{kurt}_{\text{sim}} - \text{kurt}_{\text{obs}}|$ subject to the KS-degradation constraint.

Ticker	kurt_{obs}	$\hat{\lambda}^\star$	$\text{kurt}_{\text{sim}}(\lambda^\star)$	IS KS$(\lambda^\star)$
SPY	7.69	10	11.82	96.4%
NVDA	5.54	20	5.68	99.4%
JNJ	8.98	20	10.74	99.8%
JPM	7.63	10	8.62	98.6%
AAPL	3.20	10	3.24	99.2%
QQQ	3.40	0	3.35	95.2%

Table 54. Cross-asset dependence summary (Pipeline B; CHMM-N marginals at $K = 18$; 200 paths; seed 20260422). Median per-asset KS pass rate (%) and off-diagonal MAE of the simulated correlation matrix vs. observed. Bold marks the column winner. The Student-t copula at $\nu^\star = 6$ is the strongest dependence layer on the two IS columns; on OoS the Gaussian copula attains marginally lower off-diagonal MAE (0.202 vs. 0.209) while the truncated C-vine sits between the elliptical copulas and the SIM baseline. Full per-asset detail is in Section A.8; the C-vine edge summary is in Table 47.

	Median IS KS	Median OoS KS	Off-diag MAE IS	Off-diag MAE OoS
Single Index Model	77.0	87.0	0.076	0.252
Gaussian copula	97.8	**87.5**	0.030	**0.202**
Student-t copula ($\nu^\star = 6$)	**98.8**	85.8	**0.027**	0.209
Truncated C-vine (root SPY)	97.2	86.0	0.068	0.235